Sectoral Labor Reallocation, Inflation and Monetary Policy

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Abstract

We study how frictional labor mobility across sectors shapes inflation dynamics and monetary policy in a multi-sector New Keynesian economy, where workers' dynamic discrete-choice decisions govern sectoral mobility. We derive analytical expressions for welfare and a generalized Phillips curve and show that imperfect labor reallocation influences sectoral inflation through two sufficient statistics: a mobility elasticity and the steady-state labor transition matrix. Using a spectral decomposition, we show that the persistence of sectoral inflation is governed not by average price rigidity or aggregate labor mobility alone, but by the price rigidity of the sectors that workers move to and from. When labor reallocation across sticky sectors is imperfect, the effects of sectoral shocks on these sectors become highly persistent. In response, monetary policy balances stabilization of standard aggregate output gaps with persistent labor and price misallocation. When these sticky sectors face a negative productivity shock, optimal monetary policy has an extra incentive to tighten to decrease relative demand and thereby incentivize labor outflows.

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1 Introduction

The multi-sector New Keynesian model is the canonical framework for studying how sectoral heterogeneity—in price rigidity, in input-output linkages, and in the shocks that hit different parts of the economy—shapes inflation and monetary policy. Yet, on the labor margin, the framework often relies on frictionless benchmarks of a unified market in which a representative worker moves freely across sectors to equalize wages, or segmented markets that abstract from reallocation entirely. In the data, sectoral labor flows exist, are small relative to employment stocks, structured by a network of bilateral reallocation flows, and slow to respond to changes in sectoral wages. Answering how monetary policy should respond to shocks requiring sectoral reallocation requires a framework in which nominal rigidity and the speed of labor adjustment are modeled jointly.

Frictional labor mobility introduces a real rigidity that operates simultaneously with the nominal friction of sticky prices. Because this nominal rigidity varies across sectors, shifts in aggregate demand affect sectoral prices unevenly. As a result, monetary policy does more than determine the aggregate price level: it shapes the relative prices that guide how workers reallocate across sectors. This generates a policy trade-off: if the central bank aggressively stabilizes inflation, it may suppress the relative price signals required to move workers toward expanding activities. Conversely, accommodating these relative price adjustments to facilitate efficient reallocation can generate persistent sectoral inflation. Which force dominates depends on the interaction of nominal rigidities and the labor flow network.

To formalize this interaction, we embed a frictional worker-reallocation block into a canonical multi-sector New Keynesian economy and ask how the sticky prices and sluggish worker flows interact to shape the transmission of sectoral shocks, the persistence of inflation, and optimal monetary policy. On the nominal side, sectors differ in price rigidity; on the real side, workers choose sectors based on forward-looking continuation values, subject to bilateral mobility costs and idiosyncratic preference shocks (Artuç et al., 2010). The model nests the standard multi-sector New Keynesian economy with integrated labor markets (e.g., Aoki, 2001) when mobility is frictionless, and the multi-sector economy with segmented labor markets when mobility costs are prohibitive. Despite the richness of the underlying micro structure, the linearized equilibrium depends on the reallocation block only through two sufficient statistics: a mobility elasticity ϕ that governs the responsiveness of worker flows to sectoral value differentials, and the steady-state transition matrix F that summarizes the pattern of gross worker flows across sector pairs.

In Section 3, we derive a generalized Phillips curve: a recursive mapping from aggregate demand and sectoral productivity to sectoral inflation, with a state tracking inherited price misalignment and employment misallocation. The eigenvalues of the associated transition matrix govern the persistence of sectoral inflation. Under homogeneous price stickiness, aggregate

demand shifts marginal costs uniformly, so the system decouples into an aggregate block that monetary policy controls and a relative block that it cannot. At the polar cases, impact and persistence move in opposite directions: segmented markets produce a larger initial inflation response but faster subsequent convergence, because the lack of marginal cost attenuation from worker inflows make firms reprice more aggressively.

In general, the eigenvalues of the transition matrix are implicit functions of the full parameter vector and have no closed-form expressions. We obtain analytical results by imposing a mild symmetry restriction on reallocation costs. Under uniform price stickiness, the system decomposes into independent components, each pairing a Phillips curve with a labor reallocation equation. The persistence of each component is governed by the interaction between nominal rigidity and the speed of labor reallocation along that dimension, with the slower-adjusting margin acting as the bottleneck.

Allowing price stickiness to vary across sectors breaks this independence but yields two key insights. First, when sectors that adjust prices slowly also sit along slow-moving reallocation dimensions, nominal and real frictions compound and adjustment stalls. Second, monetary policy has leverage over reallocation only when sticky- and flexible-price sectors load on opposite sides of a component, so that aggregate demand can tilt relative prices along it.

We formalize the optimal policy response as a dynamic Ramsey problem (Section 5). The second-order welfare loss penalizes aggregate output gaps, sectoral inflation dispersion, relative price distortions, and cross-sector employment misallocation—the last of which is new to our setting and compounds over time when labor reallocation is slow. As in any multi-sector model, the central bank faces a fundamental constraint: a single instrument can move sectoral prices only along the direction of Phillips curve slopes, which generically differs from the direction of efficient relative price adjustment.

The optimal targeting criterion nests the textbook rule when price stickiness is uniform but introduces a reallocation wedge when it is not. The wedge is governed by the interaction between the two diagnostic statistics: effective rigidity determines whether misalignment along a reallocation component is persistent enough to warrant intervention, and the demand spillover determines whether aggregate demand has leverage to correct it. When both are large along the same component, the planner actively tilts demand to redirect worker flows—tightening when efficient reallocation requires outflows from sticky sectors, loosening in the reverse case.

We bring the model to U.S. sectoral data (Section 6), estimating the worker transition matrix from industry-to-industry worker flows in the CPS and measuring sectoral price rigidity from micro price data. We evaluate the alignment between the labor flow network and the distribution of nominal rigidities to assess the quantitative relevance of these channels.

Related Literature

Our paper is most closely related to work on monetary policy during structural reallocation. Guerrieri et al. (2021) show in a two-sector model with downward wage rigidity that sectoral reallocation can shift optimal policy in either direction: contractionary when easing slows reallocation, expansionary when easing accelerates it. Ferrante et al. (2023) study the interaction of inflation and labor reallocation in a multi-sector model with convex hiring costs, showing that reallocation costs account for a large share of post-pandemic U.S. inflation; Bengui et al. (2025) characterize how the phase of reallocation shapes monetary policy transmission. We generalize the mechanisms and policy insights by micro-founding sectoral labor reallocation through workers' forward-looking sector choice, deriving sufficient statistics computable from worker flow data and sectoral price adjustment frequencies that determine both when reallocation frictions amplify the persistence of inflation and when they affect optimal policy.

We also contribute to the study of optimal monetary policy in multi-sector New Keynesian economies. Optimal stabilization puts more weight on sticky-price sectors (Aoki, 2001; Benigno, 2004), a principle that extends to heterogeneous rigidities and production networks (Pasten et al., 2020; La'O and Tahbaz-Salehi, 2022; Rubbo, 2023; Afrouzi and Bhattarai, 2025). In much of this literature, labor markets are either perfectly integrated or exogenously segmented.[1] We study the intermediate case of imperfect sectoral mobility and show that it generates propagation channels and policy wedges that are absent at either polar benchmark.

In our model, imperfect sectoral labor mobility is a source of real rigidity, connecting our framework to the literature on inflation dynamics and the Phillips curve. Nominal rigidities generate large macroeconomic effects when accompanied by real rigidities (Ball and Romer, 1990; Woodford, 2003), with wage rigidity serving as a prominent microfoundation (Blanchard and Galí, 2007); the empirical Phillips curve is quite flat (Hazell et al., 2022), consistent with substantial real rigidity. In our setting, a different labor market friction—imperfect sectoral reallocation—generates persistent endogenous cost-push dynamics: inherited employment misallocation shifts sectoral marginal costs, and these shifts decay slowly because the same frictions that created them also delay their correction.

The magnitude of this mechanism depends on the distribution of nominal frictions across sectors. Micro evidence documents substantial heterogeneity in price adjustment frequencies (Bils and Klenow, 2004; Nakamura and Steinsson, 2008), and this heterogeneity has become central to understanding inflation persistence (Altissimo et al., 2009; Pasten et al., 2020; Carvalho et al., 2021; Pastén et al., 2024). We show that the interaction between this nominal heterogeneity and the topology of worker flows—which sectors are connected, and how quickly workers move between them—determines which dimensions of misalignment are persistent and which

[1] Recent work incorporates limited labor adjustment via imperfect two-sector mobility (Cantelmo and Melina, 2023) or convex hiring costs (Ferrante et al., 2023). Instead, we model worker reallocation as an explicit dynamic transition system that shapes the Phillips curve and welfare.

matter for aggregate inflation.

Our micro-foundation for frictional labor reallocation builds on the literature on forward-looking sector choice, originating from Lilien (1982) and formalized via dynamic discrete choice by Artuç et al. (2010) with general equilibrium applications in Caliendo et al. (2019); Allen et al. (2020); Seo and Oh (2025); Kim et al. (2026). We embed this micro-foundation into a New Keynesian model and show that the reallocation block enters the Phillips curve and welfare loss through our sufficient statistics. Methodologically, we adapt spectral decomposition techniques used in spatial migration (Kleinman et al., 2023) and input-output production networks (Liu and Tsyvinski, 2024) to characterize the dynamics of cross-sector worker flows.

2 Model

We build a multi-sector New Keynesian model in which sectors differ in price rigidity and workers reallocate across sectors through a dynamic discrete choice problem. We describe the production side, household preferences, and the labor reallocation problem, then state the equilibrium and its log-linear approximation.

Time is discrete, $t = 0, 1, \ldots$ The economy consists of N sectors indexed by $i, j \in \mathcal{N} = \{1, \ldots, N\}$.

2.1 Production and Pricing

Final and sectoral aggregation. The economy produces a final good Y_t which is a Cobb-Douglas aggregate of sectoral bundles $\{Y_{it}\}_{i \in \mathcal{N}}$:

$$Y_t = \prod_{i \in \mathcal{N}} (Y_{it}/\alpha_i)^{\alpha_i}, \qquad \sum_{i \in \mathcal{N}} \alpha_i = 1, \tag{1}$$

with associated consumer price index $P_t^C = \prod_i P_{it}^{\alpha_i}$, where P_{it} is the price index of sector i. Each sectoral bundle is a CES aggregate of differentiated varieties $f \in [0, 1]$ with elasticity of substitution $\epsilon_i > 1$:

$$Y_{it} = \left(\int_0^1 Y_{it}(f)^{\frac{\epsilon_i - 1}{\epsilon_i}} \, df \right)^{\frac{\epsilon_i}{\epsilon_i - 1}},$$

yielding demand for variety f and the sectoral price index:

$$Y_{it}(f) = \left(\frac{P_{it}(f)}{P_{it}} \right)^{-\epsilon_i} Y_{it}, \qquad P_{it} = \left(\int_0^1 P_{it}(f)^{1-\epsilon_i} \, df \right)^{\frac{1}{1-\epsilon_i}}.$$

Technology. Each firm f in sector i operates a linear production technology:

$$Y_{it}(f) = A_{it} L_{it}(f), \tag{2}$$

where A_{it} is sector-specific productivity and $L_{it}(f)$ is labor input. Nominal marginal cost is $MC_{it} = W_{it}/A_{it}$, where W_{it} is the nominal wage in sector i.

Pricing. Firms set prices subject to Calvo frictions. In each period, a fraction $1 - \delta_i \in (0,1)$ of firms in sector i can reset their prices. The government provides a revenue subsidy of rate $\frac{1}{\epsilon_i - 1}$ to each sector, financed by lump-sum taxes, that offsets the monopolistic markup and ensures the flexible-price steady state is efficient. Under this subsidy, a re-optimizing firm chooses $P^*_{it}(f)$ to maximize:

$$\max_{P^*_{it}(f)} \mathbb{E}_t \sum_{s=0}^{\infty} \delta_i^s Q_{t,t+s} \left(\frac{P^*_{it}(f)}{P_{it+s}} \right)^{-\epsilon_i} Y_{it+s} \left[\frac{\epsilon_i}{\epsilon_i - 1} P^*_{it}(f) - MC_{it+s} \right], \tag{3}$$

where $Q_{t,t+s} = \beta^s (C_{t+s}/C_t)^{-\gamma} (P^C_t / P^C_{t+s})$ is the stochastic discount factor.

2.2 Households

A representative family consists of a unit mass of workers distributed across sectors. Let N_{it} denote the fraction of workers in sector i at date t, with $\sum_{i \in \mathcal{N}} N_{it} = 1$. Each worker in sector i supplies L_{it} hours during period t. The distribution $\{N_{it}\}$ evolves through a reallocation process described in Section 2.3.

Preferences. The family maximizes:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \sum_{i \in \mathcal{N}} N_{it} \zeta_i \frac{L_{it}^{1+\varphi}}{1+\varphi} - \Psi_t \right], \tag{4}$$

where $\beta \in (0,1)$ is the discount factor, $\gamma > 0$ is the inverse of the elasticity of intertemporal substitution, $\varphi > 0$ is the inverse Frisch elasticity, $\zeta_i > 0$ are sector-specific disutility shifters, and Ψ_t is the cost of labor reallocation incurred in period t (defined in Section 2.3).

Budget constraint. The family faces the nominal budget constraint:

$$P^C_t C_t + B_t \leq (1 + i_{t-1}) B_{t-1} + \sum_{i \in \mathcal{N}} W_{it} L_{it} N_{it} + \Pi_t + T_t, \tag{5}$$

where B_t are one-period nominal bonds paying gross interest $1 + i_t$ set by the central bank, Π_t are aggregate profits, and T_t are net lump-sum transfers from the government.

Optimality conditions. Maximizing (4) subject to (5), taking $\{N_{it}\}$ as given, yields the intratemporal labor supply condition for each sector i:

$$\frac{W_{it}}{P_t^C} = \zeta_i L_{it}^{\varphi} C_t^{\gamma}, \tag{6}$$

and the intertemporal Euler equation:

$$1 = \beta(1+i_t)\,\mathbb{E}_t\left[\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}\frac{P_t^C}{P_{t+1}^C}\right]. \tag{7}$$

2.3 Labor Reallocation

We model labor mobility as a family-level choice: at the start of period t, the representative household reallocates workers across sectors subject to reallocation costs Ψ_t.[2]

Reallocation cost. In each period t, the family incurs a labor reallocation cost

$$\Psi_t = \sum_{i\in\mathcal{N}} N_{i,t-1}\,\psi_i\Big(\{\mu_t^{ij}\}_{j\in\mathcal{N}}\Big),$$

where $N_{i,t-1}$ is the inherited worker distribution and μ_t^{ij} are the chosen transition probabilities from i to j satisfying $N_{jt} = \sum_i N_{i,t-1}\,\mu_t^{ij}$. We adopt the following functional form for ψ_i, standard in the dynamic discrete choice literature (Anderson et al., 1988; Artuç et al., 2010; Caliendo et al., 2019):

$$\psi_i\Big(\{\mu_t^{ij}\}_j\Big) = \sum_{j\in\mathcal{N}} \mu_t^{ij}\tau^{ij} + \frac{1}{\theta}\sum_{j\in\mathcal{N}} \mu_t^{ij}\log\mu_t^{ij},$$

where $\tau^{ij} \geq 0$ are bilateral reallocation costs ($\tau^{ii} = 0$). This aggregate penalty is equivalent to a decentralized setting in which individual workers draw i.i.d. Type-I Extreme Value preference shocks with dispersion $1/\theta$ (Anderson et al., 1992). The parameter $\theta > 0$ governs the elasticity of labor reallocation: as $\theta \to 0$, labor markets become segmented; conversely, as $\theta \to \infty$, workers move frictionlessly to equalize wages across sectors, recovering a canonical one-labor-type New Keynesian economy (e.g., Rubbo, 2023).

Value functions and optimal flows. Let V_{it} denote the marginal continuation value to the family of an additional worker located in sector i at date t. The family optimally sets transition

[2]The timing within period t is as follows: (1) the period begins with predetermined employment $\{N_{i,t-1}\}$ and prices $\{P_{i,t-1}\}$; (2) aggregate and sectoral shocks at time t are realized; (3) all period-t choices (labor reallocation $\{N_{it}\}$, Calvo price setting, labor supply L_{it}, and consumption) are determined contemporaneously.

rates to equate the marginal cost of moving workers to the realized value differentials $\{V_{jt}\}$. The cost function ψ_i yields transition probabilities of the logit form:

$$\mu_t^{ij} = \frac{\exp\big(\theta[V_{jt} - \tau^{ij}]\big)}{\sum_{j' \in \mathcal{N}} \exp\big(\theta[V_{j't} - \tau^{ij'}]\big)}. \tag{8}$$

Substituting these optimal flows back into the family's objective yields the Bellman equation:

$$V_{it} = U_{it}^{\text{flow}} + \beta\, \mathbb{E}_t \left[\frac{1}{\theta} \log\left(\sum_{j \in \mathcal{N}} \exp\big(\theta[V_{j,t+1} - \tau^{ij}]\big) \right) \right], \tag{9}$$

where the flow utility from working in sector j at time t is $U_{jt}^{\text{flow}} \equiv \frac{W_{jt}}{P_t^C} C_t^{-\gamma} L_{jt} - \zeta_j \frac{L_{jt}^{1+\varphi}}{1+\varphi}$. Using the intratemporal labor supply condition (6), this simplifies to:

$$U_{jt}^{\text{flow}} = \zeta_j \frac{\varphi}{1+\varphi} L_{jt}^{1+\varphi}.$$

Steady-state worker flows. Let F denote the *forward* transition matrix, where $F_{ij} = \bar{\mu}^{ij}$ is the steady-state probability that a worker in sector i moves to sector j. The stationary employment distribution satisfies $\sum_i \bar{N}_i F_{ij} = \bar{N}_j$. We define the *backward* transition matrix B by

$$B_{ji} \equiv \frac{\bar{N}_i}{\bar{N}_j} F_{ij}. \tag{10}$$

While F describes where workers go, B describes where they came from: Bji gives the fraction of workers currently in sector j who arrived from sector i. Both matrices are row-stochastic and share the same stationary distribution $\bar{\mathbf{N}}$. Because $\bar{\mathbf{N}}$ is itself determined by F, the backward matrix B is fully pinned down by F.

2.4 Equilibrium

Market clearing. In equilibrium, the goods market for each variety and the aggregate economy must clear:

$$Y_{it}(f) = C_{it}(f), \quad Y_t = C_t, \tag{11}$$

where C_t is household consumption. Combined with Cobb–Douglas demand, this implies that nominal sectoral output equals the expenditure share of total nominal consumption:

$$P_{it} Y_{it} = \alpha_i P_t^C C_t. \tag{12}$$

Labor market clearing in each sector requires:

$$\int_0^1 L_{it}(f)\, df = N_{it} L_{it}, \tag{13}$$

where N_{it} is the post-reallocation employment distribution. The government finances the output subsidies through lump-sum taxes, balancing its budget each period:

$$T_t + \sum_{i \in \mathcal{N}} \frac{1}{\epsilon_i - 1} P_{it} Y_{it} = 0. \tag{14}$$

The central bank sets the nominal interest rate i_t according to a policy rule specified below.

Definition 1 (Equilibrium)**.** *Given initial conditions* $(\{P_{i,-1}\}, \{N_{i,-1}\})$ *and a sequence of productivity shocks* $\{A_{it}\}$*, an equilibrium is a collection of prices* $\{P_{it}(f), P_{it}, P_t^C, W_{it}, i_t\}$*, quantities* $\{C_t, Y_{it}, L_{it}(f), L_{it}, B_t\}$*, employment shares* $\{N_{it}\}$*, transition probabilities* $\{\mu_t^{ij}\}$*, and value functions* $\{V_{it}\}$ *such that: (i) households optimize consumption, savings, labor supply, and reallocation; (ii) firms optimize pricing subject to Calvo constraints; (iii) all markets clear and the government budget constraint* (14) *holds; and (iv) the central bank follows its policy rule.*

Steady state. We impose that steady-state flow surplus is equalized across sectors. This ensures an interior steady state: without it, the frictionless limit ($\theta \to \infty$) would concentrate all workers in the highest-surplus sector. We achieve this by normalizing $\bar{L}_i = \bar{L}$ and $\zeta_i = \zeta$ for all i.[3] Combining steady-state labor supply (6) with goods market clearing (12) gives $\alpha_i \bar{C}^{1-\gamma} = \zeta \bar{N}_i \bar{L}^{1+\varphi}$, so $\bar{N}_i = \alpha_i$: steady-state employment shares equal expenditure shares.

2.5 Linearized Equilibrium

Log-linearization of the equilibrium conditions around a zero-shock steady state yields a system that organizes into three groups: cost determination, pricing, and demand. While the pricing and demand blocks are standard, the cost determination block is where frictional labor reallocation enters the system.

Notation. Throughout the analysis, we compare the sticky-price and flexible-price equilibria using the following notation. Vectors are set in bold. Lowercase letters denote log-deviations from the steady state ($x_t \equiv \log X_t - \log \bar{X}$), except for interest rates (i_t, r_t^{nat}) which are in levels. The superscript nat denotes flexible-price natural levels (Lemma 1), and tildes denote gaps from this benchmark ($\tilde{x}_t \equiv x_t - x_t^{\text{nat}}$). For sectoral variables, hats denote log-deviations relative

[3] The normalization $\bar{L}_i = \bar{L}$ is without loss of generality: N_{it} measures the mass of workers in sector i, and L_{it} can be interpreted as either per-worker hours or the fraction of workers who supply labor (Galí, 2011). The condition $\zeta_i = \zeta$ then ensures equal surplus.

to the aggregate economy. For example, relative prices and wages are $\hat{p}_{it} \equiv p_{it} - p_t^C$ and $\hat{w}_{it} \equiv w_{it} - p_t^C$, while relative productivity is $\hat{a}_{it} \equiv a_{it} - \sum_j \alpha_j a_{jt}$. Finally, $\overset{(1)}{=}$ and $\overset{(2)}{=}$ denote first- and second-order approximations, and $=$ denotes an exact identity.

2.5.1 Real Marginal Cost and Labor Reallocation

Relative prices and the dynamic reallocation of labor jointly determine real marginal cost. Relative prices follow the law of motion:

$$\hat{p}_{it} = \hat{p}_{it-1} + \pi_{it} - \pi_t^C. \tag{15}$$

Log-linearizing nominal marginal cost $MC_{it} = W_{it}/A_{it}$ and substituting the labor supply condition (6), the production function (2), and Cobb–Douglas sectoral demand,

$$y_{it} - y_t = -\hat{p}_{it}, \tag{16}$$

yields the real marginal cost for sector i:

$$mc_{it} - p_{it} \overset{(1)}{=} (\gamma + \varphi)\, y_t - (1 + \varphi)(\hat{p}_{it} + a_{it}) - \varphi\, n_{it}. \tag{17}$$

Three terms drive sectoral cost pressures: aggregate demand y_t, the relative price–productivity gap $\hat{p}_{it} + a_{it}$, and current employment n_{it}. The third is the novel reallocation channel: an overstaffed sector ($n_{it} > 0$) has lower hours per worker and hence lower marginal cost, while understaffing creates upward cost pressure.

The employment term n_{it} is governed by the labor reallocation block. Normalizing value and flow-utility deviations by $\varphi\zeta\bar{L}^{1+\varphi}$:

$$\mathbf{n}_t \overset{(1)}{=} B\,\mathbf{n}_{t-1} + \phi\,(I - BF)\,\mathbf{v}_t, \tag{18}$$

$$\mathbf{v}_t \overset{(1)}{=} \mathbf{u}_t + \beta\, F\, \mathbb{E}_t[\mathbf{v}_{t+1}], \tag{19}$$

$$u_{it} \overset{(1)}{=} y_t - \hat{p}_{it} - a_{it} - n_{it}, \tag{20}$$

where B, F are the steady-state transition matrices, and $\phi \equiv \varphi\theta\zeta\bar{L}^{1+\varphi}$ is the *mobility elasticity*. The ratio ϕ/φ isolates the elasticity of transition probabilities to destination wages.[4]

Equation (18) is the law of motion for employment: absent value differentials, the distribution reverts toward steady state through B; value gaps at date t induce worker flows contemporaneously through the matrix $(I - BF)$ scaled by ϕ. In the frictionless limit $\phi \to \infty$, $(I - BF)\mathbf{v}_t \to \mathbf{0}$

[4]Normalized deviations are $v_{it} \equiv (V_{it} - \bar{V}_i)/(\varphi\zeta\bar{L}^{1+\varphi})$ and $u_{it} \equiv (U_{it}^{\text{flow}} - \bar{U}^{\text{flow}})/(\varphi\zeta\bar{L}^{1+\varphi})$. Combining steady-state conditions gives $\phi = \varphi\theta\bar{C}^{1-\gamma}$. Appendix D.2 maps this to the dispersion parameter ν in Artuç et al. (2010); Caliendo et al. (2019), giving $\phi = \varphi/\nu$.

forces value differentials to vanish, giving $n_{it} = -(\hat{p}_{it} + \hat{a}_{it})$. Equation (19) is the Bellman recursion: iterating forward,

$$\mathbf{v}_t \overset{(1)}{=} \sum_{s=0}^{\infty} \beta^s F^s \, \mathbb{E}_t[\mathbf{u}_{t+s}],$$

so the continuation value in each sector reflects the discounted present value of the expected path of flow utilities.

2.5.2 Pricing

Given real marginal costs, log-linearizing the firms' optimal reset price under Calvo pricing yields the sectoral New Keynesian Phillips curve:

$$\pi_{it} \overset{(1)}{=} \kappa_i \left[mc_{it} - p_{it}\right] + \beta \, \mathbb{E}_t \pi_{it+1}, \tag{21}$$

where $\kappa_i \equiv (1-\delta_i)(1-\beta\delta_i)/\delta_i$ is the slope of sector i's Phillips curve. The derivation is standard and identical to the one-sector benchmark (Woodford, 2003).

2.5.3 Demand

The IS curve is:

$$\tilde{y}_t \overset{(1)}{=} \mathbb{E}_t \tilde{y}_{t+1} - \frac{1}{\gamma}\left(i_t - \mathbb{E}_t \pi^C_{t+1} - r^{\text{nat}}_t\right), \tag{22}$$

where $\pi^C_t = \sum_i \alpha_i \pi_{it}$ is CPI inflation and $r^{\text{nat}}_t = -\log\beta + \gamma(\mathbb{E}_t y^{\text{nat}}_{t+1} - y^{\text{nat}}_t)$ is the natural real rate. The central bank sets the nominal interest rate according to:

$$i_t = \psi_\pi \, \pi_t(\omega) + \psi_y \, \tilde{y}_t, \tag{23}$$

for some inflation index $\pi_t(\omega) = \sum_i \omega_i \pi_{it}$ with weights summing to one. Sectoral demand y_{it} follows from Cobb–Douglas preferences, giving Equation (16) used above.

The IS curve (22) and Taylor rule (23) are standard. Richer demand structures (heterogeneous agents, nonhomothetic preferences) would modify (22) and (16) but leave the rest of the cost determination block unchanged.

3 Dynamics of Inflation and Labor Reallocation

This section characterizes the joint dynamics of sectoral inflation and labor reallocation. Section 3.1 shows that the linearized reallocation block depends on only two sufficient statistics:

a mobility elasticity and the steady-state worker transition matrix. Section 3.2 uses these to derive a generalized Phillips curve in which employment misallocation acts as an endogenous cost-push force.

We then characterize the persistence of these dynamics (Section 3.3), showing that the persistence of a sectoral shock depends on its exposure to the labor-flow network rather than the shocked sector's own price rigidity. Finally, to isolate the role of mobility frictions, Section 3.4 contrasts the polar cases of segmented and integrated labor markets, demonstrating an inverse relationship between impact inflation and inflation persistence.

Throughout this section, we condition on a given path of aggregate demand $\{y_t\}$; Section 5 studies the optimal choice of that path.

3.1 Sufficient Statistics and the State

Two features of the reallocation block govern the linearized dynamics: how strongly workers respond to value differentials, and which sectors they flow between. Proposition 1 shows that these are the only features that matter.

Proposition 1 (Sufficient Statistics)**.** *The linearized system depends on the labor reallocation structure only through:*

(i) the mobility elasticity $\phi = \varphi\theta\zeta\bar{L}^{1+\varphi}$, and

(ii) the forward transition matrix F (and the backward matrix B derived from it via (10)).

The limiting cases are: $\phi \to 0$ (segmented markets), in which $\mathbf{n}_t = \mathbf{0}$ for all t; and $\phi \to \infty$ (free mobility), in which continuation values equalize across sectors ($\mathbf{v}_t \propto \mathbf{1}$) and n_{it} offsets the relative price–productivity gap ($n_{it} = -(\hat{p}_{it} + \hat{a}_{it})$) at every date $t \geq 0$.

Proof. See Appendix C.1. □

This parallels findings in dynamic spatial models, where the linearized system similarly reduces to a transition matrix and a mobility elasticity (Kleinman et al., 2023; Seo and Oh, 2025). The $\phi \to \infty$ limit recovers the standard multi-sector NK model with a single labor type.

Next, we identify the state variables of the linearized system. We first characterize the flexible-price benchmark. We define the natural level at date t as the solution to the flex-price equilibrium conditional on past variables $(\hat{\mathbf{p}}_{t-1}, \mathbf{n}_{t-1})$, with prices flexible from t onward and the reallocation friction held fixed.[5]

Lemma 1 (Natural Allocation)**.** *The flexible-price allocation satisfies:*

[5] Following Woodford (2003), we re-anchor this counterfactual each period. When computing y^{nat}_{t+1}, the predetermined input is the realized sticky-price state $\mathbf{n}_t$. Consequently, the sequence $\{y^{\text{nat}}_t, \mathbf{n}^{\text{nat}}_t, \dots\}_{t\geq 0}$ captures the period-by-period flex-price response to the actual history.

(i) Aggregate natural output and the natural real rate depend only on aggregate productivity:

$$y_t^{\text{nat}} = \frac{1+\varphi}{\gamma+\varphi}\sum_i \alpha_i a_{it}, \qquad r_t^{\text{nat}} = -\log\beta + \gamma\,\mathbb{E}_t(y_{t+1}^{\text{nat}} - y_t^{\text{nat}}),$$

independent of the predetermined labor distribution $\mathbf{n}_{t-1}$.

(ii) Sectoral natural quantities are state-dependent: the predetermined $\mathbf{n}_{t-1}$ *yields an endogenous natural employment distribution* $\mathbf{n}_t^{nat}$*, which determines sectoral hours and natural relative prices:*

$$l_{it}^{\text{nat}} = \frac{1-\gamma}{1+\varphi}\,y_t^{\text{nat}} - \frac{1}{1+\varphi}\,n_{it}^{\text{nat}}, \tag{24}$$

$$\hat{p}_{it}^{\text{nat}} = \hat{p}_{it}^{\text{nat},0} - \frac{\varphi}{1+\varphi}\,n_{it}^{\text{nat}}, \tag{25}$$

where $\hat{p}_{it}^{\text{nat},0} = \sum_j \alpha_j a_{jt} - a_{it}$ *is the natural relative price when* $\mathbf{n}_t^{\text{nat}} = \mathbf{0}$*. A naturally-overstaffed sector (*$n_{it}^{\text{nat}} > 0$*) has lower hours and a lower natural relative price.*

Proof. See Appendix C.1. □

Parts (i) and (ii) establish that while aggregate natural output depends only on aggregate productivity, yielding a standard one-sector IS curve (22), sectoral natural quantities depend on the endogenous employment distribution $\mathbf{n}_t^{\text{nat}}$. Labor reallocation therefore enters the system entirely through cost determination.

To characterize the joint price and employment dynamics, we define the *relative price distortion* as the deviation of relative prices from their steady-state benchmark:

$$\chi_{it} \equiv \hat{p}_{it} + \hat{a}_{it}. \tag{26}$$

Because $\hat{p}_{it}$ and $\hat{a}_{it}$ always appear together in the marginal cost and flow utility, χ_{it} acts as a sufficient statistic for pricing dynamics.[6] Substituting this into (17), marginal cost becomes:

$$mc_{it} - p_{it} = (\gamma+\varphi)\,\tilde{y}_t - (1+\varphi)\,\chi_{it} - \varphi\,n_{it}, \tag{27}$$

and from (15), the distortion evolves as

$$\boldsymbol{\chi}_t = \boldsymbol{\chi}_{t-1} + \hat{\boldsymbol{\pi}}_t + \Delta\hat{\mathbf{a}}_t. \tag{28}$$

Thus, two variables summarize the pricing and labor allocation dynamics: price misalignment ($\boldsymbol{\chi}_t$) and employment misallocation ($\mathbf{n}_t$). Because both adjust contemporaneously, their inherited

[6] Equation (25) gives $\hat{p}_{it}^{\text{nat}} = \hat{p}_{it}^{\text{nat},0} - \frac{\varphi}{1+\varphi}\,n_{it}^{\text{nat}}$, so $\hat{p}_{it} - \hat{p}_{it}^{\text{nat}} = \chi_{it} + \frac{\varphi}{1+\varphi}\,n_{it}^{\text{nat}}$. Defining χ_{it} relative to the steady-state benchmark keeps it a function of observable sticky-price quantities and makes χ_{it} and n_{it} enter additively in the marginal cost equation.

components form the predetermined state, $\mathbf{s}_t = (\chi_{t-1}, \mathbf{n}_{t-1})$. These two margins interact: sticky prices distort the relative-price signals that drive worker transitions, while the resulting misallocation feeds back into marginal cost. The next subsection characterizes the dynamics of this joint adjustment.

3.2 Transition Dynamics and the Generalized Phillips Curve

The marginal cost equation (27), the sectoral pricing rule (21), and the reallocation block (18)–(20) jointly define a *generalized Phillips curve*: a mapping from the paths of aggregate demand and productivity into sectoral inflation,

$$\{\pi_{it}\}_{i,t} = \mathcal{K}(\{y_t\}_t,\ \{a_{it}\}_{i,t})\,. \tag{29}$$

We now show that this mapping admits a recursive representation in the state $\mathbf{s}_t = (\chi_{t-1}, \mathbf{n}_{t-1})$: price misalignment and employment misallocation inherited from the previous period.[7] We write $\mathrm{Cov}_\alpha(x_i, y_i) \equiv \sum_i \alpha_i (x_i - \bar{x})(y_i - \bar{y})$, with $\bar{x} = \sum_i \alpha_i x_i$, for the α-weighted covariance across sectors.

Proposition 2 (Generalized Phillips Curve)**.** *Under determinacy, the mapping $\mathcal{K}$ in (29) admits a recursive representation:*

(i) ***Transition.*** *The state evolves as*

$$\mathbf{s}_{t+1} = \mathbf{P}\,\mathbf{s}_t + \mathbf{b}_t^y + \mathbf{b}_t^a, \tag{30}$$

with $\mathbf{b}_t^y = \sum_{j\geq 0} \mathbf{H}_j^y\, \mathbb{E}_t[\tilde{y}_{t+j}]$, $\mathbf{b}_t^a = \sum_{j\geq 0} \mathbf{H}_j^a\, \mathbb{E}_t[\Delta\hat{\mathbf{a}}_{t+j}]$, and $\rho(\mathbf{P}) < 1$.

(ii) ***Relative inflation.*** *Current relative inflation $\hat{\pi}_{it} = \pi_{it} - \pi_t^C$ satisfies*

$$\hat{\boldsymbol{\pi}}_t = \Theta_{11}\,\chi_{t-1} + \Theta_{12}\,\mathbf{n}_{t-1} + \sum_{j\geq 0} \Pi_j^y\, \mathbb{E}_t[\tilde{y}_{t+j}] + \sum_{j\geq 0} \Pi_j^a\, \mathbb{E}_t[\Delta\hat{\mathbf{a}}_{t+j}], \tag{31}$$

where all coefficient sequences decay exponentially. The matrix Θ_{12} vanishes in the free-mobility limit; under finite mobility, it is the channel through which inherited employment misallocation enters sectoral inflation.

(iii) ***Aggregate inflation.*** *CPI inflation satisfies*

$$\pi_t^C = \bar{\kappa}(\gamma + \varphi)\,\tilde{y}_t - (1+\varphi)\ \mathrm{Cov}_\alpha(\kappa_i,\, \chi_{it}) - \varphi\ \mathrm{Cov}_\alpha(\kappa_i,\, n_{it}) + \beta\,\mathbb{E}_t \pi_{t+1}^C, \tag{32}$$

[7] Both χ_{t-1} and $\mathbf{n}_{t-1}$ satisfy $\sum_i \alpha_i \chi_{i,t-1} = 0$ and $\sum_i \alpha_i n_{i,t-1} = 0$, so the state is $2(N-1)$-dimensional. Let $\mathcal{R} \equiv \{x \in \mathbb{R}^N : \boldsymbol{\alpha}^\top x = 0\}$ denote the relative subspace; the state lives in $\mathcal{R} \times \mathcal{R}$.

with $\bar{\kappa} \equiv \sum_i \alpha_i \kappa_i$. Sectoral inflation is recovered as $\boldsymbol{\pi}_t = \hat{\boldsymbol{\pi}}_t + \pi_t^C \mathbf{1}$. Sectoral price and employment distortions affect aggregate inflation only through their covariances with the sectoral Phillips slopes κ_i.

Proof. See Appendix C.1. Part (iii) follows from weighting (21) by α_i and substituting (27). □

Proposition 2 is the central object of the section. Given the inherited state $\mathbf{s}_t$ and the expected paths of aggregate demand and productivity, equations (30)–(32) recursively determine current inflation and the next period's state, fully summarizing the path and persistence of inflation.

Equation (32) makes precise how sectoral distortions feed into aggregate inflation: the two covariance terms act as endogenous cost-push shifters, affecting aggregate inflation only when they interact with heterogeneity in price rigidity. The first, $\mathrm{Cov}_\alpha(\kappa_i, \chi_{it})$, is inflationary when sticky-price sectors have excessively high relative prices. The second, $\mathrm{Cov}_\alpha(\kappa_i, n_{it})$, is inflationary when sticky-price sectors are overstaffed; flexible-price sectors are understaffed and facing high marginal costs. These flexible sectors rapidly pass their cost increases into prices, while the overstaffed sticky sectors fail to offset this aggregate pressure with timely price cuts.

Equivalently, the cost-push term can be written as

$$u_t^\pi = \sum_i \alpha_i \left(\bar{\kappa} - \kappa_i\right)\left[(1+\varphi)\,\chi_{it} + \varphi\, n_{it}\right]. \tag{33}$$

The bracketed term is the negative of sector i's relative marginal-cost pressure from (27): it is positive when the sector's relative price is high, or it has high labor supply, or both. Price-rigidity heterogeneity determines whether a given sectoral misalignment *passes through* to CPI; the reallocation block determines how *persistent* that misalignment is.

Equation (32) nests the standard multi-sector New Keynesian benchmark. Under free mobility ($\phi \to \infty$), Proposition 1 gives $n_{it} = -\chi_{it}$, and the two covariances collapse to the single cost-push term $\mathrm{Cov}_\alpha(\kappa_i, \chi_{it})$ familiar from multi-sector NK models. Away from this limit, the two covariances are independent: $\mathrm{Cov}_\alpha(\kappa_i, n_{it})$ is the novel channel, operating even when relative prices are aligned.

Under uniform stickiness, both covariances vanish and the relative and aggregate blocks decouple entirely:

Corollary 1 (Block Separation)**.** *If $\kappa_i = \kappa$ for all i, then $\mathbf{H}_j^y = \mathbf{0}$ and $\Pi_j^y = \mathbf{0}$ for all j: aggregate demand drops out of both the transition equation* (30) *and the observation equation* (31)*, and the state drops out of the aggregate Phillips curve* (32)*. The relative block evolves independently of monetary policy, and the aggregate economy reduces to a standard one-sector New Keynesian model in $(\pi_t^C, \tilde{y}_t)$.*

Proof. See Appendix C.1. □

When $\kappa_i = \bar{\kappa}$, monetary policy can stabilize the aggregate economy but cannot influence sectoral reallocation: aggregate demand shifts all sectoral marginal costs equally, leaving relative prices and reallocation incentives unchanged.[8]

Under heterogeneous stickiness, the covariances are generically nonzero, and the relative block spills over to the aggregate Phillips curve whenever misalignment is correlated with the distribution of nominal rigidities.

Persistence and the transition matrix. The persistence of these cost-push dynamics is governed by the eigenvalues of $\mathbf{P}$. Under free mobility, $\mathbf{P}$ reduces to the standard system in χ_{t-1} with eigenvalues governed by pricing alone. Finite mobility adds a second block tracking $\mathbf{n}_{t-1}$, governed by the reallocation matrices F and B. The two blocks couple through price rigidity $\mathbf{K}$. The resulting eigenvalues combine pricing and reallocation persistence, and their interaction can generate slow-decay components absent from either limit. The next subsection characterizes these eigenvalues through a spectral decomposition of $\mathbf{P}$.

3.3 Spectral Analysis

To analytically characterize the persistence of misalignment and the dynamic propagation of shocks, we eigendecompose the transition matrix $\mathbf{P}$.[9] Write $\mathbf{P} = \mathbf{U}\,\mathbf{\Lambda}\,\mathbf{V}$, where $\mathbf{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_{2(N-1)})$ contains the eigenvalues in decreasing order of absolute value, $\mathbf{U}$ collects the right eigenvectors $\mathbf{u}_h$ as columns, and $\mathbf{V} = \mathbf{U}^{-1}$ collects the left eigenvectors $\mathbf{v}_h^\top$ as rows, with $\mathbf{v}_j^\top \mathbf{u}_h = \mathbf{1}_{j=h}$. Since $\rho(\mathbf{P}) < 1$, all eigenvalues satisfy $|\lambda_h| < 1$. Each right eigenvector $\mathbf{u}_h = (\mathbf{u}_h^\chi, \mathbf{u}_h^n)$ identifies a *corridor*: a joint pattern of inherited price misalignment and employment misallocation that decays by the factor λ_h each period through $\mathbf{P}$, orthogonal to the other corridors.

To characterize how each corridor translates into inflation, we separate its contribution into price-misalignment and employment-misallocation channels. For corridor h, define the inflation response

$$\mathbf{q}_h \;\equiv\; \Theta \mathbf{u}_h \;=\; \underbrace{\Theta_{11}\mathbf{u}_h^\chi}_{\mathbf{q}_h^\chi} \;+\; \underbrace{\Theta_{12}\mathbf{u}_h^n}_{\mathbf{q}_h^n}, \tag{34}$$

so that q_{ih} is sector i's relative inflation response associated with corridor h. The channel $\mathbf{q}_h^\chi$ is inflation generated by inherited relative-price misalignment; $\mathbf{q}_h^n$ is the labor-reallocation channel,

[8] When κ_i varies across sectors, flexible sectors respond more to aggregate demand than sticky ones, generating relative price movements that tilt reallocation incentives. Section 4 characterizes when this channel is quantitatively important.

[9] This adapts the spectral approach to spatial transition matrices in Kleinman et al. (2023) and to input-output networks in Liu and Tsyvinski (2024) to the joint dynamics of prices and employment in a New Keynesian setting.

introducing a cost-push pressure from inherited employment misallocation. It vanishes under free mobility ($\Theta_{12} = \mathbf{0}$).

For a unit productivity shock to sector j ($\Delta\hat{a}_{i0} = \delta_{ij} - \alpha_j$), define the *shock exposure* of corridor h as

$$c_h^{(j)} \;\equiv\; \mathbf{v}_h^\top \, \mathbf{H}_0^a \, (\mathbf{e}_j - \alpha_j \, \mathbf{1}). \tag{35}$$

The exposure $c_h^{(j)}$ measures the initial impact of the shock on corridor h, determined jointly by the shocked sector's position in the labor-flow network and the sectoral distribution of price rigidities. Combined with the eigenvalues $\{\lambda_h\}$ and inflation patterns $\{\mathbf{q}_h\}$, these exposures allow the full transition path of relative inflation to be expressed as a weighted sum of corridor contributions, with each corridor decaying at rate λ_h:

Proposition 3 (Persistence of Relative Prices)**.** *Consider a one-time permanent unit productivity shock to sector j at $t = 0$, starting from steady state, with $\tilde{y}_t = 0$ for all t.*

(i) ***Transition path.*** *Impact relative inflation is $\hat{\boldsymbol{\pi}}_0^{(j)} = \Pi_0^a \, \mathbf{x}_j$. For $t \geq 1$, relative inflation decomposes into corridor contributions:*

$$\hat{\boldsymbol{\pi}}_t^{(j)} \;=\; \sum_{h=1}^{2(N-1)} \lambda_h^{t-1} \, c_h^{(j)} \, \mathbf{q}_h. \tag{36}$$

(ii) ***Half-life.*** *Corridor h has half-life*

$$\tau_h^{1/2} \;=\; \left\lceil \frac{\log(1/2)}{\log|\lambda_h|} \right\rceil, \tag{37}$$

governed by $|\lambda_h|$ alone, independent of the shock.

(iii) ***Long-run neutrality.*** *The cumulative relative inflation in sector i is*

$$\sum_{t=0}^{\infty} \hat{\pi}_{it}^{(j)} \;=\; \alpha_j - \delta_{ij}. \tag{38}$$

Nominal rigidities and reallocation frictions reshape the transition path but do not alter the long-run relative price.

Proof. See Appendix C.1. □

Proposition 3 characterizes the transition path via three ingredients: initial shock exposure ($c_h^{(j)}$), persistence of the corridor (λ_h), and sectoral incidence (q_{ih}). The long-run identity in part (iii) establishes a long-run neutrality result: while nominal rigidities and reallocation frictions

reshape the timing of relative price adjustments, they do not alter the cumulative relative inflation required to reach the flexible-price equilibrium.

The labor-flow network plays two distinct roles. The shock exposure $c_h^{(j)}$ captures the *static incidence* of the shock: how a productivity change maps into contemporaneous distortions across corridors. The eigenvalues $\{\lambda_h\}$ and inflation patterns $\{\mathbf{q}_h\}$ capture *dynamic propagation*: the rate at which activated corridors decay and how their decay shows up in inflation. Importantly, inflation persistence following a shock to sector j is not a function of sector j's own price rigidity in isolation. Rather, it depends on the rigidities of the sectors with which j is linked through labor flows. A flexible-price sector can exhibit highly persistent inflation if it loads onto a "slow" corridor anchored by sticky-price neighbors, and vice versa.

Long-run neutrality is a statement about relative prices. It does not carry over to the aggregate price level: the aggregate Phillips curve has no force that mechanically reverses past inflation, so any transitory cost-push sequence cumulates into aggregate inflation unless policy explicitly targets it.

Proposition 4 (Cumulative Aggregate Inflation)**.** *Following the sector-j productivity shock in Proposition 3, the cumulative aggregate inflation is*

$$p_\infty^{C,(j)} - p_{-1}^C \;=\; \sum_{t=0}^{\infty} \pi_t^C \;=\; \pi_0^C \;+\; \sum_{h=1}^{2(N-1)} \frac{q_h^C\, c_h^{(j)}}{(1-\lambda_h)(1-\beta\lambda_h)}, \tag{39}$$

where $q_h^C \equiv \sum_i \alpha_i\, q_{ih}$ is the $t = 0$ aggregate inflation response of corridor h.

(i) Under uniform stickiness, $q_h^C = 0$ for all h and $p_\infty^{C,(j)} - p_{-1}^C = 0$.

(ii) Under heterogeneous stickiness, $p_\infty^{C,(j)} - p_{-1}^C \neq 0$ generically.

Proof. See Appendix C.1. □

Proposition 4 is the aggregate counterpart to the relative-price neutrality of Proposition 3: relative prices fully revert, but aggregate inflation along the transition path cumulates into a permanent change in the price level.[10]

The formula identifies which sectoral shocks generate the most persistent aggregate inflation. A corridor contributes only if two conditions hold: the shock loads on it ($c_h^{(j)} \neq 0$), and the corridor's inflation pattern is misaligned with the distribution of sectoral rigidities ($q_h^C \neq 0$). Slow corridors amplify their contribution through two compounding factors: $1/(1-\lambda_h)$ is a *duration* factor, reflecting how many periods cost-push pressure lasts; $1/(1-\beta\lambda_h)$ is an *expectations* factor, reflecting how forward-looking price setters front-load the present value of future pressure into today's reset prices. For a corridor with $\lambda_h = 0.95$ and $\beta = 0.99$, their

[10]This is conditional on the path of aggregate demand $\{\tilde{y}_t\}$ taken as given in this section. Monetary policy can undo the price-level change by engineering an appropriate output gap; Section 5 formalizes this tradeoff.

product is roughly 336, demonstrating how heavily slow-moving labor frictions amplify initially modest cost-push shocks.

As such, a shock that looks modest today can load heavily on a slow corridor and generate persistence far exceeding what the shocked sector's own rigidity would suggest. Given observed labor-flow data (F) and the sectoral distribution of price rigidities ($\{\kappa_i\}$), the corridor decomposition is computable; Section 6 quantifies this channel for the U.S. economy.

The corridor decomposition also characterizes how aggregate demand enters these dynamics. Projecting the transition equation (30) onto corridor h,

$$s^*_{ht+1} = \lambda_h s^*_{ht} + \mathbf{v}_h^\top (\mathbf{b}_t^y + \mathbf{b}_t^a), \qquad s^*_{ht} \equiv \mathbf{v}_h^\top \mathbf{s}_t, \tag{40}$$

shows that each corridor evolves as an independent AR(1). The loading $\mathbf{v}_h^\top \mathbf{b}_t^y$ measures policy's leverage over corridor h: under heterogeneous stickiness, aggregate demand tilts cross-sectoral reallocation by activating corridors differentially rather than uniformly.

3.4 Segmented versus Integrated Markets

To clarify how labor reallocation affects the persistence of the system, we examine the eigenvalues of $\mathbf{P}$ at the two polar extremes of labor mobility.

When labor markets are segmented ($\phi = 0$), employment never moves ($\mathbf{n}_t = \mathbf{0}$ for all $t \geq 0$, starting from $\mathbf{n}_{-1} = \mathbf{0}$). The shock-generated dynamics reduce to a single-variable system in χ_{t-1}, and firms face the full intensive-margin slope $\kappa_i(1+\varphi)$ in marginal cost. Under uniform stickiness, the restriction of $\mathbf{P}$ to this subspace has stable root $\lambda_{\text{seg}} \in (0,1)$ solving

$$\beta\lambda^2 - [\kappa(1+\varphi) + \beta + 1]\lambda + 1 = 0. \tag{41}$$

When labor is perfectly mobile ($\phi \to \infty$), employment offsets price distortions contemporaneously: $n_{it} = -\chi_{it}$ at every date $t \geq 0$, including impact. The shock-generated dynamics again reduce to a single-variable system, now in the subspace where $\mathbf{n}_t = -\boldsymbol{\chi}_t$, and the effective slope drops to κ_i: reallocation absorbs the intensive-margin cost pressure. Under uniform stickiness, the restriction of $\mathbf{P}$ to this subspace has stable root $\lambda_{\text{int}} \in (0,1)$ solving

$$\beta\lambda^2 - (\kappa + \beta + 1)\lambda + 1 = 0. \tag{42}$$

The two characteristic equations differ only in the coefficient on λ: $\kappa(1+\varphi) + \beta + 1$ versus $\kappa + \beta + 1$. Since $\kappa\varphi > 0$, the segmented-market equation has a larger middle coefficient, which pushes the stable root down: $\lambda_{\text{seg}} < \lambda_{\text{int}}$. Thus we have:

Proposition 5 (Segmented vs. Integrated Markets)**.** *Under uniform stickiness ($\kappa_i = \kappa$), consider a one-time permanent productivity shock with $\tilde{y}_t = 0$:*

(i) *Segmented labor markets ($\phi = 0$) produce a larger initial inflation response and faster subsequent decay at rate λ_{seg}.*

(ii) *Integrated labor markets ($\phi \to \infty$) produce a smaller initial inflation response and slower subsequent decay at rate $\lambda_{\text{int}} > \lambda_{\text{seg}}$.*

(iii) *Cumulative relative inflation is $\sum_{t=0}^{\infty} \hat{\pi}_{it} = -\hat{a}_i$ in both cases.*

Proof. See Appendix C.1. □

The mechanism is the effective marginal-cost slope. Under integration, contemporaneous reallocation offsets part of the current cost pressure, reducing the effective slope from $\kappa(1+\varphi)$ to κ. The lower slope attenuates impact inflation but also weakens the force driving distortions back to zero, so convergence is slower. Under segmentation, current cost pressures are unmitigated and firms reprice aggressively: the initial response is larger, but convergence is faster.

Part (iii) is an accounting identity: long-run relative prices must fully reflect the productivity shock regardless of mobility. Mobility reshapes the path but not the destination.

For intermediate ϕ, both margins are active and the eigenvalues of $\mathbf{P}$ depend on their interaction. In general, these eigenvalues can only be computed numerically, may be complex, and offer no direct interpretation of how price adjustment and labor reallocation jointly determine persistence. Section 4 obtains closed-form expressions under an additional mild restriction on the reallocation cost structure.

4 Closed-Form Results under Quasi-symmetry

The eigenvalues of $\mathbf{P}$ are implicit functions of the full parameter vector $(\varphi, \beta, \phi, F, B, \mathbf{K})$ and have no closed-form expressions in general. In this section, we analytically characterize them by restricting bilateral reallocation costs to be quasi-symmetric:

Assumption 1 (Quasi-Symmetric Reallocation Costs)**.** *The bilateral reallocation costs satisfy $\tau^{ii} = 0$ and $\tau^{ij} = \tau_i^{out} + \tau_j^{in} + \hat{\tau}^{ij}$ for $i \neq j$, where $\hat{\tau}^{ij} = \hat{\tau}^{ji}$ for all i, j.*

This is standard in the dynamic discrete choice literature (Artuç et al., 2010; Allen et al., 2020; Seo and Oh, 2025), nesting symmetric costs and additive origin-destination costs. Its key implication is that F is reversible ($B = F$), with real eigenvalues and an orthogonal eigenbasis:

Lemma 2 (Diagonalization of F)**.** *Under Assumption 1:*

(i) *The steady-state transition matrix satisfies $\bar{N}_i F_{ij} = \bar{N}_j F_{ji}$, so $B = F$.*

(ii) *The matrix $Q \equiv A^{1/2} F A^{-1/2}$, where $A = \text{diag}(\alpha_i)$, is real symmetric with eigenvalues $1 = \rho_1 > \rho_2 \geq \cdots \geq \rho_N > -1$ and orthonormal eigenvectors $\{s_k\}$.*

Proof. See Appendix C.2. □

Reversibility is a close approximation in U.S. data: the asymmetry measure $\|Q - Q^\top\| / \|Q\|$ is 8×10^{-4} (Appendix D.1). Under uniform stickiness (Section 4.1), the diagonalization of F block-diagonalizes the full supply-side system into $N-1$ independent components, each pairing a Phillips curve with a reallocation equation. Section 4.2 relaxes uniform stickiness and derives two diagnostic statistics that characterize when reallocation frictions amplify persistence and when monetary policy has leverage.[11]

Table 1 collects the key objects from Sections 3.3–4.2 for reference.

Object	Space	Role
λ_h	**P**-eigenvalue	persistence of corridor h
$c_h = \mathbf{v}_h^\top \mathbf{s}_1$	scalar	how much a shock projects onto corridor h
$\Theta \mathbf{u}_h$	$\mathbb{R}^{N-1}$	relative inflation along corridor h
ρ_k	F-eigenvalue	labor mixing speed along component k
$\Delta\kappa_k^{\text{eff}}$	scalar	whether component k connects sticky sectors
d_k	scalar	whether demand tilts relative prices along component k

Table 1: Key objects. The first three are defined in Section 3.3 for any F; the last three require quasi-symmetry (Assumption 1).

4.1 Independent Components under Uniform Stickiness

We first study the special case of uniform price stickiness ($\kappa_i = \kappa$). Under quasi-symmetry, block separation (Corollary 1) cuts the aggregate block from the relative block, and the diagonalization of F (Lemma 2) further decomposes the relative block into $N-1$ independent component-level systems. Each component reduces to a scalar dynamical system whose persistence is a single eigenvalue λ_k^*, solving a quartic in $(\kappa, \beta, \varphi, \phi, \rho_k)$. These eigenvalues isolate how nominal and reallocation frictions interact along each labor-flow component, and serve as the baseline against which Section 4.2 measures the effect of heterogeneous stickiness.

Lemma 3 (Decoupling and Component Persistence)**.** *Under Assumption 1 and uniform stickiness ($\kappa_i = \kappa$):*

(i) ***Decoupling.*** *The coordinate transformation $x_{kt}^* = s_k^\top A^{1/2} \hat{\mathbf{x}}_t$ for $x \in \{\chi, n, v, \hat{a}\}$ decomposes the*

[11] We reserve *corridor* (index h) for the eigenvectors of the full transition matrix **P**, which are joint price-employment patterns in $\mathbb{R}^{2(N-1)}$ (Section 3.3), and *component* (index k) for the eigenvectors of F, which are pure labor-flow patterns in $\mathbb{R}^N$. Under uniform stickiness, each component generates a pair of corridors through the quartic (43).

relative block into $N-1$ independent three-variable systems, one for each component $k \geq 2$:

$$\begin{aligned}
[\kappa(1+\varphi)+\beta+1]\,\chi^*_{kt} &= \beta\,\mathbb{E}_t[\chi^*_{kt+1}] + \chi^*_{kt-1} - \kappa\varphi\, n^*_{kt} + \Delta\hat{a}^*_{kt} - \beta\,\mathbb{E}_t[\Delta\hat{a}^*_{kt+1}],\\
n^*_{kt} &= \rho_k\, n^*_{k,t-1} + \phi(1-\rho_k^2)\, v^*_{kt},\\
v^*_{kt} &= -\chi^*_{kt} - n^*_{kt} + \beta\rho_k\,\mathbb{E}_t[v^*_{kt+1}].
\end{aligned}$$

(ii) ***Component persistence.*** *The eigenvalues of* $\mathbf{P}$ *associated with component k solve the quartic*

$$\begin{aligned}
&\Big[(\lambda-\rho_k)(1-\beta\rho_k\lambda)+\phi(1-\rho_k^2)\lambda\Big]\Big[\beta\lambda^2-(\kappa(1+\varphi)+\beta+1)\lambda+1\Big]\\
&\qquad + \kappa\varphi\phi(1-\rho_k^2)\lambda^2 = 0. \qquad (43)
\end{aligned}$$

Exactly two roots satisfy $|\lambda|<1$; let λ^*_k denote the larger in absolute value. Then λ^*_k is continuous in ϕ, with

$$\lim_{\phi\to 0}\lambda^*_k = \max\{\lambda_{\text{seg}},\,|\rho_k|\}, \qquad \lim_{\phi\to\infty}\lambda^*_k = \lambda_{\text{int}},$$

where λ_{seg} and λ_{int} are the stable roots of (41) *and* (42).

Proof. See Appendix C.2. □

Each component k pairs a Phillips curve with a reallocation block. The Phillips curve is a second-order difference equation in χ^*_{kt}, with employment misallocation n^*_{kt} entering as an endogenous cost-push shifter. The reallocation block drives workers to close the gap $-\chi^*_{kt} - n^*_{kt}$: inherited employment $n^*_{k,t-1}$ mean-reverts toward zero at rate ρ_k, and contemporaneous value differentials v^*_{kt} tilt this mean reversion with elasticity ϕ. Under uniform stickiness, the dynamics along component k depend on F only through ρ_k.

At $\phi = 0$, the quartic factors into the price and labor margins separately, and persistence is the slower of the two: $\max\{\lambda_{\text{seg}}, |\rho_k|\}$. Positive ϕ couples them, and the limiting cases formalize the tradeoff from Section 3.4: moving from segmented to integrated markets raises persistence when prices are the slower margin ($\lambda_{\text{seg}} > |\rho_k|$) and lowers it when labor is ($|\rho_k| > \lambda_{\text{int}}$).

4.2 Heterogeneous Rigidity: Persistence and Policy Leverage

When κ_i varies across sectors, the independent-component structure of Section 4.1 breaks down: eigenvalues shift from their uniform-κ baselines, and the demand channel $\mathbf{b}^y_t$ becomes active, breaking Corollary 1. We first characterize the first-order effect of heterogeneous stickiness on persistence using a perturbation argument.

Proposition 6 (Effective Rigidity). *Write $\kappa_i = \bar{\kappa} + \varepsilon\,\Delta\kappa_i$ with $\sum_i \alpha_i \Delta\kappa_i = 0$ and $|\varepsilon|$ small. For each component $k \geq 2$, define the effective rigidity*

$$\Delta\kappa_k^{eff} \;\equiv\; \sum_i \alpha_i\,\Delta\kappa_i\,v_{k,i}^2, \tag{44}$$

where $v_{k,i} = \alpha_i^{-1/2} s_{k,i}$ is sector i's loading on component k. To first order in ε, the dynamics along component k behave as if governed by a uniform slope $\bar{\kappa} + \varepsilon\,\Delta\kappa_k^{eff}$, and the dominant eigenvalue of $\mathbf{P}$ along component k shifts to

$$\lambda_k^*(\varepsilon) = \lambda_k^* + \varepsilon\,\Delta\kappa_k^{eff} \cdot \Gamma_k + O(\varepsilon^2),$$

where λ_k^* is the baseline eigenvalue under uniform $\bar{\kappa}$ (Lemma 3) and $\Gamma_k < 0$ is a constant. A negative $\Delta\kappa_k^{eff}$ strictly increases persistence.

Proof. See Appendix C.2. □

The effective rigidity $\Delta\kappa_k^{\text{eff}}$ is a weighted average of sectoral stickiness deviations, with weights $v_{k,i}^2$ proportional to each sector's squared loading on component k. It is negative when the sectors that load heavily on component k also tend to have stickier prices. When $\Delta\kappa_k^{\text{eff}} < 0$, nominal and real frictions compound: sluggish price adjustment in the sectors connected by component k dampens the wage differentials that drive reallocation, while slow labor flows sustain the marginal-cost gaps that drive inflation. The effect is strongest for slow-mixing components (high ρ_k), where labor frictions are already the bottleneck.

Heterogeneous κ_i also breaks the block separation of Corollary 1: aggregate demand spills into relative dynamics.

Proposition 7 (Demand Spillover). *A unit change in $\tilde{y}_t$ shifts relative inflation in component $k \geq 2$ by $(\gamma + \varphi)\,d_k$, where*

$$d_k \;\equiv\; \sum_i \alpha_i\,\Delta\kappa_i\,v_{k,i} = \mathrm{Cov}_\alpha(\Delta\kappa_i,\, v_{k,i}). \tag{45}$$

Under uniform κ, $d_k = 0$ for all k, recovering block separation.

Proof. Project the demand term $(\gamma + \varphi)\,M\mathbf{K1}\,\tilde{y}_t$ from the demeaned Phillips curve (C.5) onto the Q-eigenbasis. The k-th component receives $(\gamma + \varphi)\,s_k^\top A^{1/2} M\mathbf{K1}\,\tilde{y}_t = (\gamma + \varphi)\,d_k\,\tilde{y}_t$, since $s_k^\top A^{1/2}\mathbf{1} = 0$ for $k \geq 2$ implies $s_k^\top A^{1/2} M = s_k^\top A^{1/2}$. □

The spillover d_k is exact for arbitrary heterogeneity in κ_i. It is large when sticky and flexible sectors sit on opposite sides of a reallocation component: aggregate demand moves their prices at different speeds, tilting reallocation incentives along that component. For example, if component k separates a flexible sector (positive loading) from a sticky sector (negative loading), then $d_k > 0$:

expansionary policy raises the flexible sector's relative price, drawing workers toward the sticky sector.

Note that the two statistics capture distinct channels. The effective rigidity $\Delta\kappa_k^{\text{eff}}$ weights stickiness by the squared loading $v_{k,i}^2$ and governs persistence of misallocation and inflation along the component. The spillover d_k weights stickiness by the signed loading $v_{k,i}$ and governs monetary policy's leverage over reallocation. The two are independent: in a symmetric two-sector economy with $\kappa_1 \neq \kappa_2$, the spillover is maximal but the effective rigidity is exactly zero.

Both statistics are computable from the labor transition matrix F and sectoral price adjustment frequencies. The effective rigidity is a first-order approximation; whether it correctly identifies the most persistent corridors when dispersion in κ_i is large is a quantitative question addressed in Section 6. The spillover is exact. Together, they determine when and how monetary policy interacts with sectoral reallocation (Section 5).

4.3 Example: A Three-Sector Illustration

We illustrate how the alignment of price rigidity with the labor-flow network governs inflation persistence and monetary-policy leverage. The example compares two economies with identical average mobility, identical labor-flow eigenvalues, and the same shock. They differ only in which sectors are connected by the slowest labor-flow component.

Setup. Consider three equal-sized sectors ($\alpha_i = 1/3$). Sectors A and B are sticky ($\kappa_A = \kappa_B < \bar{\kappa}$), while sector C is flexible ($\kappa_C > \bar{\kappa}$). A permanent negative productivity shock hits sector A.

We compare two labor networks with transition probabilities m and ℓ, where $0 < m < \ell \ll \frac{1}{2}$. In network AB, workers move slowly between A and B; in network AC, between A and C:

$$F^{AB} = \begin{bmatrix} 1-m-\ell & m & \ell \\ m & 1-m-\ell & \ell \\ \ell & \ell & 1-2\ell \end{bmatrix}, \qquad F^{AC} = \begin{bmatrix} 1-m-\ell & \ell & m \\ \ell & 1-2\ell & \ell \\ m & \ell & 1-m-\ell \end{bmatrix}.$$

One matrix is a permutation of the other, so they share the same eigenvalues:

$$\rho_1 = 1, \qquad \rho_2 = 1 - 2m - \ell, \qquad \rho_3 = 1 - 3\ell.$$

Since $m < \ell$, component 2 is the slow labor-mixing bottleneck ($\rho_2 > \rho_3$) in both economies. If persistence were determined by average mobility or sectoral stickiness per se, the two economies would behave identically.

Different bottlenecks. What differs is which sectors the slow component connects. The eigenvectors of the slow mode are

$$s_2^{AB} = \tfrac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \qquad s_2^{AC} = \tfrac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

Under F^{AB}, the slow mode separates the two sticky sectors A and B. Under F^{AC}, it separates the sticky sector A from the flexible sector C.

Sufficient statistics. We evaluate $\Delta\kappa_k^{\text{eff}}$ and d_k for the slow component. With equal shares, sector i's loading simplifies to $v_{k,i} = \sqrt{3}\, s_{k,i}$. Write $\tilde{\kappa}_i \equiv \kappa_i - \bar{\kappa}$, so $\tilde{\kappa}_A = \tilde{\kappa}_B = -\varepsilon$ and $\tilde{\kappa}_C = 2\varepsilon$.

Under network AB, the slow mode loads on A and B with $v_{2,A} = \sqrt{3/2}$, $v_{2,B} = -\sqrt{3/2}$, $v_{2,C} = 0$. The signed terms cancel while the squared terms reinforce:

$$\Delta\kappa_2^{\text{eff},AB} = \sum_i \alpha_i\, \tilde{\kappa}_i\, v_{2,i}^2 = \tfrac{1}{3}\big[(-\varepsilon)\tfrac{3}{2} + (-\varepsilon)\tfrac{3}{2} + 0\big] = -\varepsilon,$$
$$d_2^{AB} = \sum_i \alpha_i\, \tilde{\kappa}_i\, v_{2,i} = \tfrac{1}{3}\big[(-\varepsilon)\sqrt{\tfrac{3}{2}} + (-\varepsilon)(-\sqrt{\tfrac{3}{2}}) + 0\big] = 0.$$

Under network AC, the slow mode loads on A and C with $v_{2,A} = -\sqrt{3/2}$, $v_{2,B} = 0$, $v_{2,C} = \sqrt{3/2}$. Both statistics are nonzero:

$$\Delta\kappa_2^{\text{eff},AC} = \sum_i \alpha_i\, \tilde{\kappa}_i\, v_{2,i}^2 = \tfrac{1}{3}\big[(-\varepsilon)\tfrac{3}{2} + 0 + 2\varepsilon \cdot \tfrac{3}{2}\big] = +\tfrac{\varepsilon}{2},$$
$$d_2^{AC} = \sum_i \alpha_i\, \tilde{\kappa}_i\, v_{2,i} = \tfrac{1}{3}\big[(-\varepsilon)(-\sqrt{\tfrac{3}{2}}) + 0 + 2\varepsilon\sqrt{\tfrac{3}{2}}\big] = \sqrt{\tfrac{3}{2}}\,\varepsilon.$$

	Network AB	Network AC
Effective rigidity $\Delta\kappa_2^{\text{eff}}$	$-\varepsilon$	$+\varepsilon/2$
Demand spillover d_2	0	$\sqrt{3/2}\,\varepsilon$

Interpretation. Under F^{AB}, the slow component connects two equally sticky sectors. The effective rigidity is negative: neither sector adjusts prices quickly, muting the wage signal that would redirect workers out of A—so persistence rises above the uniform-κ baseline ($\lambda_{2,AB}^* > \lambda_2^*$). The demand spillover is zero: aggregate demand moves A's and B's prices symmetrically, leaving the central bank no leverage over reallocation along this component.

Under F^{AC}, the slow component connects a sticky sector to a flexible one. Sector C's rapid price adjustment generates a sharp wage differential that accelerates reallocation out of A: the

effective rigidity is positive and persistence falls below the baseline ($\lambda^*_{2,AC} < \lambda^*_2$). The demand spillover is positive: expansionary policy raises C's price faster than A's, tilting reallocation along the slow component.

Sector A is equally sticky and faces the same shock in both economies. What differs is its network position, which determines both inflation persistence and the central bank's leverage over reallocation.

5 Optimal Monetary Policy

Sections 3–4 characterized the generalized Phillips curve $\mathcal{K}$ (29) and transition dynamics conditional on the path of aggregate demand. We now consider the optimal choice of that path. This section derives the welfare loss function and the Ramsey problem (Section 5.1), characterizes the optimal targeting criterion (Section 5.2), and identifies conditions under which optimal policy deviates from the standard rule (Section 5.3).

5.1 The Ramsey Problem

We adopt the timeless perspective of Woodford (2003). The efficient steady-state subsidy eliminates first-order terms, so the problem admits a linear-quadratic approximation.

Proposition 8 (Ramsey Problem under Commitment)**.** *Under commitment in the timeless perspective, the central bank chooses a sequence* $\{\tilde{y}_t, \boldsymbol{\pi}_t, \mathbf{n}_t, \boldsymbol{\chi}_t, \mathbf{v}_t\}_{t=0}^{\infty}$ *to minimize the intertemporal welfare loss*

$$\min \quad \mathcal{W} = \sum_{t=0}^{\infty} \beta^t \, \mathcal{L}_t, \tag{46}$$

where the per-period loss is

$$\mathcal{L}_t \overset{(2)}{=} \frac{\gamma + \varphi}{2} \tilde{y}_t^2 + \sum_i \frac{\alpha_i \epsilon_i}{2\kappa_i} \pi_{it}^2 + \sum_i \alpha_i \left[\frac{1}{2} \chi_{it}^2 + \frac{\varphi}{2} (\chi_{it} + n_{it})^2 \right], \tag{47}$$

subject to, for each sector i and date t:

$$\pi_{it} = \kappa_i [(\gamma + \varphi) \, \tilde{y}_t - (1 + \varphi) \, \chi_{it} - \varphi \, n_{it}] + \beta \, \mathbb{E}_t \pi_{it+1}, \tag{48}$$

$$\chi_{it} = \chi_{it-1} + \pi_{it} - \pi_t^C + \Delta \hat{a}_{it}, \tag{49}$$

$$\mathbf{n}_t = B \, \mathbf{n}_{t-1} + \phi (I - BF) \, \mathbf{v}_t, \tag{50}$$

$$\mathbf{v}_t = -\boldsymbol{\chi}_t - \mathbf{n}_t + \beta F \, \mathbb{E}_t [\mathbf{v}_{t+1}], \tag{51}$$

taking as given initial conditions $(\hat{\mathbf{p}}_{-1}, \mathbf{n}_{-1})$ *and the exogenous productivity process* $\{\mathbf{a}_t\}$.

Proof. The per-period loss (47) follows from a second-order expansion of household welfare around the efficient steady state (Appendix B). The constraints restate the marginal cost (27) and NKPC (21), relative price identity, and reallocation block from Section 2.5. □

The sectoral loss (47) has a clean interpretation. The output gap and sectoral inflation terms are standard. The first misallocation term, $\frac{1}{2}\chi_{it}^2$, penalizes relative-price misalignment as in the multisector New Keynesian optimal policy literature (Rubbo, 2023). The novel piece is the second sectoral term: it penalizes $\chi_{it} + n_{it}$, the deviation from the full-mobility benchmark $n_{it} = -\chi_{it}$. Labor-market frictions matter for welfare through this term, and shrink as $\phi \to \infty$. Equivalently, the Ramsey problem is an LQ control problem in which the planner chooses $\tilde{y}_t$ to steer the state $\mathbf{s}_t = (\boldsymbol{\chi}_{t-1}, \mathbf{n}_{t-1})$ through the generalized Phillips curve, with slow-decaying corridors of the transition matrix $\mathbf{P}$ disproportionately costly.

5.2 The Targeting Criterion

The first-order conditions of the Ramsey problem involve multipliers $\boldsymbol{\lambda}_t$ (Phillips curves), $\boldsymbol{\eta}_t$ (relative price identities), $\boldsymbol{\psi}_t$ (employment law of motion), and $\boldsymbol{\xi}_t$ (continuation values); the full system is in Appendix C.3. These collapse into a targeting criterion:

Proposition 9 (Monetary Policy Target)**.** *Under commitment, optimal monetary policy is characterized by the targeting rule*

$$\tilde{y}_t - \tilde{y}_{t-1} + \sum_i \alpha_i \epsilon_i \, \pi_{it} \;=\; (MK\mathbf{1})^\top \boldsymbol{\eta}_t = \sum_i (\kappa_i - \bar{\kappa}) \eta_{it} \tag{52}$$

where $\boldsymbol{\eta}_t$ is determined by the system of first-order conditions in Appendix C.3.

Proof. See Appendix C.3. □

The left-hand side is the standard targeting criterion. The right-hand side is a *reallocation wedge*: it is nonzero when reallocation needs (η_{it}) are systematically correlated with Phillips curve slopes (κ_i). When all sectors share the same slope, the wedge vanishes and the criterion reduces to

$$\tilde{y}_t - \tilde{y}_{t-1} + \sum_i \alpha_i \epsilon_i \, \pi_{it} = 0, \tag{53}$$

which, if additionally $\epsilon_i = \epsilon$, becomes the textbook rule $\tilde{y}_t - \tilde{y}_{t-1} + \epsilon \, \pi_t^C = 0$.

The reallocation wedge. The multiplier η_{it} is the shadow value of raising sector i's relative price. Rearranging the relative-price FOC (Equation (F_χ) in Appendix C.3) and iterating forward:

$$\boldsymbol{\eta}_t \;=\; -\sum_{s=0}^{\infty} \beta^s \, \mathbb{E}_t \, \mathbf{g}_{t+s}, \tag{54}$$

where $\mathbf{g}_t$ collects the marginal welfare costs of the current relative-price configuration — direct misallocation, cost-push pressure in sectoral Phillips curves, and distortion of reallocation incentives; its explicit form is derived in Appendix C.3. The shadow value η_{it} is large when sector i's relative price is persistently too low: slow-decaying corridors of $\mathbf{P}$ generate large cumulative distortions, raising $|\boldsymbol{\eta}_t|$ and enlarging the wedge. Section 5.3 makes this link precise.

The sign of the wedge. Aggregate demand moves flexible-sector prices more than sticky-sector prices. The planner exploits this asymmetry to tilt relative prices toward the side of the economy that needs to shed workers. Consider a negative productivity shock in a sticky sector. Workers must reallocate outward, so the planner wants to raise that sector's relative price ($\eta_{it} > 0$). Because the sector has $\kappa_i < \bar{\kappa}$, the wedge is negative, and optimal policy tightens: contraction lowers flexible-sector prices more than sticky-sector prices, raising the stuck sector's relative price and unlocking reallocation within the same period. The planner accepts a negative output gap to induce reallocation that sticky prices would otherwise muffle. The logic reverses when workers must leave flexible sectors.

Benchmark: closing the output gap. An aggregation result following Rubbo (2023) provides a useful benchmark. Consider the weighted inflation $\pi_t^R \equiv \sum_i \omega_i^R \, \pi_{it}$ with weights $\omega_i^R = (\alpha_i/\kappa_i) / \sum_j (\alpha_j/\kappa_j)$: these weights tilt the inflation index toward sticky sectors (low κ_j). Summing the sectoral NKPC (21) with weights ω_i^R and using $\sum_i \alpha_i n_{it} = 0$ gives

$$\pi_t^R = (\gamma + \varphi) \, \tilde{y}_t + \beta \, \mathbb{E}_t \pi_{t+1}^R, \tag{55}$$

independent of $\mathbf{n}_t$, $\hat{\mathbf{p}}_t$, or F. Stabilizing $\pi_t^R = 0$ therefore closes the output gap at all dates. This is nonetheless suboptimal: the targeting criterion (52) shows that the planner accepts output gap costs whenever the reallocation wedge is nonzero.

5.3 When Does the Wedge Matter?

A pattern of misalignment matters for monetary policy under two conditions: the pattern must be slow to self-correct, so that cumulative welfare losses are large; and aggregate demand must be able to tilt relative prices along that pattern, so that the central bank has a lever. When either

condition fails, the reallocation wedge in (52) is small and the standard targeting rule (53) is a good approximation.

The corridor decomposition from Section 3.3 makes both conditions concrete. Equation (40) shows that each corridor evolves as a scalar AR(1) with persistence λ_h, driven by aggregate demand through the term $\mathbf{v}_h^\top \mathbf{b}_t^y$. A slow corridor generates persistent misalignment along $(\chi_{t-1}, \mathbf{n}_{t-1})$, which the per-period loss (47) penalizes directly. Through the present-value representation (54), this persistence raises the shadow value $|\boldsymbol{\eta}_t|$, enlarging the reallocation wedge. But the wedge is large only if $\mathbf{v}_h^\top \mathbf{b}_t^y$ is also nonzero, so that the planner can steer the corridor by choosing $\tilde{y}_t$.

Under quasi-symmetric costs (Assumption 1), the statistics from Section 4 make both conditions precise. Persistence along component k is governed by the eigenvalue λ_k^* (Lemma 3) and its shift under heterogeneous stickiness, $\Delta\kappa_k^{\text{eff}}$ (Proposition 6): a negative $\Delta\kappa_k^{\text{eff}}$ raises λ_k^*, slowing self-correction and increasing the shadow value $|\eta_{kt}^*|$ through the present-value representation (54). Demand leverage along component k is governed by the spillover coefficient d_k (Proposition 7): when $|d_k|$ is large, a unit change in $\tilde{y}_t$ materially tilts relative prices along that component. Both statistics are entries of the matrix

$$\mathcal{G}_{kj} \equiv \sum_i \kappa_i \, \alpha_i \, v_{k,i} \, v_{j,i}, \qquad d_k = \mathcal{G}_{k1}, \qquad \bar{\kappa} + \Delta\kappa_k^{\text{eff}} = \mathcal{G}_{kk},$$

so that persistence and demand leverage arise from the diagonal and off-diagonal structure of a single object: the sectoral Phillips-curve slopes viewed in the F-eigenbasis.

Projecting the targeting criterion (52) onto the F-eigenbasis (Lemma 2) decomposes the wedge component by component. Since $MK\mathbf{1} = K\mathbf{1} - \bar{\kappa}\mathbf{1}$ is in the relative subspace, it can be expanded as

$$MK\mathbf{1} \;=\; \sum_{k=2}^{N} A^{-1/2} s_k \, d_k, \qquad d_k \;\equiv\; s_k^\top A^{1/2} \, MK\mathbf{1} \;=\; \sum_i \alpha_i^{1/2} \, s_{k,i} \, (\kappa_i - \bar{\kappa}). \tag{56}$$

Substituting into (52) decomposes the reallocation wedge into a sum of component-level contributions, each determined by the demand spillover and the shadow value of misalignment along that component:

Proposition 10 (Spectral Targeting Rule)**.** *Under Assumption 1, the targeting criterion (52) decomposes as*

$$\tilde{y}_t - \tilde{y}_{t-1} + \sum_i \alpha_i \epsilon_i \, \pi_{it} \;=\; \sum_{k=2}^{N} d_k \, \eta_{kt}^*, \qquad \eta_{kt}^* \equiv s_k^\top A^{1/2} \, \boldsymbol{\eta}_t, \tag{57}$$

where η_{kt}^* is the shadow value of tilting relative-price gaps in component k. The reallocation wedge along component k is large when $|d_k|$ is large, so that aggregate demand tilts relative prices along k, and $|\eta_{kt}^*|$ is

large, which by (54) *requires persistent misalignment along k (large $|\Delta\kappa_k^{\text{eff}}|$). When $d_k \approx 0$ for all k, the wedge vanishes to first order and optimal policy approximates the standard rule* (53).

Proof. Expand $MK\mathbf{1}$ in the F-eigenbasis using (56) and substitute into $(MK\mathbf{1})^\top \boldsymbol{\eta}_t$. □

The sign of each component's contribution is $d_k\, \eta_{kt}^*$, which is invariant to the sign normalization of the eigenvectors. The planner expands when $d_k\, \eta_{kt}^* > 0$ and contracts when $d_k\, \eta_{kt}^* < 0$. The logic is the same as in the concrete case of Section 5.2: aggregate demand moves flexible-sector prices more than sticky-sector prices, so the planner chooses the sign of the output gap that tilts relative prices toward the side that should shed workers.

When $d_k \approx 0$ for all k, the one-sector Ramsey problem

$$\min_{\{\tilde{y}_t\}} \sum_{t=0}^{\infty} \beta^t \left[\frac{\gamma+\varphi}{2}\, \tilde{y}_t^2 + \frac{\epsilon}{2\kappa}\, (\pi_t^C)^2 \right] \quad \text{s.t.} \quad \pi_t^C = \kappa(\gamma+\varphi)\, \tilde{y}_t + \beta\, \mathbb{E}_t \pi_{t+1}^C + u_t \tag{58}$$

serves as a close first-order approximation, with u_t an exogenous cost-push term inherited from sectoral misalignment that the central bank cannot influence at first order. The planner cannot do better at first order in stickiness heterogeneity because aggregate demand has no leverage over the relative prices that drive reallocation.

6 Quantitative Evidence

We bring the model to U.S. sectoral data to assess whether the analytical channels identified in Section 3 are quantitatively relevant. We estimate the transition matrix F, B from worker flows, measure sectoral price rigidity from micro price data, and characterize the generalized Phillips curve and assess the quantitative importance of reallocation frictions for inflation persistence and monetary policy.

6.1 Data, Calibration, and Flow Facts

We set the model period to one quarter. We set $\gamma = 1$ (log utility over consumption), $\varphi = 2$ (Frisch elasticity of 0.5), and $\beta = 0.99$ (quarterly discount factor), following standard calibrations in the New Keynesian literature (e.g., Rubbo, 2023).

We measure sector-level price adjustment frequencies using the estimates of Pasten et al. (2020), who estimate monthly price adjustment probabilities at the industry level. We convert these to quarterly Calvo parameters: for a sector with monthly adjustment probability f_i, the quarterly probability of non-adjustment is $\delta_i = (1-f_i)^3$, yielding the Phillips curve slope $\kappa_i = (1-\delta_i)(1-\beta\delta_i)/\delta_i$.

We estimate the sectoral transition matrix F from industry-to-industry worker flows in the March CPS, converting to quarterly frequency following Artuç et al. (2010).[12] The backward matrix B is constructed from F and the stationary employment shares via (10), and the expenditure shares $\{\alpha_i\}$ are set equal to the stationary distribution of F.

The mobility elasticity ϕ is pinned down by the elasticity of worker flows to sectoral wages. In Appendix D.2, we show that linearizing the logit transition probabilities (8) around steady state yields a semi-elasticity of log relative transition probabilities to log destination wages of ϕ/φ. In Caliendo et al. (2019), where the flow payoff is $\log w_j$ and preference shocks have dispersion ν, the corresponding semi-elasticity is $1/\nu$; thus $\phi = \varphi/\nu$, and Caliendo et al. (2019) estimate $\nu = 5.34$ at quarterly frequency. With $\varphi = 2$, this yields $\phi \approx 0.37$.

Sectoral classification. We aggregate industries into 14 broad sectors, chosen to balance granularity with reliable cell sizes in the CPS worker-flow data.[13]

Table 2 reports the expenditure share, Phillips curve slope, and implied price duration for each sector, sorted from stickiest to most flexible. Price rigidity varies by nearly two orders of magnitude across sectors, from $\kappa = 0.008$ (Professional & Business Services) to $\kappa = 0.758$ (Agriculture, Mining & Extraction), with an expenditure-weighted mean of $\bar{\kappa} = 0.112$. The correlation between expenditure share and Phillips curve slope is -0.27: sticky sectors are not systematically larger or smaller.

Worker flows. The estimated transition matrix implies that 6.75% of workers change their sector each quarter. The asymmetry ratio (the maximum deviation from the detailed-balance condition $\alpha_i F_{ij} = \alpha_j F_{ji}$) is 0.08%, confirming that quasi-symmetry (Assumption 1) is a reasonable empirical approximation. The slowest-mixing component of the transition matrix has eigenvalue $\rho_2 = 0.955$, corresponding to a half-life of 15 quarters; the fastest has eigenvalue $\rho_{13} = 0.898$ with a half-life of 6 quarters. Whether this pace of labor reallocation is fast enough to matter for quarterly inflation dynamics is the question we turn to next.

6.2 Reallocation Frictions and the Spectrum of Persistence

We construct the transition matrix $\mathbf{P}$ from Section 3.2 and characterize its eigenvalues under four configurations: *Data* (calibrated F, ϕ, heterogeneous κ_i), *Uniform* κ (all $\kappa_i = \bar{\kappa}$), *Segmented*

[12] Respondents report their current industry and the industry of their longest-held job in the previous year. Recollection bias implies an effective horizon of approximately 5 months (Artuç et al., 2010); we treat the raw matrix as F_{5m} and convert via $F = F_{5m}^{3/5}$.

[13] The 14 sectors are: Agriculture, Mining & Extraction (AGM); Construction (CON); Nondurable Manufacturing (NDM); Durable & Advanced Manufacturing (DUR); Utilities (UTL); Trade, Wholesale & Retail (TRD); Transportation & Warehousing (TRN); Leisure, Hospitality & Other Services (LSR); Information & Communications (INF); Finance, Insurance & Real Estate (FIR); Professional & Business Services (PBS); Government (GOV); Education (EDU); and Healthcare (HLT).

Table 2: Sectoral price rigidity and expenditure shares

Code	Sector	α_i	κ_i	Duration (q)
PBS	Professional & Business Services	0.091	0.008	12.4
HLT	Healthcare	0.118	0.011	10.7
INF	Information & Communications	0.041	0.012	9.9
DUR	Durable & Advanced Manufacturing	0.085	0.030	6.4
LSR	Leisure, Hospitality & Other Services	0.061	0.039	5.7
EDU	Education	0.096	0.041	5.6
GOV	Government	0.053	0.056	4.8
TRN	Transportation & Warehousing	0.048	0.059	4.7
FIR	Finance, Insurance & Real Estate	0.101	0.093	3.9
NDM	Nondurable Manufacturing	0.025	0.157	3.1
UTL	Utilities	0.012	0.194	2.8
TRD	Trade (Wholesale & Retail)	0.180	0.235	2.6
CON	Construction	0.067	0.304	2.4
AGM	Agriculture, Mining & Extraction	0.021	0.758	1.8
Expenditure-weighted mean		—	0.112	4.8

Notes. Phillips curve slopes $\kappa_i = (1-\delta_i)(1-\beta\delta_i)/\delta_i$ computed on a quarterly horizon. Duration is median price duration in quarters. Sectors sorted by κ_i ascending (stickiest first).

Table 3: Corridor thickening under alternative configurations

	Dominant $\lvert\lambda\rvert$	Half-life (q)	Corridors HL $>$ 4q	Corridors HL $>$ 8q
Segmented	0.853	4.4	1	0
Uniform κ	0.897	6.4	11	0
Data	0.918	8.1	12	1
Integrated	0.914	7.7	3	0

Notes. Each configuration has $2(N-1) = 26$ eigenvalues. Data: calibrated F, $\phi = 0.37$, heterogeneous κ_i. Uniform κ: same F and ϕ, all $\kappa_i = \bar{\kappa}$. Segmented: $\phi = 0$. Integrated: $\phi \to \infty$.

($\phi = 0$), and *Integrated* ($\phi \to \infty$).

The labor-flow eigenbasis. The labor transition matrix F has 13 non-trivial eigenvalues ranging from $\rho_2 = 0.955$ (half-life 3.8 years) to $\rho_{13} = 0.898$ (half-life 1.6 years). The slowest component separates Construction and Agriculture (flexible) from Education and Healthcare (sticky), with $\Delta\kappa_2^{\text{eff}} = +0.031$: nominal and real frictions partially offset. The second-slowest ($\rho_3 = 0.948$) separates Education and Construction from Healthcare and Government, with $\Delta\kappa_3^{\text{eff}} = -0.078$: here sticky sectors dominate the slow-mixing side, and the two frictions compound. Overall, 6 of 13 components have $\Delta\kappa_k^{\text{eff}} < 0$. The full table and details are in the Appendix.

Corridor thickening. Table 3 reports the eigenvalues of $\mathbf{P}$ under each configuration.

Under segmentation, only one corridor exceeds a four-quarter half-life. Introducing cali-

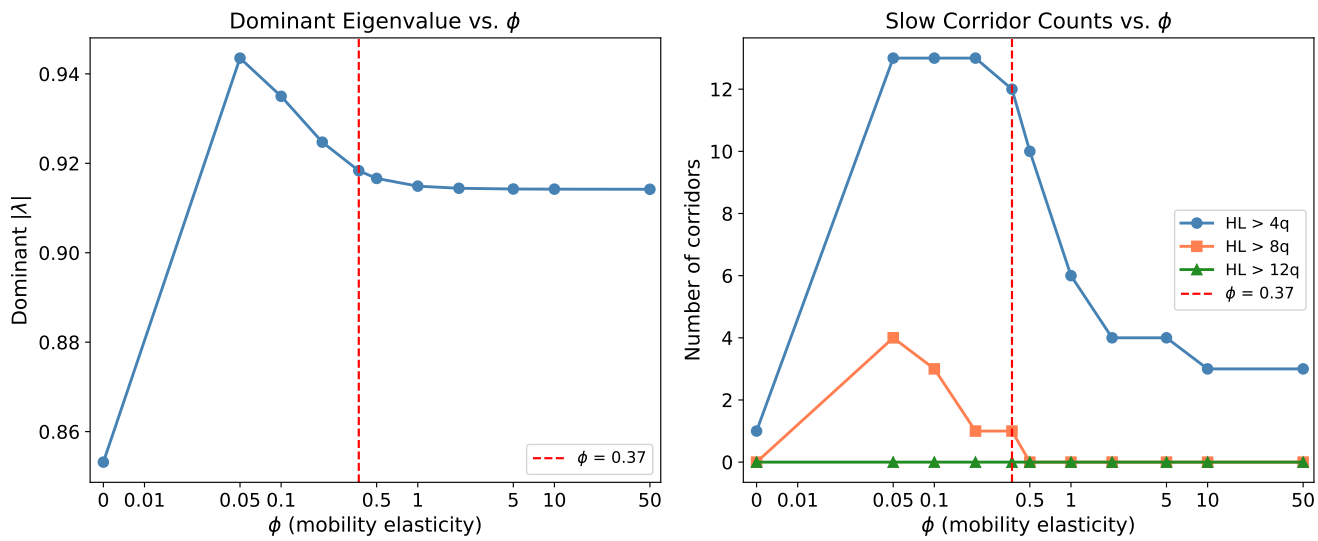

Figure 1: Dominant eigenvalue and corridor counts as a function of ϕ.

Notes. Vertical dashed line: calibrated $\phi = 0.37$. The x-axis is log-scale ϕ, with $\phi = 0$ segmented labor market.

brated mobility raises the count to 12: finite mobility populates the spectrum with slow-decaying corridors of price-employment misalignment. Heterogeneous κ_i further raises the dominant half-life from 6.4 to 8.1 quarters, consistent with Proposition 6.

As a robustness test, we eigendecompose $\mathbf{P}$ for various levels of ϕ. The activation is sharp in ϕ: at $\phi = 0.05$, already 13 corridors exceed the four-quarter threshold (Figure 1).

Diagnostics. We can check that the perturbation-based results from Proposition 6 hold well: the predicted relationship between $\Delta\kappa_k^{\text{eff}}$ and actual eigenvalue shifts of $\mathbf{P}$ holds with a correlation of -0.51 when eigenvectors are matched by overlap. To assess whether the observed alignment of price rigidity to the labor network is generic, we permute κ across sectors while holding F fixed (1000 draws). In the U.S. data, the observed price-rigidity assignment produces a dominant eigenvalue near the median of random reassignments, but a slow-corridor count exceeding 93% of them.

6.3 The Demand Jacobian of the Phillips Curve

The generalized Phillips curve (29) maps aggregate demand and sectoral productivity into inflation. We characterize each channel in turn, starting with demand. Define the cumulative inflation response

$$S_H \equiv \sum_{h=0}^{H} \frac{\partial \pi_{t+h}^C}{\partial \tilde{y}_t}, \tag{59}$$

Table 4: Cumulative price level response S_H

	S_0	S_4	S_8	S_{20}	S_{40}
Data	0.217	0.110	0.094	0.084	0.082
Segmented	0.215	0.103	0.088	0.082	0.082
Integrated	0.254	0.132	0.102	0.085	0.082
Uniform κ	0.335	0.335	0.335	0.335	0.335

Notes. $S_H = \sum_{h=0}^{H} \partial \pi_{t+h}^{C} / \partial \tilde{y}_t$. Under uniform κ, block separation (Corollary 1) implies $S_H = \bar{\kappa}(\gamma + \varphi) = 0.335$ for all H.

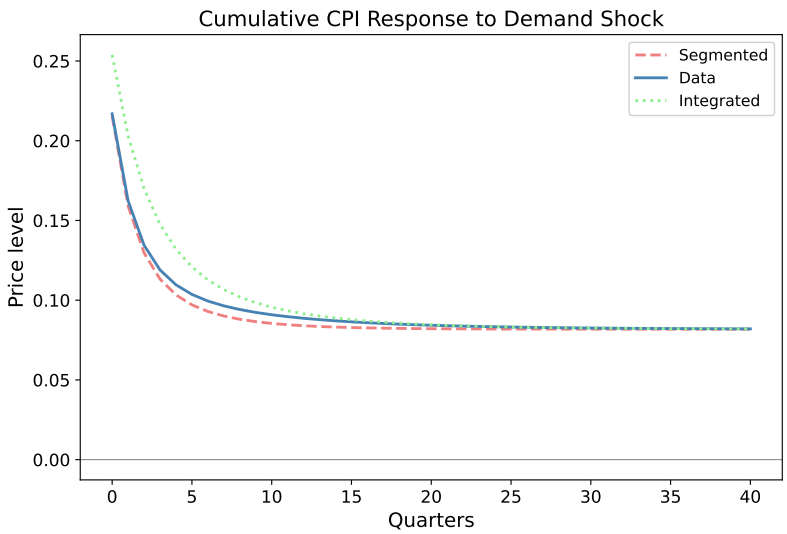

Figure 2: Cumulative price level response S_H under data, integrated, segmented configurations.

the total CPI change through horizon H induced by a one-time unit demand shock. Table 4 reports S_H at key horizons; Figure 2 plots the full paths.

Under uniform price rigidity, S_H is perfectly flat: a unit output gap raises the price level by $\bar{\kappa}(\gamma + \varphi) = 0.335$ at every horizon. Aggregate demand shifts all sectoral costs equally, so relative prices and reallocation incentives are unaffected (Corollary 1).

Heterogeneous κ_i breaks this decoupling. At impact, flexible-price sectors transmit the demand shock disproportionately into the price level. As the economy adjusts, these sectors correct: their high κ_i means they track current marginal cost closely, so prices that jumped at impact revert once demand has passed, generating mild deflation at medium horizons. The remaining misalignment concentrates in sticky sectors, and the marginal price level response decays: by $H = 20$, S_H has lost more than half its impact value. This decay is driven primarily by the distribution of κ_i, not by labor mobility—the Data and Segmented paths are close throughout.

Where mobility matters is in the comparison with the integrated benchmark. Under frictionless labor markets, the price level response is uniformly higher: $S_0 = 0.254$ versus 0.217 at impact, $S_8 = 0.102$ versus 0.094 at two years. When workers can freely reallocate, contemporaneous reallocation attenuates current marginal-cost pressure, reducing the effective supply

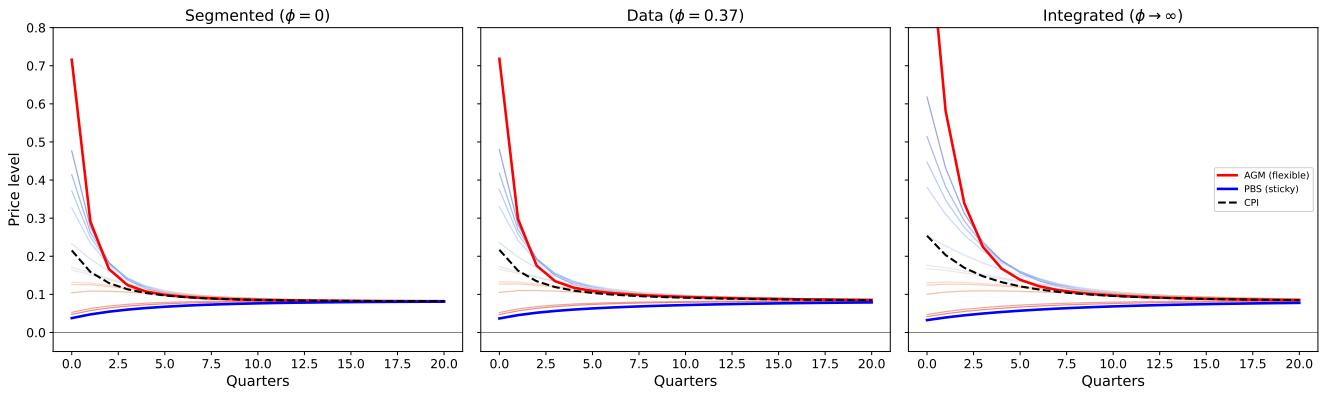

Figure 3: Sectoral demand Jacobian: cumulative inflation $\sum_{h=0}^{H} \partial \pi_{it+h} / \partial \tilde{y}_t$ for each sector. Left: Segmented ($\phi = 0$). Center: Data ($\phi = 0.37$). Right: Integrated ($\phi \to \infty$). Colors range from blue (stickiest, PBS) to red (most flexible, AGM). Dashed black line: CPI (α-weighted average). All sectors converge to the same long-run value.

slope (Proposition 5); the lower slope means a larger pass-through of demand into prices at every horizon. In the calibrated economy, sluggish reallocation leaves current cost pressures less attenuated, dampening the initial response but sustaining it for longer.

Despite these differences in the path, the long-run price level effect S_∞ is identical across Data, Segmented, and Integrated (0.082). Mobility reshapes *when* demand passes through into prices, but not the long-run price level effect.

Sectoral transmission. Figure 3 disaggregates the demand Jacobian to the sector level, plotting the sectoral price level response $\sum_{h=0}^{H} \partial \pi_{it+h} / \partial \tilde{y}_t$ under three mobility configurations. Flexible sectors (AGM, $\kappa = 0.758$, red) pass through the demand impulse almost entirely on impact, then revert within a few quarters as their prices track the fading cost pressure. Sticky sectors (PBS, $\kappa = 0.008$, blue) barely respond on impact but accumulate price adjustment gradually over years. Despite these sharply different paths, all 14 sectors converge to the same long-run value (≈ 0.082) by $H = 40$: the permanent price level effect of a demand shock is uniform across sectors, as relative prices eventualy return to the pre-shock level.

The three panels show how mobility reshapes the fan of sectoral paths. Under segmentation, the fan is narrow and closes by $H = 10$. Under calibrated mobility, anticipated worker flows dampen price adjustment in both flexible and sticky sectors, widening the fan and extending convergence to $H \approx 20$. Under integration, the fan is widest at medium horizons, reflecting the larger initial pass-through that frictionless mobility permits. The aggregate price level path (dashed) is the α-weighted average of these sectoral paths, and its shape mirrors the S_H curves in Table 4.

6.4 The Supply Jacobian: Sectoral Productivity Shocks

We now turn to the supply side of the generalized Phillips curve. The state-space representation (30) is driven by two forcing variables: the output gap $\tilde{y}_t$ and relative productivity growth $\Delta\hat{\mathbf{a}}_t$. Section 6.3 characterized the price level response to the former; we now characterize the response to the latter. We feed a unit impulse in $\Delta\hat{a}_{it}$—equivalently, a permanent unit productivity increase in sector i—holding $\tilde{y}_t = 0$, and trace the CPI price level response $\sum_{h=0}^{H} \pi^C_{t+h}$. This isolates the supply-side propagation channel: how sectoral shocks interact with the labor flow network and heterogeneous price rigidity.

Sectoral transmission. Figure 4 plots the CPI inflation response $\partial\pi^C_{t+h}/\partial\Delta\hat{a}_{it}$ to a permanent unit productivity increase in each sector, under three mobility configurations. The sign of the response is governed by own-price rigidity. A positive productivity shock to a flexible sector (TRD, red; AGM, purple) lowers CPI on impact: the shocked sector passes through its cost reduction quickly, pulling the price level down. A positive productivity shock to a sticky sector (PBS, blue) raises CPI: the shocked sector barely adjusts its own price, while the demeaned productivity decline in other sectors raises their marginal costs, and the flexible ones among them pass this through into higher prices. The result is persistent positive CPI inflation that decays only as the sticky sector's price gradually adjusts downward. Intermediate sectors (DUR, EDU) exhibit sign reversals, with initial inflation giving way to deflation as the cross-sectoral adjustment unfolds.

The three panels reveal how mobility reshapes these dynamics. Under frictionless integration, the inflation response to any sectoral shock is small but persistent: workers reallocate to offset productivity differences, compressing the initial impact, but the gradual reallocation sustains price misalignment over a longer horizon. Calibrated mobility produces larger impact responses—reallocation frictions prevent workers from immediately offsetting the shock—but smaller persistence. Under full segmentation, the fan is widest at impact but closes quickly: without worker flows to sustain misalignment, inflation responses die out within 10 quarters.

The role of labor reallocation in pricing. A natural question is how much of this persistence comes from labor reallocation rather than from price misalignment alone. To answer this, we decompose the expected path of marginal costs that drives firms' optimal pricing decisions. Under Calvo pricing, a resetting firm in sector i sets its price as a function of the discounted path of future marginal costs, with weights $(\beta\delta_i)^s$. The marginal cost equation (27) has two state-dependent terms: the relative-price distortion $-(1+\varphi)\chi_{it}$ and the employment-misallocation

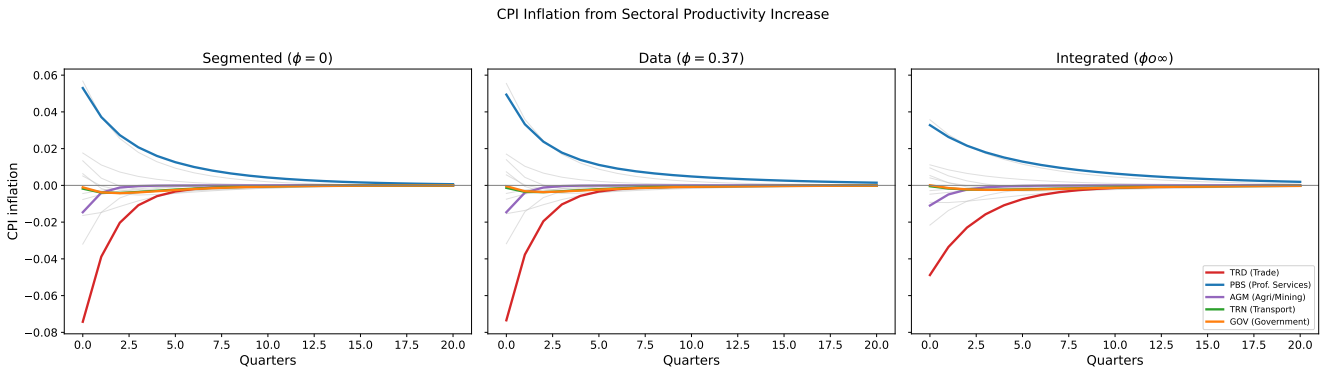

Figure 4: CPI inflation response to sectoral supply shocks (positive productivity increase) under three mobility configurations. Each line is one sector's unit shock. Highlighted: TRD (red), PBS (blue), AGM (purple), TRN (green), GOV (orange). Gray: remaining sectors.

term $-\varphi n_{it}$. We define the pricing-relevant reallocation share (PRS) as

$$\text{PRS}_{ij} \equiv \frac{\sum_{s\geq 0}(\beta\delta_i)^s \left|\varphi\, n_{i,t+s}^{(j)}\right|}{\sum_{s\geq 0}(\beta\delta_i)^s \left[\left|(1+\varphi)\,\chi_{i,t+s}^{(j)}\right| + \left|\varphi\, n_{i,t+s}^{(j)}\right|\right]}, \tag{60}$$

which isolates the fraction of the discounted marginal-cost path for sector i, following a shock to sector j, that is driven by inherited employment misallocation rather than relative-price distortions. Under segmented labor markets, this share must be zero; under integrated labor markets, Proposition Y implies $n_{it} = -\chi_{it}$, yielding an upper bound of $\varphi/(1+2\varphi) = 0.40$. Across most sector pairs, the PRS lies in the range of 35 to 40%, with a mean of 37.8%. Thus, more than one-third of the expected marginal cost dynamics relevant for price-setting reflects employment reallocation.

Thus labor reallocation is quantitatively important for price-setting. Even though worker flows are gradual, they accumulate over the effective horizon of a sticky price, so employment misallocation accounts for a large share of the marginal-cost variation firms internalize when resetting prices.

Sectoral versus aggregate effects. Contrasting Figures 4 and 5, labor reallocation drives 37% of the marginal cost path, close to the 40% frictionless benchmark. Yet aggregate CPI paths under calibrated mobility are much closer to the segmented market benchmark. The aggregate Phillips curve (32) provides an explanation: since $\sum_i \alpha_i n_{it} = 0$, employment misallocation enters CPI inflation only through its covariance with sectoral Phillips curve slopes. Section 6.2 confirms this covariance is modest in U.S. data. Reallocation frictions are quantitatively important for understanding which sectors face persistent cost pressure, but does not affect aggregates significantly.

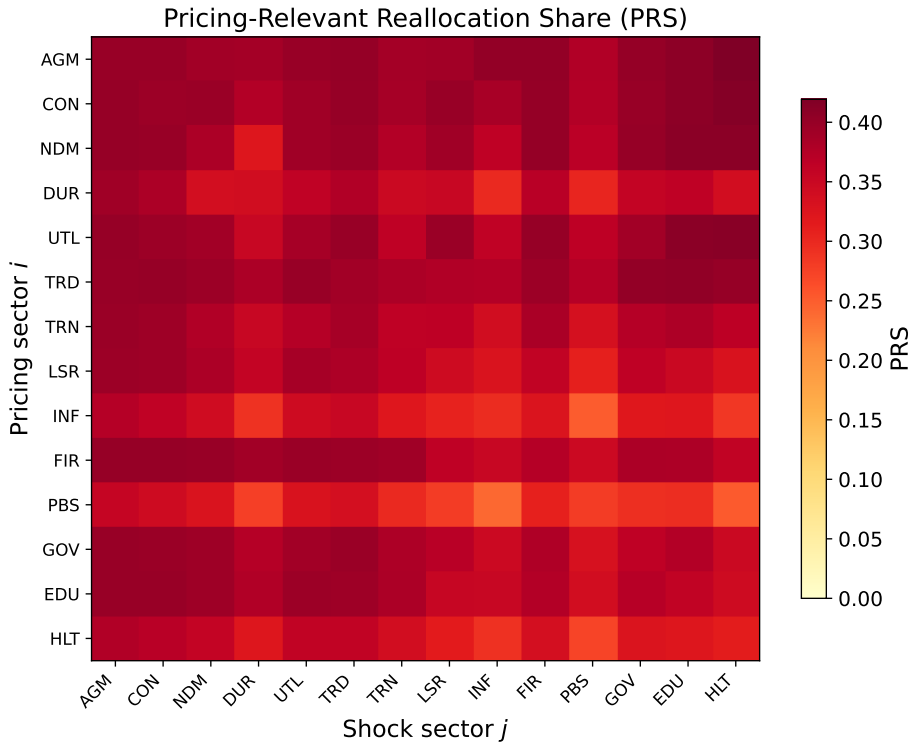

Figure 5: Pricing-relevant reallocation share (PRS) across sector pairs. Entry (i, j) is the share of sector i's discounted marginal-cost path attributable to employment misallocation following a shock to sector j.

6.5 Monetary Policy and the Reallocation Wedge

Sections 6.3–6.4 characterized the generalized Phillips curve implied by sectoral reallocation frictions. We now close the model with the IS curve (22) and compare the Ramsey-optimal policy (Appendix D.3) to four simple rules: *zero-gap* ($\tilde{y}_t = 0$), a standard *Taylor rule* ($\psi_\pi = 1.5$, $\psi_y = 0.5$), an *inflation-only Taylor rule* ($\psi_\pi = 1.5$, $\psi_y = 0$), and *strict inflation targeting* ($\pi_t^C = 0$). Table 5 reports welfare losses and on-impact responses of the economy under each policy, computed as $\boldsymbol{\alpha}$-weighted averages across $N = 14$ scenarios, each featuring a unit negative productivity shock to that sector.

Ramsey attains the minimum and the simple rules rank by how aggressively they fight aggregate inflation. Strict IT, which enforces $\pi_0^C = 0$ by construction, is the most costly policy and requires an average output-gap response ($|\tilde{y}_0|_\alpha = 0.051$) more than double Ramsey's (0.023); zero-gap, the opposite corner with $\tilde{y}_0 = 0$ in every scenario, comes within 1.5% of Ramsey. The two Taylor variants fall in between, with the output-responsive rule dominating the inflation-only rule. The Ramsey impact responses are nonzero in both dimensions—$|\pi_0^{C,*}|_\alpha = 0.021$, $|\tilde{y}_0^*|_\alpha = 0.023$—neither at the strict-IT nor the zero-gap corner. On average across scenarios, optimal policy accommodates roughly two-thirds of the aggregate-inflation magnitude that zero-gap permits, using a small output gap to damp CPI, consistent with the targeting criterion (52)

Table 5: Welfare and impact responses across policy closures (TRD shock)

Policy	$\mathcal{W}$	$\lvert\pi_0^C\rvert$	$\lvert\tilde{y}_0\rvert$	$\mathcal{W}_\pi$	$\mathcal{W}_y$	$\mathcal{W}_\chi$	$\mathcal{W}_r$
Ramsey	1.191	0.021	0.023	1.077	0.003	0.057	0.055
Zero-gap	1.208	0.031	0.000	1.101	0.000	0.055	0.053
Taylor	1.239	0.020	0.020	1.128	0.002	0.055	0.054
Infl. Taylor	1.297	0.013	0.030	1.183	0.004	0.056	0.054
Strict IT	1.465	0.000	0.051	1.342	0.011	0.057	0.056

Notes. Each column reports the $\boldsymbol{\alpha}$-weighted average of the corresponding quantity across $N = 14$ single-sector shock scenarios, where scenario i features a one-standard-deviation negative productivity shock to sector i alone. Total welfare loss satisfies $\mathcal{W}_\alpha = \mathcal{W}_\pi + \mathcal{W}_y + \mathcal{W}_\chi + \mathcal{W}_r$ (normalized by shock variance). Components: $\mathcal{W}_\pi$ Calvo price dispersion, $\mathcal{W}_y$ output-gap loss, $\mathcal{W}_\chi$ relative-price distortion, $\mathcal{W}_r$ employment misallocation. Impact columns report the $\boldsymbol{\alpha}$-average of absolute period-0 responses.

Table 6: Impact responses to a negative productivity shock in a flexible vs. sticky sector

	TRD ($\kappa = 0.235$, flexible)			PBS ($\kappa = 0.008$, sticky)		
Policy	π_0^C	$\tilde{y}_0$	$\mathcal{W}$	π_0^C	$\tilde{y}_0$	$\mathcal{W}$
Ramsey	-0.051	$+0.056$	1.025	$+0.028$	-0.031	2.061
Zero-gap	$+0.072$	0	1.057	-0.047	0	2.107
Taylor	$+0.048$	-0.048	1.094	-0.025	$+0.025$	2.228
Infl. Taylor	$+0.034$	-0.073	1.196	-0.014	$+0.038$	2.401
Strict IT	0	-0.128	1.708	0	$+0.056$	2.656

in which the reallocation wedge $(MK\mathbf{1})^\top \boldsymbol{\eta}_t$ generically pushes $\pi_t^{C,*}$ away from zero.

The averaged magnitudes hide a sign pattern that only emerges at the sector shock level. Table 6 contrasts impact responses to a negative productivity shock in a flexible sector (tradables, $\kappa = 0.235$) and in a sticky sector (professional and business services, $\kappa = 0.008$).

Zero-gap reproduces the natural direction of sectoral inflation: a negative shock to a TRD shock drives CPI up by 0.072; the PBS shock drives it down by 0.047. The non-Ramsey benchmark rules all fight these movements in the conventional direction: contracting against the TRD-induced inflation, expanding against the PBS-induced deflation, with strict IT at the extreme ($\tilde{y}_0 = -0.128$ for TRD, $+0.056$ for PBS). Ramsey does the opposite. It expands when tradables are hit ($\tilde{y}_0 = +0.056$), driving CPI inflation negative ($\pi_0^C = -0.051$); it contracts when the sticky sector is hit ($\tilde{y}_0 = -0.031$), driving CPI inflation positive ($\pi_0^C = +0.028$). In both cases Ramsey moves aggregate demand and aggregate inflation in the opposite direction. This shows that optimal policy uses aggregate demand to assist the cross-sector reallocation rather than to offset its nominal consequences.

7 Conclusion

We have developed a multi-sector New Keynesian model in which frictional labor reallocation, summarized by a mobility elasticity and a steady-state transition matrix, reshapes the transmission of sectoral shocks and the design of optimal monetary policy. Three results stand out. First, labor mobility creates countervailing effects on sectoral inflation: it dampens impact but slows convergence. Second, the persistence of shocks depends not on average mobility or average stickiness alone, but on whether displaced workers flow into flexible or sticky sectors. Third, optimal policy under commitment deviates from standard sticky-price targeting precisely when price rigidity aligns with the slowest-mixing component of the reallocation network, as the planner exploits flexible sectors' price responsiveness to redirect worker flows.

Several extensions are natural. Our model assumes that aggregate demand affects demand for all sectors symmetrically: in practice, interest rate changes have heterogeneous incidence across sectors—durables respond more than services, for instance—which would change the incidence of aggregate demand onto sectoral demand. Incorporating production networks alongside the labor network would allow both input-output and worker-flow linkages to shape persistence and policy simultaneously.

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Appendix

A Log-Linearization Derivations

This appendix details the log-linearization of the model around the steady state. Because all output is consumed, we impose goods market clearing $y_t = c_t$ and $y_{it} = c_{it}$ throughout, using y to denote both output and consumption.

Notation. For any variable X_t, we write $x_t \equiv \log X_t - \log \bar{X}$ for its log-deviation from steady state. We use the following conventions throughout:

- *Natural (flexible-price) levels*: x_t^{nat} denotes the value under flexible prices, defined conditional on the actual predetermined state at t; Lemma 1 gives the explicit construction.
- *Gaps*: $\tilde{x}_t \equiv x_t - x_t^{\text{nat}}$ denotes the deviation from the natural level.
- *Relative prices*: $\hat{p}_{it} \equiv p_{it} - p_t^C$ is the log sectoral price relative to the CPI.
- *Relative price distortion*: $\chi_{it} \equiv \hat{p}_{it} + \hat{a}_{it}$. Equivalently, $\chi_{it} = \hat{p}_{it} - \hat{p}_{it}^{\text{nat},0}$ where $\hat{p}_{it}^{\text{nat},0} \equiv -\hat{a}_{it}$ is the relative price when prices are flexible and inherited employment is at its steady-state distribution (Lemma 1).

We write $\overset{(1)}{=}$ for equalities that hold up to a first-order approximation, and $\overset{(2)}{=}$ for second-order. Plain $=$ denotes exact relationships or definitions.

A.1 The Macro Block

A.1.1 Sectoral Marginal Cost

Real marginal cost in sector i (in log-deviations) is given by $mc_{it} - p_{it} = w_{it} - a_{it} - p_{it}$. Using the intratemporal labor supply condition (6) and substituting aggregate consumption with output ($c_t = y_t$), the sectoral wage satisfies:

$$w_{it} - p_t^C = \varphi l_{it} + \gamma y_t. \tag{A.1}$$

Substituting this into the marginal cost expression yields:

$$mc_{it} - p_{it} = \gamma y_t + \varphi l_{it} - a_{it} - \hat{p}_{it},$$

where $\hat{p}_{it} \equiv p_{it} - p_t^C$ is the relative sectoral price. To eliminate l_{it}, we use the log-linearized production function ($l_{it} = y_{it} - a_{it} - n_{it}$) and substitute out sectoral output y_{it} using the demand

curve ($y_{it} = y_t - \hat{p}_{it}$). Note that n_{it} here is the *contemporaneous*, post-reallocation employment share in sector i — this is the mass of workers physically producing in sector i at date t, and hence the relevant margin for marginal cost. Grouping terms directly yields Equation (17) from the main text:

$$mc_{it} - p_{it} \overset{(1)}{=} (\gamma + \varphi)y_t - (1 + \varphi)(\hat{p}_{it} + a_{it}) - \varphi n_{it}. \tag{A.2}$$

A.1.2 Phillips Curve

Standard log-linearization of the Calvo pricing problem (3) yields the sectoral New Keynesian Phillips curve. A re-optimizing firm in sector i sets its price as a markup over a weighted average of current and expected future marginal costs. Log-linearizing the optimal reset price around the zero-inflation steady state, and combining with the Calvo price index evolution $P_{it} = [\delta_i P_{it-1}^{1-\epsilon_i} + (1 - \delta_i)(P_{it}^*)^{1-\epsilon_i}]^{1/(1-\epsilon_i)}$, yields:

$$\pi_{it} \overset{(1)}{=} \kappa_i(mc_{it} - p_{it}) + \beta\, \mathbb{E}_t \pi_{it+1}, \tag{A.3}$$

where $\kappa_i \equiv (1 - \delta_i)(1 - \beta\delta_i)/\delta_i$. This is Equation (21).

A.1.3 The IS Curve and Natural Real Rate

Note on convention: As is standard, we make an exception to the log-deviation convention for interest rates. We let i_t and r_t^{nat} denote the levels of the net nominal and natural real interest rates, respectively.

The Euler equation (7) is

$$1 = \beta(1 + i_t)\, \mathbb{E}_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \frac{P_t^C}{P_{t+1}^C} \right].$$

Taking logs and using the first-order approximation $\log(1 + i_t) \overset{(1)}{=} i_t$:

$$0 \overset{(1)}{=} \log \beta + i_t - \gamma\, \mathbb{E}_t(y_{t+1} - y_t) - \mathbb{E}_t \pi_{t+1}^C,$$

where we have substituted $c_t = y_t$ (goods market clearing) and $\pi_{t+1}^C = p_{t+1}^C - p_t^C$. Rearranging yields:

$$y_t \overset{(1)}{=} \mathbb{E}_t y_{t+1} - \frac{1}{\gamma}(i_t - \mathbb{E}_t \pi_{t+1}^C + \log \beta). \tag{A.4}$$

The same equation holds under flexible prices with y_t^{nat} in place of y_t and i_t^{nat} in place of i_t (and

$\pi_t^C = 0$ under flexible prices). We define the natural real rate (in levels) as:

$$r_t^{\text{nat}} \equiv -\log\beta + \gamma(\mathbb{E}_t y_{t+1}^{\text{nat}} - y_t^{\text{nat}}).$$

Subtracting the flexible-price Euler equation from (A.4) gives the IS curve, Equation (22):

$$\tilde{y}_t \overset{(1)}{=} \mathbb{E}_t \tilde{y}_{t+1} - \frac{1}{\gamma}\left(i_t - \mathbb{E}_t \pi_{t+1}^C - r_t^{\text{nat}}\right). \tag{A.5}$$

A.1.4 Goods Market Clearing and Relative Prices

Cobb–Douglas demand gives $P_{it}C_{it} = \alpha_i P_t^C C_t$, so sectoral output satisfies $P_{it}Y_{it} = \alpha_i P_t^C C_t$. In logs:

$$y_{it} - y_t = -\hat{p}_{it}. \tag{A.6}$$

Log aggregate output is $y_t = \sum_i \alpha_i a_{it} + \sum_i \alpha_i (n_{it} + l_{it})$, using $y_{it} = a_{it} + n_{it} + l_{it}$ (from the production function and labor market clearing) weighted by consumption shares.

The CPI definition $P_t^C = \prod_i P_{it}^{\alpha_i}$ implies $p_t^C = \sum_i \alpha_i p_{it}$, so CPI inflation is $\pi_t^C = \sum_i \alpha_i \pi_{it}$. Relative prices evolve as:

$$\hat{p}_{it} - \hat{p}_{it-1} = (p_{it} - p_{it-1}) - (p_t^C - p_{t-1}^C) = \pi_{it} - \pi_t^C, \tag{A.7}$$

which is Equation (15).

A.2 Dynamic Labor Reallocation Block

The dynamic reallocation block consists of the flow utility, the Bellman equation, and the employment evolution equation.

A.2.1 Flow Utility

Using the intratemporal labor supply condition (6), flow utility simplifies to $U_{it}^{\text{flow}} = \zeta \frac{\varphi}{1+\varphi} L_{it}^{1+\varphi}$. Define the normalized deviation $u_{it} \equiv (U_{it}^{\text{flow}} - \bar{U}^{\text{flow}})/(\varphi\zeta\bar{L}^{1+\varphi})$:

$$\frac{U_{it}^{\text{flow}} - \bar{U}^{\text{flow}}}{\varphi\zeta\bar{L}^{1+\varphi}} = \frac{\zeta\frac{\varphi}{1+\varphi}(L_{it}^{1+\varphi} - \bar{L}^{1+\varphi})}{\varphi\zeta\bar{L}^{1+\varphi}} = \frac{L_{it}^{1+\varphi} - \bar{L}^{1+\varphi}}{(1+\varphi)\bar{L}^{1+\varphi}}.$$

A first-order expansion of $L_{it}^{1+\varphi}$ around $\bar{L}$ gives $L_{it}^{1+\varphi} \approx \bar{L}^{1+\varphi}(1+(1+\varphi)l_{it})$, so $u_{it} \overset{(1)}{=} l_{it}$. Substituting $l_{it} = y_t - \hat{p}_{it} - a_{it} - n_{it}$ yields Equation (20):

$$u_{it} = y_t - \hat{p}_{it} - a_{it} - n_{it}. \tag{A.8}$$

A.2.2 Value Function Recursion

The Bellman equation (9) is:

$$V_{it} = U_{it}^{\text{flow}} + \beta\, \mathbb{E}_t \left[\frac{1}{\theta} \log \left(\sum_{j \in \mathcal{N}} \exp\big(\theta[V_{j,t+1} - \tau^{ij}]\big) \right) \right].$$

In steady state, $\bar{V}_i = \bar{U}^{\text{flow}} + \beta \Lambda_i(\bar{\mathbf{z}})$ where $\Lambda_i(\mathbf{z}) \equiv \frac{1}{\theta} \log\Big(\sum_j \exp(\theta z_j)\Big)$ and $\bar{z}_j = \bar{V}_j - \tau^{ij}$. The first-order Taylor expansion of Λ_i around $\bar{\mathbf{z}}$ uses the standard logit property:

$$\left.\frac{\partial \Lambda_i}{\partial z_j}\right|_{\bar{z}} = \frac{\exp(\theta \bar{z}_j)}{\sum_{j'} \exp(\theta \bar{z}_{j'})} = F_{ij},$$

where $F_{ij} = \bar{\mu}^{ij}$ is the steady-state transition probability from sector i to sector j (Equation (8) evaluated at steady state). Therefore, with $z_{j,t+1} = V_{j,t+1} - \tau^{ij}$:

$$\Lambda_i(\mathbf{z}_{t+1}) - \Lambda_i(\bar{\mathbf{z}}) \overset{(1)}{=} \sum_j F_{ij}(V_{j,t+1} - \bar{V}_j).$$

Subtracting the steady-state Bellman from the time-t Bellman, taking $\mathbb{E}_t$ through the linear term, and normalizing by $\varphi\zeta\bar{L}^{1+\varphi}$:

$$v_{it} = u_{it} + \beta \sum_j F_{ij}\, \mathbb{E}_t[v_{j,t+1}].$$

In matrix notation, this is Equation (19):

$$\mathbf{v}_t = \mathbf{u}_t + \beta F\, \mathbb{E}_t[\mathbf{v}_{t+1}]. \tag{A.9}$$

A.2.3 Employment Evolution

Employment evolves according to $N_{jt} = \sum_i \mu_t^{ij} N_{i,t-1}$, where the transition probabilities are given by the logit form (8):

$$\mu_t^{ij} = \frac{\exp\big(\theta[V_{jt} - \tau^{ij}]\big)}{\sum_{j'} \exp\big(\theta[V_{j't} - \tau^{ij'}]\big)}.$$

The transition probability μ_t^{ij} has the logit form, so its first-order expansion around the steady state $(F_{ij}, \bar{V}_j)$ is:

$$\begin{aligned}\mu_t^{ij} - F_{ij} &\overset{(1)}{=} F_{ij}\,\theta\left((V_{jt} - \bar{V}_j) - \sum_{j'} F_{ij'}(V_{j't} - \bar{V}_{j'})\right) \\ &= F_{ij}\,\phi\left(v_{jt} - \sum_{j'} F_{ij'}\,v_{j't}\right),\end{aligned}$$

where the second line uses the normalization $v_{jt} = (V_{jt} - \bar{V}_j)/(\varphi\zeta\bar{L}^{1+\varphi})$ and $\phi \equiv \varphi\theta\zeta\bar{L}^{1+\varphi}$, so that $\theta(V_{jt} - \bar{V}_j) = \phi\,v_{jt}$. That is:

$$\mu_t^{ij} - F_{ij} \overset{(1)}{=} \phi\,F_{ij}\big(v_{jt} - [F\,\mathbf{v}_t]_i\big)\,. \tag{A.10}$$

Deriving the employment law of motion. Log-linearize $N_{jt} = \sum_i \mu_t^{ij} N_{i,t-1}$ around $\bar{N}_j = \alpha_j$. Since $\bar{N}_j = \sum_i F_{ij}\bar{N}_i$ (stationarity of $\bar{\mathbf{N}}$ under F), taking total differentials:

$$\bar{N}_j\,n_{jt} \overset{(1)}{=} \sum_i F_{ij}\bar{N}_i\,n_{i,t-1} + \sum_i \bar{N}_i(\mu_t^{ij} - F_{ij}).$$

We handle the two terms in turn. For the first term, the definition of the backward matrix $B_{ji} \equiv \bar{N}_i F_{ij}/\bar{N}_j$ (Equation (10)) gives $\bar{N}_i F_{ij} = \bar{N}_j B_{ji}$, so:

$$\frac{1}{\bar{N}_j}\sum_i F_{ij}\bar{N}_i\,n_{i,t-1} = \sum_i B_{ji}\,n_{i,t-1} = [B\,\mathbf{n}_{t-1}]_j.$$

For the second term, substituting (A.10):

$$\sum_i \bar{N}_i(\mu_t^{ij} - F_{ij}) = \phi\sum_i \bar{N}_i F_{ij}\big(v_{jt} - [F\,\mathbf{v}_t]_i\big)\,.$$

Applying $\bar{N}_i F_{ij} = \bar{N}_j B_{ji}$ and dividing by $\bar{N}_j$:

$$\begin{aligned}(\text{second term})/\bar{N}_j &= \phi\sum_i B_{ji}\big(v_{jt} - [F\,\mathbf{v}_t]_i\big) \\ &= \phi\big(v_{jt} - [BF\,\mathbf{v}_t]_j\big)\,,\end{aligned}$$

where the second line uses $\sum_i B_{ji} = 1$ (row-stochasticity of B).

Combining both terms:

$$n_{jt} = [B\,\mathbf{n}_{t-1}]_j + \phi[(I - BF)\,\mathbf{v}_t]_j\,.$$

In vector form, this is Equation (18):

$$\mathbf{n}_t = B\,\mathbf{n}_{t-1} + \phi(I - BF)\,\mathbf{v}_t. \tag{A.11}$$

Limiting cases. When $\phi = 0$ (segmented markets), the second term vanishes and $\mathbf{n}_t = B\,\mathbf{n}_{t-1}$. Starting from $\mathbf{n}_{-1} = \mathbf{0}$, we have $\mathbf{n}_t = \mathbf{0}$ for all $t \geq 0$: no reallocation occurs.

When $\phi \to \infty$, boundedness of $\mathbf{n}_t$ requires $(I - BF)\,\mathbf{v}_t \to \mathbf{0}$. Since BF is row-stochastic and primitive, $\ker(I - BF) = \text{span}\{\mathbf{1}\}$, so $\mathbf{v}_t \to c_t\,\mathbf{1}$: continuation values equalize across sectors *within each period*. Substituting into the Bellman recursion (19) and using $F\,\mathbf{1} = \mathbf{1}$ gives $\mathbf{u}_t = m_t\,\mathbf{1}$, where $m_t \equiv c_t - \beta\,\mathbb{E}_t[c_{t+1}]$. By (20), $u_{it} = y_t - \hat{p}_{it} - a_{it} - n_{it}$, so $n_{it} = y_t - \hat{p}_{it} - a_{it} - m_t$. Imposing $\sum_i \alpha_i n_{it} = 0$ and $\sum_i \alpha_i \hat{p}_{it} = 0$ yields $m_t = y_t - \sum_j \alpha_j a_{jt}$, and substituting back gives $n_{it} = -(\hat{p}_{it} + \hat{a}_{it}) = -\chi_{it}$. This identity holds for *all* $t \geq 0$ because reallocation is contemporaneous.

B Second-Order Expansion Details

This appendix derives the quadratic welfare loss function in 8. We expand household utility around the efficient (natural) allocation to second order, term by term.

B.1 Proposition 8: Dynamic LQ

We derive the per-period welfare loss under the timeless perspective:

$$\mathcal{L}_t \overset{(2)}{=} \frac{\gamma + \varphi}{2}\,\tilde{y}_t^2 + \sum_i \frac{\alpha_i \epsilon_i}{2\kappa_i}\,\pi_{it}^2 + \sum_i \alpha_i \left[\frac{1+\varphi}{2}\,\chi_{it}^2 + \varphi\,\chi_{it}\,n_{it} + \frac{\varphi}{2}\,n_{it}^2\right].$$

The derivation proceeds in two steps. Step 1 expands the consumption-minus-labor-disutility component $u(C_t) - \sum_i N_{it}\,v(L_{it})$ to second order in steady-state deviations. Step 2 shows that the mobility cost Ψ_t is a term whose discounted sum telescopes into a boundary term that enters the Ramsey Lagrangian only through the date-0 first-order conditions. Under the timeless perspective, the per-period loss is therefore the second-order expansion of $u - \sum Nv$ alone.

Notation. All variables denote log-deviations from the efficient steady state, as defined in Appendix A. We write $h_{it} \equiv l_{it} - l_t^\alpha$ for cross-sector hours and normalize by $U_C\bar{C} = \bar{C}^{1-\gamma} = \zeta\bar{L}^{1+\varphi}$. Three identities are used throughout:

(i) *Production:* $y_t = \bar{a}_t + l_t^\alpha + \sum_i \alpha_i\,n_{it} + \sum_i \alpha_i\,\xi_{it}$, where $\xi_{it} \equiv -\log\Delta_{it} \leq 0$ is the within-sector price-dispersion wedge. This is exact under Cobb-Douglas utility.

(ii) *Cross-section:* $h_{it} \overset{(1)}{=} -(\chi_{it} + n_{it})$, from goods market clearing and the production function.

(iii) *Adding-up:* $\sum_i \alpha_i\, n_{it} \overset{(2)}{=} -\frac{1}{2}\sum_i \alpha_i\, n_{it}^2$, from $\sum_i \alpha_i\, e^{n_{it}} = 1$.

B.1.1 Second-Order Expansion of $u(C_t) - \sum_i N_{it}\, v(L_{it})$

A standard expansion of $u(C)$ gives:

$$\frac{u(C_t) - u(\bar{C})}{U_C \bar{C}} \overset{(2)}{=} y_t + \frac{1-\gamma}{2}\, y_t^2. \tag{B.1}$$

For labor disutility, expanding $N_{it}\, v(L_{it}) = \frac{\alpha_i \zeta \bar{L}^{1+\varphi}}{1+\varphi} \exp\big(n_{it} + (1+\varphi) l_{it}\big)$ to second order, summing over i, dividing by $U_C\bar{C}$, and using identity (iii) to cancel the n_{it} and n_{it}^2 terms:

$$\frac{\sum_i N_{it}\, v(L_{it}) - \sum_i \alpha_i\, v(\bar{L})}{U_C \bar{C}} \overset{(2)}{=} l_t^\alpha + \sum_i \alpha_i\, n_{it}\, h_{it} + \frac{1+\varphi}{2}\Big[(l_t^\alpha)^2 + \mathcal{M}_t\Big], \tag{B.2}$$

where $\mathcal{M}_t \equiv \sum_i \alpha_i\, h_{it}^2$. Subtracting (B.2) from (B.1):

$$\frac{\big[u(C_t) - \sum_i N_{it}\, v(L_{it})\big] - \text{SS}}{U_C \bar{C}} \overset{(2)}{=} \underbrace{(y_t - l_t^\alpha)}_{\text{first-order}} + \underbrace{\frac{1-\gamma}{2}\, y_t^2 - \frac{1+\varphi}{2}\,(l_t^\alpha)^2}_{\text{aggregate quadratic}} - \underbrace{\frac{1+\varphi}{2}\,\mathcal{M}_t + \sum_i \alpha_i\, n_{it}\, h_{it}}_{\text{cross-sector}}. \tag{B.3}$$

We evaluate each group of terms in turn.

Aggregate quadratic. Since $l_t^\alpha \overset{(1)}{=} y_t - \bar{a}_t$ (from identity (i)), the aggregate quadratic in (B.3) is:

$$\frac{1-\gamma}{2}\, y_t^2 - \frac{1+\varphi}{2}\,(y_t - \bar{a}_t)^2 \;=\; -\frac{\gamma+\varphi}{2}\, y_t^2 + (1+\varphi)\, y_t\, \bar{a}_t - \frac{1+\varphi}{2}\, \bar{a}_t^2. \tag{B.4}$$

Substituting $y_t = y_t^{\text{nat}} + \tilde{y}_t$ with $y_t^{\text{nat}} = \frac{1+\varphi}{\gamma+\varphi}\, \bar{a}_t$, the coefficient of $\tilde{y}_t$ is $-(\gamma+\varphi)\, y_t^{\text{nat}} + (1+\varphi)\, \bar{a}_t = 0$. Hence:

$$(\text{B.4}) \;=\; -\frac{\gamma+\varphi}{2}\, \tilde{y}_t^2 + (\text{t.i.p}), \tag{B.5}$$

with t.i.p denoting terms independent of policy.[14]

[14] The vanishing of the cross-term is the multi-sector counterpart of the standard result: the efficient-subsidy condition ensures that the first variation of $u - \sum N v$ with respect to aggregate output vanishes at the natural level.

Cross-sector terms. Using identity (ii), the cross-sector contribution in (B.3) is, for each sector i (with the minus sign from the subtraction):

$$-\frac{1+\varphi}{2}\left(\chi_{it}+n_{it}\right)^2+n_{it}\left(\chi_{it}+n_{it}\right) \;=\; -\frac{1+\varphi}{2}\,\chi_{it}^2-\varphi\,\chi_{it}\,n_{it}-\frac{\varphi-1}{2}\,n_{it}^2. \tag{B.6}$$

Output–labor gap. The first-order term $y_t - l_t^{\alpha}$ in (B.3) has a nonzero second-order component. From the exact production identity (i) and the exogeneity of $\bar{a}_t$:

$$y_t - l_t^{\alpha} \;=\; \bar{a}_t+\sum_i \alpha_i\,n_{it}+\sum_i \alpha_i\,\xi_{it} \;\overset{(2)}{=}\; \text{t.i.p.}-\frac{1}{2}\sum_i \alpha_i\,n_{it}^2+\sum_i \alpha_i\,\xi_{it}, \tag{B.7}$$

using identity (iii). The Calvo price-dispersion recursion implies the present-value equivalence (assuming $\xi_{i,-1}=0$):

$$\sum_{t=0}^{\infty}\beta^t\sum_i \alpha_i\,\xi_{it} \;\overset{(2)}{=}\; -\sum_{t=0}^{\infty}\beta^t\sum_i \frac{\alpha_i\epsilon_i}{2\kappa_i}\,\pi_{it}^2. \tag{B.8}$$

Assembly. Substituting (B.5), (B.6), and (B.7) into (B.3), the period return relative to steady state is, in discounted-sum equivalence:

$$\begin{aligned}\frac{\left[u-\sum Nv\right]-\text{SS}}{U_C\bar{C}} \;\overset{(2)}{=}\; &\text{t.i.p.}-\frac{\gamma+\varphi}{2}\,\tilde{y}_t^2-\sum_i \frac{\alpha_i\epsilon_i}{2\kappa_i}\,\pi_{it}^2\\ &-\sum_i \alpha_i\left[\frac{1+\varphi}{2}\,\chi_{it}^2+\varphi\,\chi_{it}\,n_{it}+\frac{\varphi}{2}\,n_{it}^2\right],\end{aligned} \tag{B.9}$$

where the coefficient $\frac{\varphi}{2}$ on n_{it}^2 combines $\frac{\varphi-1}{2}$ from the cross-sector terms (B.6) and $\frac{1}{2}$ from the output–labor gap (B.7). The former reflects the convexity of the disutility of hours; the latter reflects the concavity of the Cobb–Douglas aggregator, which penalizes uneven employment even holding total labor input fixed.

Negating (B.9) and dropping terms independent of policy:

$$\mathcal{L}_t^{u-Nv} \;\overset{(2)}{=}\; \frac{\gamma+\varphi}{2}\,\tilde{y}_t^2+\sum_i \frac{\alpha_i\epsilon_i}{2\kappa_i}\,\pi_{it}^2+\sum_i \alpha_i\left[\frac{1+\varphi}{2}\,\chi_{it}^2+\varphi\,\chi_{it}\,n_{it}+\frac{\varphi}{2}\,n_{it}^2\right], \tag{B.10}$$

in discounted-sum equivalence.[15]

[15] Because the π_{it}^2 term arises from the present-value equivalence (B.8), equation (B.10) is the per-period term whose discounted sum equals the normalized welfare loss to second order.

B.1.2 Mobility Cost Telescopes into a Startup Term

Step 1 expanded $u(C_t) - \sum_i N_{it}\, v(L_{it})$. The full per-period utility also includes the mobility cost $-\Psi_t$, incurred at the start of period t on the inherited mass $N_{i,t-1}$. We now construct a sector-specific net continuation value for the *pre-reallocation* stage that strips out future flow utility, yielding a boundary term weighted by the predetermined initial distribution $\{N_{i,-1}\}$.

Pre-reallocation value function. Let

$$V_{it}^{pre} \;\equiv\; \frac{1}{\theta} \log \sum_j \exp\big(\theta[V_{jt} - \tau^{ij}]\big) \tag{B.11}$$

denote the value of a worker labeled in sector i at the start of period t, before drawing a preference shock and reallocating. By the envelope theorem applied to the logit optimization,

$$V_{it}^{pre} \;=\; \sum_j \mu_t^{ij} V_{jt} - \psi_i\Big(\{\mu_t^{ij}\}_j\Big)\,. \tag{B.12}$$

The Bellman equation (9) for post-reallocation V_{jt} reads $V_{jt} = U_{jt}^{flow} + \beta\, \mathbb{E}_t V_{j,t+1}^{pre}$.

Sector-specific net continuation value. Define the flow-utility continuation at the pre-reallocation stage

$$F_{it}^{pre} \;\equiv\; \sum_j \mu_t^{ij} \Big[U_{jt}^{flow} + \beta\, \mathbb{E}_t F_{j,t+1}^{pre}\Big]\,, \tag{B.13}$$

and the net pre-reallocation continuation value

$$\check{V}_{it}^{pre} \;\equiv\; V_{it}^{pre} - F_{it}^{pre}. \tag{B.14}$$

F_{it}^{pre} captures the expected discounted flow utility a pre-reallocation worker in sector i would earn along the realized transition path; $\check{V}_{it}^{pre}$ isolates the option value of reallocation net of mobility costs.

Lemma B.1. *Let* $\check{\mathcal{B}}_t \equiv \sum_i N_{i,t-1}\, \check{V}_{it}^{\text{pre}}$. *Then:*

$$-\Psi_t \;=\; \check{\mathcal{B}}_t - \beta\, \mathbb{E}_t\, \check{\mathcal{B}}_{t+1}. \tag{B.15}$$

Consequently, if $\lim_{T\to\infty} \beta^T\, \mathbb{E}_0 |\check{\mathcal{B}}_T| = 0$, *then*

$$\mathcal{W}_0 \;=\; \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Big[u(C_t) - \sum_i N_{it}\, v(L_{it})\Big] \;+\; \check{\mathcal{B}}_0. \tag{B.16}$$

Proof. Combining the envelope (B.12) with $V_{jt} = U_{jt}^{\text{flow}} + \beta \mathbb{E}_t V_{j,t+1}^{\text{pre}}$:

$$V_{it}^{\text{pre}} \;=\; \sum_j \mu_t^{ij} \Big[U_{jt}^{\text{flow}} + \beta\, \mathbb{E}_t V_{j,t+1}^{\text{pre}} \Big] - \psi_i\Big(\{\mu_t^{ij}\}_j \Big) .$$

Subtracting the F_{it}^{pre} recursion (B.13) and using $\check{V} = V - F$ term by term:

$$\check{V}_{it}^{\text{pre}} \;=\; \beta \sum_j \mu_t^{ij}\, \mathbb{E}_t\, \check{V}_{j,t+1}^{\text{pre}} - \psi_i\Big(\{\mu_t^{ij}\}_j \Big) . \tag{B.17}$$

Multiplying by $N_{i,t-1}$, summing over i, and using the flow identity $\sum_i N_{i,t-1}\, \mu_t^{ij} = N_{jt}$:

$$\sum_i N_{i,t-1}\, \check{V}_{it}^{\text{pre}} \;=\; \beta\, \mathbb{E}_t \sum_j N_{jt}\, \check{V}_{j,t+1}^{\text{pre}} - \sum_i N_{i,t-1}\, \psi_i\Big(\{\mu_t^{ij}\}_j \Big) .$$

The left-hand side is $\check{\mathcal{B}}_t$; the first right-hand-side term is $\beta\, \mathbb{E}_t \check{\mathcal{B}}_{t+1}$ since $N_{jt} = N_{j,(t+1)-1}$; the second is Ψ_t by definition. This yields (B.15). Multiplying by β^t, summing, and applying the transversality condition telescopes the mobility cost into $-\check{\mathcal{B}}_0$. Substituting into $\mathcal{W}_0 = \mathbb{E}_0 \sum_{t \geq 0} \beta^t [u - \sum N v - \Psi_t]$ yields (B.16). □

Equation (B.16) is exact: the entire contribution of the mobility block to lifetime welfare is captured by $\check{\mathcal{B}}_0 = \sum_i N_{i,-1}\, \check{V}_{i0}^{\text{pre}}$. The key property of this construction is that $\check{\mathcal{B}}_0$ is a *pure startup term*: it depends on the future allocation only through the initial pre-reallocation net continuation values $\{\check{V}_{i0}^{\text{pre}}\}$, which are themselves determined by the recursion (B.17), weighted by the predetermined initial distribution $\{N_{i,-1}\}$. In the Ramsey problem with $\{\check{V}_{it}^{\text{pre}}\}$ treated as auxiliary variables subject to this recursion, $\check{\mathcal{B}}_0$ affects only the date-0 first-order conditions—it is the source of the well-known tension between the optimal long-run policy rule and the incentive to deviate at the initial date.

Under the timeless perspective (Woodford, 2003), the planner commits to a policy rule that would have been optimal had it been adopted far in the past. This means requiring that the same targeting criterion hold at every date, including $t = 0$, rather than exploiting the special initial conditions. $\check{\mathcal{B}}_0$ affects only the exceptional startup conditions of the date-0 Ramsey problem. Under the timeless perspective, these startup conditions are replaced by the same self-consistent targeting conditions that apply for $t \geq 1$, and the per-period loss for the Ramsey problem under the timeless perspective is therefore $\mathcal{L}_t^{u-Nv}$ from (B.10).

Combining Steps 1 and 2 and dropping terms independent of policy, the per-period loss for the Ramsey problem is:

$$\mathcal{L}_t \;\overset{(2)}{=}\; \frac{\gamma + \varphi}{2}\, \tilde{y}_t^2 + \sum_i \frac{\alpha_i \epsilon_i}{2\kappa_i}\, \pi_{it}^2 + \sum_i \alpha_i \left[\frac{1+\varphi}{2}\, \chi_{it}^2 + \varphi\, \chi_{it}\, n_{it} + \frac{\varphi}{2}\, n_{it}^2 \right] , \tag{B.18}$$

in discounted-sum equivalence, which is equation (47). □

C Proofs of Propositions and Lemmas

C.1 Proofs for Section 3

Proof of Proposition 1. Inspection of (18)–(20) shows that the transition rates $\{\mu^{ij}\}$ enter only through F, B, and ϕ.

When $\phi = 0$, (18) gives $\mathbf{n}_t = B\,\mathbf{n}_{t-1}$ with $\mathbf{n}_{-1} = \mathbf{0}$, so $\mathbf{n}_t = \mathbf{0}$ for all $t \geq 0$.

When $\phi \to \infty$, boundedness of $\mathbf{n}_t$ in (18) requires $(I - BF)\,\mathbf{v}_t \to \mathbf{0}$. Since BF is row-stochastic and primitive, $\ker(I - BF) = \text{span}\{\mathbf{1}\}$, so $\mathbf{v}_t = c_t\,\mathbf{1}$ for some scalar c_t. Substituting into the Bellman recursion (19) and using $F\,\mathbf{1} = \mathbf{1}$ gives $\mathbf{u}_t = m_t\,\mathbf{1}$, where $m_t \equiv c_t - \beta\,\mathbb{E}_t[c_{t+1}]$. By (20), $u_{it} = y_t - \hat{p}_{it} - a_{it} - n_{it}$, so $n_{it} = y_t - \hat{p}_{it} - a_{it} - m_t$. Imposing the aggregate constraints $\sum_i \alpha_i n_{it} = 0$ and $\sum_i \alpha_i \hat{p}_{it} = 0$ yields $m_t = y_t - \sum_j \alpha_j a_{jt}$. Substituting back:

$$n_{it} = -\hat{p}_{it} - \left(a_{it} - \textstyle\sum_j \alpha_j a_{jt}\right) = -(\hat{p}_{it} + \hat{a}_{it})$$

for all $t \geq 0$. □

Proof of Lemma 1. Under flexible prices with the efficient subsidy, $mc_{it}^{\text{nat}} = p_{it}^{\text{nat}}$, so the log-linearized marginal cost (17) evaluated at the natural allocation gives

$$(1+\varphi)(\hat{p}_{it}^{\text{nat}} + a_{it}) = (\gamma + \varphi)\,y_t^{\text{nat}} - \varphi\,n_{it}^{\text{nat}}. \tag{C.1}$$

Part (i). Weighting (C.1) by α_i and summing: the left-hand side becomes $(1+\varphi)\sum_i \alpha_i a_{it}$ (using $\sum_i \alpha_i \hat{p}_{it}^{\text{nat}} = 0$), and the right-hand side becomes $(\gamma+\varphi)\,y_t^{\text{nat}}$ (using $\sum_i \alpha_i n_{it}^{\text{nat}} = 0$). Solving gives

$$y_t^{\text{nat}} = \frac{1+\varphi}{\gamma+\varphi} \sum_i \alpha_i a_{it}.$$

This depends only on aggregate productivity: neither $\mathbf{n}_{t-1}$ nor $\mathbf{n}_t^{\text{nat}}$ appears. The natural real rate follows from the Euler equation (7) evaluated at the natural allocation.

Part (ii). Solving (C.1) for $\hat{p}_{it}^{\text{nat}}$ and substituting y_t^{nat}:

$$\hat{p}_{it}^{\text{nat}} = \sum_j \alpha_j a_{jt} - a_{it} - \frac{\varphi}{1+\varphi}\,n_{it}^{\text{nat}} = \hat{p}_{it}^{\text{nat},0} - \frac{\varphi}{1+\varphi}\,n_{it}^{\text{nat}}.$$

Sectoral hours follow from the production function and goods market clearing ($l_{it}^{\text{nat}} = y_t^{\text{nat}} -$

$\hat{p}_{it}^{\text{nat}} - a_{it} - n_{it}^{\text{nat}}$):

$$l_{it}^{\text{nat}} = \frac{1-\gamma}{1+\varphi}\, y_t^{\text{nat}} - \frac{1}{1+\varphi}\, n_{it}^{\text{nat}}.$$

It remains to characterize $\mathbf{n}_t^{\text{nat}}$. The flex-price flow utility, from (20) evaluated at the natural allocation, is $u_{it}^{\text{nat}} = y_t^{\text{nat}} - \hat{p}_{it}^{\text{nat}} - a_{it} - n_{it}^{\text{nat}}$, which coincides with l_{it}^{nat} above. The Bellman recursion (19) evaluated at the natural allocation gives

$$\mathbf{v}_t^{\text{nat}} = \mathbf{u}_t^{\text{nat}} + \beta F\, \mathbb{E}_t \mathbf{v}_{t+1}^{\text{nat}},$$

and the law of motion (18) gives

$$\mathbf{n}_t^{\text{nat}} = B\, \mathbf{n}_{t-1} + \phi(I - BF)\, \mathbf{v}_t^{\text{nat}},$$

with $\mathbf{n}_{t-1}$ the actual sticky-price predetermined distribution. Together, these determine $\mathbf{n}_t^{\text{nat}}$ given $\mathbf{n}_{t-1}$ and the path of productivity. □

Proof of Proposition 2. We write the conditional supply block as a linear rational-expectations system, solve it via Blanchard and Kahn (1980), and reduce it to a law of motion for the predetermined state.

All variables are N-vectors constrained to the relative subspace $\mathcal{R} \equiv \{x \in \mathbb{R}^N : \boldsymbol{\alpha}^\top x = 0\}$; equivalently, after choosing a basis for $\mathcal{R}$, all matrices below act on $\mathcal{R}$-valued vectors. Let $M \equiv I - \mathbf{1}\boldsymbol{\alpha}^\top$ and $\mathbf{K} \equiv \operatorname{diag}(\kappa_i)$.

First, let's reduce $\mathbf{n}_t$. Under the contemporaneous reallocation timing, the employment law of motion (18) gives

$$\mathbf{n}_t = B\, \mathbf{n}_{t-1} + \phi(I - BF)\, \mathbf{v}_t, \tag{C.2}$$

and the value recursion (19) gives

$$\mathbf{v}_t = \mathbf{u}_t + \beta F\, \mathbb{E}_t[\mathbf{v}_{t+1}]. \tag{C.3}$$

Equation (C.2) pins down $\mathbf{n}_t$ as a static function of the predetermined $\mathbf{n}_{t-1}$ and the contemporaneous forward-looking $\mathbf{v}_t$: $\mathbf{n}_t$ is not a separate forward-looking variable but a jump that inherits its forward-looking content from $\mathbf{v}_t$.

Next, we can build the system in $(\hat{\boldsymbol{\pi}}_t, \mathbf{v}_t)$. Flow utility on the relative subspace is $\mathbf{u}_t = -\boldsymbol{\chi}_t - \mathbf{n}_t$. Substituting $\boldsymbol{\chi}_t = \boldsymbol{\chi}_{t-1} + \hat{\boldsymbol{\pi}}_t + \Delta\hat{\mathbf{a}}_t$ and (C.2) into (C.3) and rearranging,

$$[I + \phi(I - BF)]\, \mathbf{v}_t - \beta F\, \mathbb{E}_t[\mathbf{v}_{t+1}] = -\boldsymbol{\chi}_{t-1} - \hat{\boldsymbol{\pi}}_t - B\, \mathbf{n}_{t-1} - \Delta\hat{\mathbf{a}}_t. \tag{C.4}$$

The demeaned sectoral Phillips curve is $\hat{\boldsymbol{\pi}}_t = M\mathbf{K}[(\gamma+\varphi)\tilde{y}_t\mathbf{1} - (1+\varphi)\boldsymbol{\chi}_t - \varphi\, \mathbf{n}_t] + \beta\, \mathbb{E}_t \hat{\boldsymbol{\pi}}_{t+1}$.

Substituting χ_t and $\mathbf{n}_t$ as above,

$$\begin{aligned}[I + (1+\varphi)M\mathbf{K}]\,\hat{\boldsymbol{\pi}}_t - \beta\,\mathbb{E}_t\hat{\boldsymbol{\pi}}_{t+1} = &-(1+\varphi)\,M\mathbf{K}\,\chi_{t-1} - \varphi\,M\mathbf{K}\,B\,\mathbf{n}_{t-1} \\ &- \varphi\phi\,M\mathbf{K}(I-BF)\,\mathbf{v}_t \\ &+ (\gamma+\varphi)\,M\mathbf{K}\mathbf{1}\,\tilde{y}_t - (1+\varphi)\,M\mathbf{K}\,\Delta\hat{\mathbf{a}}_t. \end{aligned} \tag{C.5}$$

Stack predetermined variables as $\mathbf{x}_t = (\chi_{t-1}, \mathbf{n}_{t-1})$ and forward-looking variables as $\mathbf{z}_t = (\hat{\boldsymbol{\pi}}_t, \mathbf{v}_t)$. Equations (C.4)–(C.5) together with $\chi_t = \chi_{t-1} + \hat{\boldsymbol{\pi}}_t + \Delta\hat{\mathbf{a}}_t$ and (C.2) yield

$$\mathcal{A}\,\mathbb{E}_t\begin{bmatrix}\mathbf{x}_{t+1}\\ \mathbf{z}_{t+1}\end{bmatrix} = \mathcal{B}\begin{bmatrix}\mathbf{x}_t\\ \mathbf{z}_t\end{bmatrix} + \mathcal{C}_y\,\tilde{y}_t + \mathcal{C}_a\,\Delta\hat{\mathbf{a}}_t, \tag{C.6}$$

with

$$\mathcal{A} = \begin{bmatrix} I & 0 & 0 & 0\\ 0 & I & 0 & 0\\ 0 & 0 & \beta I & 0\\ 0 & 0 & 0 & \beta F\end{bmatrix}, \qquad \mathcal{B} = \begin{bmatrix} I & 0 & I & 0\\ 0 & B & 0 & \phi(I-BF)\\ (1+\varphi)M\mathbf{K} & \varphi M\mathbf{K}\,B & I+(1+\varphi)M\mathbf{K} & \varphi\phi M\mathbf{K}(I-BF)\\ I & B & I & I+\phi(I-BF)\end{bmatrix},$$

$$\mathcal{C}_y = \begin{bmatrix} 0\\ 0\\ -(\gamma+\varphi)M\mathbf{K}\mathbf{1}\\ 0\end{bmatrix}, \qquad \mathcal{C}_a = \begin{bmatrix} I\\ 0\\ (1+\varphi)M\mathbf{K}\\ I\end{bmatrix}.$$

The Blanchard–Kahn solution expresses the forward-looking variables as linear functions of the predetermined state. Write $\hat{\boldsymbol{\pi}}_t = \Theta_{11}\,\chi_{t-1} + \Theta_{12}\,\mathbf{n}_{t-1} + (\text{exog})$ and $\mathbf{v}_t = \Theta_{21}\,\chi_{t-1} + \Theta_{22}\,\mathbf{n}_{t-1} + (\text{exog})$, where $\{\Theta_{k\ell}\}$ are determined by the stable eigenspace of $(\mathcal{B}, \mathcal{A})$. Plugging these into the state equations:

$$\begin{aligned}\chi_t &= \chi_{t-1} + \hat{\boldsymbol{\pi}}_t + \Delta\hat{\mathbf{a}}_t = (I+\Theta_{11})\,\chi_{t-1} + \Theta_{12}\,\mathbf{n}_{t-1} + (\text{exog}),\\ \mathbf{n}_t &= B\,\mathbf{n}_{t-1} + \phi(I-BF)\,\mathbf{v}_t = \phi(I-BF)\,\Theta_{21}\,\chi_{t-1} + [B + \phi(I-BF)\,\Theta_{22}]\,\mathbf{n}_{t-1} + (\text{exog}).\end{aligned}$$

Shifting forward one period, $\mathbf{x}_{t+1} = \mathbf{P}\,\mathbf{x}_t + \mathbf{b}_t^y + \mathbf{b}_t^a$ with

$$\mathbf{P} = \begin{bmatrix} I+\Theta_{11} & \Theta_{12}\\ \phi(I-BF)\,\Theta_{21} & B + \phi(I-BF)\,\Theta_{22}\end{bmatrix}. \tag{C.7}$$

The observation equation (31) is the first row of the policy function $(\hat{\boldsymbol{\pi}}_t, \mathbf{v}_t) = \Theta(\chi_{t-1}, \mathbf{n}_{t-1}) + \text{exog}$. Stability of $\mathbf{P}$ follows from the Blanchard–Kahn condition on $(\mathcal{A}, \mathcal{B})$.

Because $\mathbf{P}$ is a function of the stable eigenspace of $(\mathcal{B}, \mathcal{A})$, it inherits their parameter depen-

dence: $\mathbf{P}$ depends on $(\mathbf{K}, \varphi, \beta, \phi, F, B)$ but not on γ, which appears only in the aggregate demand loading $\mathcal{C}_y$ and therefore affects only the forcing sequences $\{\mathbf{H}_j^y\}$. □

Proof of Corollary 1. If $\kappa_i = \kappa$ for all i, then $M\mathbf{K}\mathbf{1} = \kappa\, M\mathbf{1} = \mathbf{0}$, so $\tilde{y}_t$ drops out of the demeaned Phillips curve (C.5). The value recursion (C.4) contains no $\tilde{y}_t$ either. Therefore $\tilde{y}_{t+j}$ does not appear in any expectational equation of the relative block, and $\mathbf{H}_j^y = \mathbf{0}$ for all $j \geq 0$. □

Proof of Proposition 3. The permanent productivity shock translates to $\Delta\hat{\mathbf{a}}_t = \mathbf{x}_j$ at $t = 0$ and $\mathbf{0}$ for $t \geq 1$, where $\mathbf{x}_j \equiv \mathbf{e}_j - \alpha_j\mathbf{1}$.

Part (i): Transition path. By Proposition 2 with $\tilde{y}_t = 0$ and $\mathbf{s}_0 = \mathbf{0}$, the state at $t = 1$ is

$$\mathbf{s}_1 = \mathbf{H}_0^a\, \mathbf{x}_j.$$

For $t \geq 1$, the state evolves as $\mathbf{s}_{t+1} = \mathbf{P}\,\mathbf{s}_t$. Expanding $\mathbf{s}_1$ in the eigenbasis using $\mathbf{s}_1 = \sum_h (\mathbf{v}_h^\top \mathbf{s}_1)\mathbf{u}_h = \sum_h c_h^{(j)} \mathbf{u}_h$:

$$\mathbf{s}_t = \mathbf{P}^{t-1}\mathbf{s}_1 = \sum_{h=1}^{2(N-1)} \lambda_h^{t-1}\, c_h^{(j)}\, \mathbf{u}_h.$$

Relative inflation for $t \geq 1$ follows from $\hat{\boldsymbol{\pi}}_t = \Theta\, \mathbf{s}_t$ (with no contemporaneous forcing):

$$\hat{\boldsymbol{\pi}}_t^{(j)} = \sum_h \lambda_h^{t-1}\, c_h^{(j)}\, \Theta\mathbf{u}_h = \sum_h \lambda_h^{t-1}\, c_h^{(j)}\, \mathbf{q}_h.$$

Impact inflation $\hat{\boldsymbol{\pi}}_0^{(j)} = \Pi_0^a\, \mathbf{x}_j$ follows from (31) evaluated at $t = 0$ with $\mathbf{s}_0 = \mathbf{0}$.

Part (ii): Half-life. Along corridor h, the scalar coefficient of $\mathbf{u}_h$ in $\mathbf{s}_t$ is $\lambda_h^{t-1} c_h^{(j)}$ for $t \geq 1$. Its magnitude relative to $t = 1$ is $|\lambda_h|^{t-1}$. The half-life is the smallest $m \geq 1$ such that $|\lambda_h|^{m-1} \leq 1/2$, giving $m - 1 \geq \log(1/2)/\log|\lambda_h|$ (the inequality flips because $|\lambda_h| < 1$ makes the log negative), hence $\tau_h^{1/2} = \lceil \log(1/2)/\log|\lambda_h| \rceil$.

Part (iii): Long-run neutrality. Consider the flex-price counterfactual with the same shock. Under flexible prices, relative prices equal $\hat{p}_{it}^{\text{nat}}$ and satisfy $\hat{p}_{it}^{\text{nat}} + a_{it} = \sum_j \alpha_j a_{jt} + O(\mathbf{n}_t^{\text{nat}})$ (Lemma 1); for t large, $\mathbf{n}_t^{\text{nat}} \to \mathbf{0}$ and relative prices converge to $\hat{p}_{i\infty}^{\text{nat}} = \alpha_j - \delta_{ij}$ (the long-run flex-price value given the permanent shock).

Under sticky prices, $\hat{p}_{it} = \hat{p}_{i,t-1} + \hat{\pi}_{it}$, and from Part (i), $\hat{\boldsymbol{\pi}}_t^{(j)} \to \mathbf{0}$ as $t \to \infty$ since $|\lambda_h| < 1$ for all h. Hence $\hat{p}_{it} \to \hat{p}_{i\infty}$ for some limit, and $\sum_{t=0}^{\infty} \hat{\pi}_{it}^{(j)} = \hat{p}_{i\infty} - \hat{p}_{i,-1} = \hat{p}_{i\infty}$ (steady-state initial

condition). The long-run identity $\hat{p}_{i\infty} = \alpha_j - \delta_{ij}$ then follows because relative prices must equal their flex-price long-run level: any persistent gap would violate $\hat{\boldsymbol{\pi}}_t \to \mathbf{0}$. □

Proof of Proposition 4. Set $\tilde{y}_t = 0$. From the aggregate Phillips curve (32), $\pi_t^C = u_t^\pi + \beta\,\mathbb{E}_t \pi_{t+1}^C$ where $u_t^\pi = -(1+\varphi)\,\mathrm{Cov}_\alpha(\kappa_i, \chi_{it}) - \varphi\,\mathrm{Cov}_\alpha(\kappa_i, n_{it})$. Using $\sum_i \alpha_i \chi_{it} = \sum_i \alpha_i n_{it} = 0$, $u_t^\pi = \boldsymbol{\ell}_\pi^\top \mathbf{s}_{t+1}$ with $\boldsymbol{\ell}_\pi = -(\mathbf{D}_\alpha \tilde{\boldsymbol{\kappa}}(1+\varphi),\ \mathbf{D}_\alpha \tilde{\boldsymbol{\kappa}} \varphi)$ and $\tilde{\boldsymbol{\kappa}} = \boldsymbol{\kappa} - \bar{\kappa}\mathbf{1}$.

From Proposition 3, $\mathbf{s}_{t+1} = \sum_h \lambda_h^t c_h^{(j)} \mathbf{u}_h$ for $t \geq 0$, so

$$u_t^\pi = \sum_h \lambda_h^t\, c_h^{(j)}\, \eta_h, \qquad \eta_h \equiv \boldsymbol{\ell}_\pi^\top \mathbf{u}_h = q_h^C,$$

where the identity $\boldsymbol{\ell}_\pi^\top \mathbf{u}_h = q_h^C$ follows by α-weighting the sectoral Phillips curve and matching to $\mathbf{q}_h = \Theta \mathbf{u}_h$.

Iterating $\pi_t^C = u_t^\pi + \beta\,\mathbb{E}_t \pi_{t+1}^C$ forward: for any $t \geq 0$,

$$\pi_t^C = \pi_0^C \mathbf{1}_{\{t=0,\text{impact}\}} + \sum_{s=0}^{\infty} \beta^s\, u_{t+s}^\pi = \sum_h q_h^C c_h^{(j)} \frac{\lambda_h^t}{1-\beta\lambda_h} + (\text{impact at } t=0).$$

Summing over t and using $\sum_{t\geq 0} \lambda_h^t = 1/(1-\lambda_h)$:

$$p_\infty^{C,(j)} - p_{-1}^C = \sum_{t=0}^{\infty} \pi_t^C = \pi_0^C + \sum_h \frac{q_h^C\, c_h^{(j)}}{(1-\lambda_h)(1-\beta\lambda_h)},$$

where π_0^C collects the impact contribution $\boldsymbol{\alpha}^\top \Pi_0^a \mathbf{x}_j$.

Parts (i)–(ii): under uniform κ, $\tilde{\boldsymbol{\kappa}} = 0$, so $\boldsymbol{\ell}_\pi = 0$ and $q_h^C = 0$ for all h; also $\pi_0^C = 0$ by direct computation. Under heterogeneous κ, $\tilde{\boldsymbol{\kappa}} \neq 0$ and generically $\eta_h \neq 0$ for some h. □

Proof of Proposition 5. Substituting $\tilde{y}_t = 0$ into the marginal cost equation (27) and the NKPC (21) gives $\pi_{it} = \kappa[-(1+\varphi)\chi_{it} - \varphi\, n_{it}] + \beta\,\mathbb{E}_t \pi_{it+1}$. Using $\chi_{it} = \chi_{it-1} + \hat{\pi}_{it} + \Delta\hat{a}_{it}$ to eliminate $\hat{\pi}_{it}$, and noting $\Delta\hat{a}_{it} = 0$ for $t \geq 1$:

$$\beta\,\mathbb{E}_t[\chi_{it+1}] - [1 + \beta + \kappa(1+\varphi)]\,\chi_{it} + \chi_{it-1} = \kappa\varphi\, n_{it}. \tag{C.8}$$

Under segmentation ($\phi = 0$), $n_{it} = 0$ for all $t \geq 0$. Conjecturing $\chi_{it} = \chi_{i0}\lambda^t$ yields the characteristic equation (41) with coefficient $\Gamma_{\text{seg}} \equiv 1 + \beta + \kappa(1+\varphi)$. Under perfect mobility ($\phi \to \infty$), $n_{it} = -\chi_{it}$ for all $t \geq 0$, so the right-hand side of (C.8) becomes $-\kappa\varphi\,\chi_{it}$, reducing the coefficient on χ_{it} to $\Gamma_{\text{int}} \equiv 1 + \beta + \kappa$. The same conjecture yields (42).

Both characteristic equations take the form $f(\lambda;\Gamma) = \beta\lambda^2 - \Gamma\lambda + 1 = 0$. Since $f(0;\Gamma) = 1 > 0$ and $f(1;\Gamma) = \beta + 1 - \Gamma < 0$ (which holds in both cases), there is a unique stable root $\lambda \in (0,1)$, and the implicit function theorem gives $d\lambda/d\Gamma = \lambda/(2\beta\lambda - \Gamma) < 0$. Since $\Gamma_{\text{seg}} - \Gamma_{\text{int}} = \kappa\varphi > 0$, $\lambda_{\text{seg}} < \lambda_{\text{int}}$, establishing the persistence claims in parts (i) and (ii).

Impact. Evaluate (C.8) at $t = 0$ with $\chi_{i,-1} = 0$ and $\Delta\hat{a}_{i0} = \hat{a}_i$. Under segmentation, $n_{i0} = 0$ and we have $\chi_{it} = \chi_{i0}\lambda_{\text{seg}}^t$, so $\mathbb{E}_0[\chi_{i1}] = \lambda_{\text{seg}}\chi_{i0}$ and

$$\chi_{i0}^{\text{seg}} = \frac{\hat{a}_i}{\Gamma_{\text{seg}} - \beta\lambda_{\text{seg}}} = \hat{a}_i\,\lambda_{\text{seg}},$$

using the characteristic-equation identity $\Gamma - \beta\lambda = 1/\lambda$. Under perfect mobility, $n_{i0} = -\chi_{i0}$ and the impact equation uses coefficient Γ_{int}; the analogous calculation gives $\chi_{i0}^{\text{int}} = \hat{a}_i\lambda_{\text{int}}$. Impact inflation is $\hat{\pi}_{i0} = \chi_{i0} - \hat{a}_i = \hat{a}_i(\lambda - 1)$, so $|\hat{\pi}_{i0}^{\text{seg}}| = |\hat{a}_i|(1 - \lambda_{\text{seg}}) > |\hat{a}_i|(1 - \lambda_{\text{int}}) = |\hat{\pi}_{i0}^{\text{int}}|$, establishing the impact.

For the cumulative identity, relative inflation satisfies $\hat{\pi}_{it} = \Delta\chi_{it} - \Delta\hat{a}_{it}$; summing from $t = 0$ to ∞ and telescoping with $\chi_{i,-1} = \hat{a}_{i,-1} = 0$, $\chi_{it} \to 0$, and $\hat{a}_{it} = \hat{a}_i$ for $t \geq 0$ gives $\sum_{t=0}^{\infty} \hat{\pi}_{it} = -\hat{a}_i$, independent of ϕ. □

C.2 Proofs for Section 4

Proof of Lemma 2. We prove each part in turn.

Part (i): Detailed balance. Evaluating the logit transition probabilities (8) at steady state:

$$F_{ij} = \frac{\exp\big(\theta[\bar{V}_j - \tau^{ij}]\big)}{\sum_k \exp\big(\theta[\bar{V}_k - \tau^{ik}]\big)}.$$

Under Assumption 1, $\tau^{ij} = \tau_i^{\text{out}} + \tau_j^{\text{in}} + \hat{\tau}^{ij}$ for $i \neq j$. The origin-specific component τ_i^{out} cancels between numerator and denominator, giving

$$F_{ij} = \frac{\tilde{\theta}_j \,/\, e^{\theta\hat{\tau}^{ij}}}{\varphi_i}, \qquad \tilde{\theta}_j \equiv e^{\theta(\bar{V}_j - \tau_j^{\text{in}})}, \qquad \varphi_i \equiv \sum_k \frac{\tilde{\theta}_k}{e^{\theta\hat{\tau}^{ik}}}.$$

We claim $\bar{N}_i \propto \tilde{\theta}_i\,\varphi_i$. Computing $\sum_i \bar{N}_i F_{ij}$:

$$\sum_i \bar{N}_i F_{ij} = c\sum_i \tilde{\theta}_i\,\varphi_i \cdot \frac{\tilde{\theta}_j \,/\, e^{\theta\hat{\tau}^{ij}}}{\varphi_i} = c\,\tilde{\theta}_j \sum_i \tilde{\theta}_i\, e^{-\theta\hat{\tau}^{ij}} = c\,\tilde{\theta}_j\,\varphi_j = \bar{N}_j,$$

where the third equality uses $\hat{\tau}^{ij} = \hat{\tau}^{ji}$. Therefore $\bar{N}_i = \tilde{\theta}_i \, \varphi_i / \sum_k \tilde{\theta}_k \, \varphi_k$ is the stationary distribution, and detailed balance holds:

$$\frac{\bar{N}_i \, F_{ij}}{\bar{N}_j \, F_{ji}} = \frac{\tilde{\theta}_i \, \varphi_i}{\tilde{\theta}_j \, \varphi_j} \cdot \frac{\tilde{\theta}_j \, e^{-\theta \hat{\tau}^{ij}} \, / \, \varphi_i}{\tilde{\theta}_i \, e^{-\theta \hat{\tau}^{ji}} \, / \, \varphi_j} = 1.$$

By (10), $B_{ji} = (\bar{N}_i / \bar{N}_j) \, F_{ij} = F_{ji}$, so $B = F$.

Part (ii): Spectral properties. Define $Q \equiv A^{1/2} F A^{-1/2}$ where $A = \mathrm{diag}(\alpha_i)$. Detailed balance in matrix form is $AF = F^\top A$, so

$$Q^\top = A^{-1/2} F^\top A^{1/2} = A^{-1/2} (A F A^{-1}) A^{1/2} = A^{1/2} F A^{-1/2} = Q,$$

and Q is real symmetric with real eigenvalues $\rho_1 \geq \cdots \geq \rho_N$ and orthonormal eigenvectors $\{s_k\}$. Since Q is similar to F, they share eigenvalues. Row-stochasticity $F\mathbf{1} = \mathbf{1}$ gives $Q(A^{1/2}\mathbf{1}) = A^{1/2}\mathbf{1}$, so $\rho_1 = 1$ with eigenvector $s_1 = A^{1/2}\mathbf{1} / \|A^{1/2}\mathbf{1}\|$. Under the logit specification, $F_{ij} > 0$ for all i, j, so Perron–Frobenius gives $1 = \rho_1 > \rho_2 \geq \cdots \geq \rho_N > -1$. □

Proof of Lemma 3. We prove each part in turn.

Part (i): Decoupling. Define the coordinate transformation $x^*_{kt} \equiv s_k^\top A^{1/2} \hat{\mathbf{x}}_t$ for each variable $x \in \{\chi, n, v, \hat{a}\}$, where $\hat{\mathbf{x}}_t$ denotes the demeaned vector (lying in $\mathcal{R}$). Since $s_1 \propto A^{1/2}\mathbf{1}$ and $\boldsymbol{\alpha}^\top \hat{\mathbf{x}}_t = 0$, we have $x^*_{1t} = 0$, so only components $k \geq 2$ are nontrivial.

For any polynomial p, the transformation diagonalizes $p(F)$:

$$s_k^\top A^{1/2} \, p(F) \, \hat{\mathbf{x}}_t = s_k^\top \, p(Q) \, A^{1/2} \hat{\mathbf{x}}_t = p(\rho_k) \, x^*_{kt},$$

using $Q = A^{1/2} F A^{-1/2}$ and $Q s_k = \rho_k s_k$. In particular, F maps to ρ_k and $I - F^2$ to $1 - \rho_k^2$. We apply $s_k^\top A^{1/2}$ to each equation of the supply-side system under $B = F$ and $\kappa_i = \kappa$.

For the NKPC (C.5) with $\tilde{y}_t = 0$ (by block separation, Corollary 1), $M\mathbf{K} = \kappa M$ acts as a scalar on $\mathcal{R}$, so $s_k^\top A^{1/2} M \mathbf{K} \hat{\mathbf{x}}_t = \kappa \, x^*_{kt}$. Combined with the relative price identity $\chi^*_{kt} = \chi^*_{k,t-1} + \hat{\pi}^*_{kt} + \Delta \hat{a}^*_{kt}$ and eliminating $\hat{\pi}^*_{kt}$, we obtain the NKPC.

Applying $s_k^\top A^{1/2}$ to the employment law of motion $\mathbf{n}_t = F \, \mathbf{n}_{t-1} + \phi (I - F^2) \, \mathbf{v}_t$ gives $n^*_{kt} = \rho_k \, n^*_{k,t-1} + \phi (1 - \rho_k^2) \, v^*_{kt}$. Applying it to the value recursion $\mathbf{v}_t = -\boldsymbol{\chi}_t - \mathbf{n}_t + \beta F \, \mathbb{E}_t[\mathbf{v}_{t+1}]$ gives $v^*_{kt} = -\chi^*_{kt} - n^*_{kt} + \beta \rho_k \, \mathbb{E}_t[v^*_{k,t+1}]$.

Each projected system for component k involves only $(\chi^*_{kt}, n^*_{kt}, v^*_{kt})$ and the forcing $\hat{a}^*_{kt}$, with no coupling across components. The uniform-κ assumption is essential: with heterogeneous κ_i, the NKPC projection would involve $s_k^\top A^{1/2} \mathbf{K} A^{-1/2} s_{k'} \neq 0$ for $k' \neq k$, coupling the components.

Part (ii): Characteristic polynomial and limits. Conjecture geometric decay: $\chi^*_{kt} = \lambda^t C$, $n^*_{kt} = \lambda^t N$, $v^*_{kt} = \lambda^t V$. Substituting into the three equations:

$$\left[\beta\lambda^2 - (\kappa(1+\varphi)+\beta+1)\lambda + 1\right] C = \kappa\varphi\lambda\, N, \tag{C.9}$$

$$(\lambda - \rho_k)\, N = \phi(1-\rho_k^2)\lambda\, V, \tag{C.10}$$

$$(1-\beta\rho_k\lambda)\, V = -(C+N). \tag{C.11}$$

Substituting (C.11) into (C.10) to eliminate V, then using the result to express N in terms of C and substituting into (C.9), yields (43) after cancelling $C \neq 0$. The polynomial is quartic: the product of the two brackets is degree four and the additive term is degree two. The component-k system has two predetermined variables ($\chi^*_{k,t-1}$ and $n^*_{k,t-1}$) and two forward-looking variables ($\hat{\pi}^*_{kt}$ and v^*_{kt}), so Blanchard–Kahn requires exactly two stable and two unstable roots.

$\phi \to 0$. Setting $\phi = 0$ in (43):

$$(\lambda - \rho_k)(1-\beta\rho_k\lambda)\left[\beta\lambda^2 - (\kappa(1+\varphi)+\beta+1)\lambda + 1\right] = 0.$$

The first factor gives $\lambda = \rho_k$ (stable). The second gives $\lambda = 1/(\beta\rho_k)$ (unstable). The quadratic is the segmented-market equation (41), whose roots have product $1/\beta > 1$: one stable root λ_{seg} and one unstable. The two stable roots are $\{\lambda_{\text{seg}}, \rho_k\}$, so $\lambda^*_k(0) = \max\{\lambda_{\text{seg}}, |\rho_k|\}$.

$\phi \to \infty$. Dividing (43) by ϕ and taking $\phi \to \infty$, the terms not proportional to ϕ vanish, leaving:

$$(1-\rho_k^2)\lambda\left[\beta\lambda^2 - (\kappa(1+\varphi)+\beta+1)\lambda + 1\right] + \kappa\varphi(1-\rho_k^2)\lambda^2 = 0.$$

Dividing by $(1-\rho_k^2)\lambda$ simplifies to $\beta\lambda^2 - (\kappa+\beta+1)\lambda + 1 = 0$, the integrated-market equation (42) with stable root λ_{int}. The root $\lambda = 0$ reflects the collapse of one state variable: as $\phi \to \infty$, employment fully tracks relative prices ($n^*_{kt} \to -\chi^*_{kt}$), eliminating one independent degree of freedom.

Roots of a polynomial depend continuously on its coefficients. Since the coefficients of (43) are continuous in ϕ and the stable roots remain bounded away from the unit circle for all finite $\phi \geq 0$, λ^*_k is continuous on $[0, \infty)$. □

Proof of Proposition 6. We proceed in three steps: we derive the effective slope, apply the implicit function theorem to find the eigenvalue shift, and finally determine the sign of the resulting coefficient.

To derive the effective slope, write $\mathbf{K} = \text{diag}(\kappa_i) = \bar{\kappa}\, I + \varepsilon\ \text{diag}(\Delta\kappa_i)$. Under uniform $\bar{\kappa}$, the NKPC projection onto component k (Lemma 3) involves the diagonal entry $s_k^\top A^{1/2}\mathbf{K}A^{-1/2}s_k = \bar{\kappa}$.

Under the perturbation, this becomes

$$s_k^\top A^{1/2}\mathbf{K}A^{-1/2}s_{k'} = \bar{\kappa}\,\mathbf{1}_{k=k'} + \varepsilon\sum_i \Delta\kappa_i\, s_{k,i}\, s_{k',i}.$$

Substituting $s_{k,i} = \alpha_i^{1/2}\, v_{k,i}$, the diagonal entry simplifies to $\bar{\kappa} + \varepsilon\,\Delta\kappa_k^{\text{eff}}$. The off-diagonal entries are $O(\varepsilon)$, which, by standard non-degenerate perturbation theory, only affect a simple eigenvalue at $O(\varepsilon^2)$. Therefore, to first order, the characteristic polynomial for component k (43) is obtained by simply replacing κ with $\bar{\kappa} + \varepsilon\,\Delta\kappa_k^{\text{eff}}$.

Next, we evaluate the eigenvalue shift. Assume λ_k^* is a simple stable root of (43) at the baseline; this holds generically throughout the parameter region of interest. Applying the implicit function theorem yields the equation

$$\lambda_k^*(\varepsilon) = \lambda_k^* + \varepsilon\,\Delta\kappa_k^{\text{eff}}\cdot\Gamma_k + O(\varepsilon^2),$$

with

$$\Gamma_k = -\frac{\partial_\kappa\, p}{\partial_\lambda\, p}\bigg|_{(\lambda_k^*,\,\bar{\kappa},\,\rho_k)} = \frac{d\lambda_k^*}{d\kappa}\bigg|_{\bar{\kappa}}.$$

It remains to determine the sign of Γ_k, which matches the sign of $d\lambda_k^*/d\kappa$. We establish that $d\lambda_k^*/d\kappa < 0$ by examining the two limits of Lemma 3 and extending the result by continuity.

In the limit $\phi = 0$, if $\lambda_{\text{seg}} > |\rho_k|$, then $\lambda_k^*(0) = \lambda_{\text{seg}}$. Implicit differentiation of the segmented-market polynomial $\beta\lambda^2 - [\kappa(1+\varphi)+\beta+1]\lambda + 1 = 0$ yields

$$\frac{d\lambda_{\text{seg}}}{d\kappa} = \frac{(1+\varphi)\lambda_{\text{seg}}}{2\beta\lambda_{\text{seg}} - [\kappa(1+\varphi)+\beta+1]} < 0,$$

where the denominator is strictly negative because it equals $-(\partial_\lambda S)(\lambda_{\text{seg}}) < 0$ (following the stable-root argument from Proposition 5). If instead $\lambda_{\text{seg}} \le |\rho_k|$, then $\lambda_k^*(0) = |\rho_k|$, which is independent of κ. A small increase in ϕ activates the coupling term $T > 0$, ensuring $d\lambda_k^*/d\kappa < 0$ by continuity.

In the limit $\phi \to \infty$, we have $\lambda_k^*(\infty) = \lambda_{\text{int}}$. Applying the same implicit differentiation to the integrated-market polynomial similarly yields $d\lambda_{\text{int}}/d\kappa < 0$.

For intermediate values of ϕ, the quantity $d\lambda_k^*/d\kappa$ varies continuously along the smooth family of simple stable roots $\lambda_k^*(\phi)$. It cannot change sign without passing through zero, which, according to $d\lambda_k^*/d\kappa = -(\partial_\kappa p)/(\partial_\lambda p)$, would require $\partial_\kappa p|_{\lambda_k^*} = 0$.

We show $\partial_\kappa p|_{\lambda_k^*} \neq 0$. Decompose $p = R\cdot S + T$, where $R(\lambda) = (\lambda - \rho_k)(1 - \beta\rho_k\lambda) + \phi(1 - \rho_k^2)\lambda$, $S(\lambda;\kappa) = \beta\lambda^2 - [\kappa(1+\varphi)+\beta+1]\lambda + 1$, and $T(\lambda;\kappa) = \kappa\varphi\phi(1-\rho_k^2)\lambda^2$. Notice that only S

and T depend on κ. Using the fact that $p(\lambda_k^*) = 0$ implies $T(\lambda_k^*) = -R(\lambda_k^*)S(\lambda_k^*)$, we find:

$$\partial_\kappa p|_{\lambda_k^*} = -(1+\varphi)\lambda_k^* R(\lambda_k^*) + \frac{T(\lambda_k^*)}{\kappa} = -\frac{R(\lambda_k^*)}{\kappa}\big[\kappa(1+\varphi)\lambda_k^* + S(\lambda_k^*)\big].$$

The term in brackets simplifies to: $\kappa(1+\varphi)\lambda_k^* + S(\lambda_k^*) = \beta(\lambda_k^*)^2 - (\beta+1)\lambda_k^* + 1 = (\lambda_k^* - 1)(\beta\lambda_k^* - 1) > 0$, since $\lambda_k^* \in (0,1)$ and $\beta < 1$. It then remains to show that $R(\lambda_k^*) \neq 0$. Evaluating p at λ_{seg} gives $p(\lambda_{\text{seg}}) = R(\lambda_{\text{seg}}) \cdot 0 + T(\lambda_{\text{seg}}) = T(\lambda_{\text{seg}}) > 0$ for $\phi > 0$, implying $\lambda_k^* \neq \lambda_{\text{seg}}$. Combining this with $T(\lambda_k^*) > 0$ and $R \cdot S = -T < 0$, we conclude that R and S must have opposite, nonzero signs at λ_k^*. Thus, $R(\lambda_k^*) \neq 0$, meaning $\partial_\kappa p|_{\lambda_k^*} \neq 0$ for all $\phi > 0$, and $d\lambda_k^*/d\kappa$ cannot change sign.

Combining these observations, $\Gamma_k = d\lambda_k^*/d\kappa < 0$ for all $\phi \geq 0$, with strict inequality when either $\phi > 0$ or $\lambda_{\text{seg}} > |\rho_k|$. Therefore, a negative $\Delta\kappa_k^{\text{eff}}$ strictly increases persistence. □

C.3 Proof of Proposition 9

This appendix derives the first-order conditions of the Ramsey problem (Proposition 8) and shows how they imply the targeting criterion (52).

Collect all N sectors into vectors $\boldsymbol{\pi}_t$, $\boldsymbol{\chi}_t$, $\mathbf{n}_t$, $\mathbf{v}_t \in \mathbb{R}^N$ and define the diagonal matrices

$$A \equiv \text{diag}(\alpha_i), \qquad K \equiv \text{diag}(\kappa_i), \qquad D_\pi \equiv \text{diag}\big(\tfrac{\alpha_i \epsilon_i}{\kappa_i}\big),$$

together with $M \equiv I - \mathbf{1}\boldsymbol{\alpha}^\top$, which removes the aggregate component: $M\boldsymbol{\pi}_t = \boldsymbol{\pi}_t - \pi_t^C\,\mathbf{1}$. Denote by $\boldsymbol{\lambda}_t$, $\boldsymbol{\eta}_t$, $\boldsymbol{\psi}_t$, and $\boldsymbol{\xi}_t$ the multipliers on the Phillips curve, relative price identity, employment law of motion, and value recursion, respectively.

The constraints (48)–(51) can be written as

$$\boldsymbol{\pi}_t - \beta\,\mathbb{E}_t\boldsymbol{\pi}_{t+1} - K[(\gamma+\varphi)\,\mathbf{1}\,\tilde{y}_t - (1+\varphi)\,\boldsymbol{\chi}_t - \varphi\,\mathbf{n}_t] = \mathbf{0}, \qquad [\boldsymbol{\lambda}_t] \tag{C.12}$$

$$\boldsymbol{\chi}_t - \boldsymbol{\chi}_{t-1} - M\boldsymbol{\pi}_t - \Delta\hat{\mathbf{a}}_t = \mathbf{0}, \qquad [\boldsymbol{\eta}_t] \tag{C.13}$$

$$\mathbf{n}_t - B\,\mathbf{n}_{t-1} - \phi(I - BF)\,\mathbf{v}_t = \mathbf{0}, \qquad [\boldsymbol{\psi}_t] \tag{C.14}$$

$$\mathbf{v}_t + \boldsymbol{\chi}_t + \mathbf{n}_t - \beta F\,\mathbb{E}_t\mathbf{v}_{t+1} = \mathbf{0}. \qquad [\boldsymbol{\xi}_t] \tag{C.15}$$

Both $\mathbf{n}_t$ and $\mathbf{v}_t$ are within-period endogenous variables: $\mathbf{n}_t$ is pinned down by (C.14) given the predetermined $\mathbf{n}_{t-1}$ and the contemporaneous $\mathbf{v}_t$, and $\mathbf{v}_t$ is pinned down by (C.15) given $\boldsymbol{\chi}_t$, $\mathbf{n}_t$, and the expected continuation.

The planner minimizes (46) subject to (C.12)–(C.15). The date-t contribution to the La-

grangian is

$$
\begin{aligned}
\mathcal{L}_t \;=\; & \frac{\gamma+\varphi}{2}\,\tilde{y}_t^2 + \tfrac{1}{2}\,\boldsymbol{\pi}_t^\top D_\pi\,\boldsymbol{\pi}_t + \tfrac{1}{2}(1+\varphi)\,\boldsymbol{\chi}_t^\top A\,\boldsymbol{\chi}_t + \varphi\,\boldsymbol{\chi}_t^\top A\,\mathbf{n}_t + \tfrac{1}{2}\varphi\,\mathbf{n}_t^\top A\,\mathbf{n}_t \\
& + \boldsymbol{\lambda}_t^\top\big[\boldsymbol{\pi}_t - \beta\,\mathbb{E}_t\boldsymbol{\pi}_{t+1} - K\big((\gamma+\varphi)\,\mathbf{1}\,\tilde{y}_t - (1+\varphi)\,\boldsymbol{\chi}_t - \varphi\,\mathbf{n}_t\big)\big] \\
& + \boldsymbol{\eta}_t^\top\big[\boldsymbol{\chi}_t - \boldsymbol{\chi}_{t-1} - M\boldsymbol{\pi}_t - \Delta\hat{\mathbf{a}}_t\big] \\
& + \boldsymbol{\psi}_t^\top\big[\mathbf{n}_t - B\,\mathbf{n}_{t-1} - \phi(I-BF)\,\mathbf{v}_t\big] \\
& + \boldsymbol{\xi}_t^\top\big[\mathbf{v}_t + \boldsymbol{\chi}_t + \mathbf{n}_t - \beta F\,\mathbb{E}_t\mathbf{v}_{t+1}\big]\,.
\end{aligned}
$$

First-order conditions. Differentiating $\sum_{t=0}^{\infty}\beta^t\,\mathcal{L}_t$ with respect to each choice variable yields:

Output gap $\tilde{y}_t$. Appears in the loss and in the Phillips curve constraint:

$$\tilde{y}_t = \mathbf{1}^\top K\,\boldsymbol{\lambda}_t. \tag{F$_{\tilde{y}}$}$$

Sectoral inflation $\boldsymbol{\pi}_t$. Appears in the date-t loss and Phillips curve, in the date-$(t-1)$ Phillips curve through $\beta\,\mathbb{E}_{t-1}\boldsymbol{\pi}_t$, and in the date-$t$ relative price identity:

$$D_\pi\,\boldsymbol{\pi}_t + \boldsymbol{\lambda}_t - \boldsymbol{\lambda}_{t-1} - M^\top\boldsymbol{\eta}_t = \mathbf{0}. \tag{F$_\pi$}$$

The term $-\boldsymbol{\lambda}_{t-1}$ is the commitment term: current inflation appears in last period's Phillips curve, and the planner internalizes this dependence.

Relative prices $\boldsymbol{\chi}_t$. Appear in the loss, the Phillips curve, the relative price identities at dates t and $t+1$, and the value recursion at t:

$$A[(1+\varphi)\,\boldsymbol{\chi}_t + \varphi\,\mathbf{n}_t] + (1+\varphi)\,K\,\boldsymbol{\lambda}_t + \boldsymbol{\eta}_t - \beta\,\mathbb{E}_t\boldsymbol{\eta}_{t+1} + \boldsymbol{\xi}_t = \mathbf{0}. \tag{F$_\chi$}$$

The first term is the direct marginal welfare cost of relative-price misalignment; the second is the cost-push effect on sectoral Phillips curves; $\boldsymbol{\eta}_t - \beta\mathbb{E}_t\boldsymbol{\eta}_{t+1}$ is the net shadow value of adjusting the relative-price state; $\boldsymbol{\xi}_t$ reflects the effect on workers' reallocation incentives through the value recursion.

Employment $\mathbf{n}_t$. Appears in the loss, the Phillips curve at t, the value recursion at t, the employment law of motion at t (as the left-hand side), and the employment law of motion at $t+1$ (as the predetermined state):

$$A\varphi(\boldsymbol{\chi}_t + \mathbf{n}_t) + \varphi\,K\,\boldsymbol{\lambda}_t + \boldsymbol{\xi}_t + \boldsymbol{\psi}_t - \beta B^\top\,\mathbb{E}_t\boldsymbol{\psi}_{t+1} = \mathbf{0}. \tag{F$_n$}$$

The term $\varphi K\boldsymbol{\lambda}_t$ captures the shadow value of the employment distribution through the sectoral

Phillips curves: because $\mathbf{n}_t$ enters marginal costs, employment misallocation matters through pricing distortions as well as directly through the welfare loss. The net shadow value $\boldsymbol{\psi}_t - \beta B^\top \mathbb{E}_t \boldsymbol{\psi}_{t+1}$ captures the dynamic effect of $\mathbf{n}_t$ on tomorrow's predetermined employment state through B.

Continuation values $\mathbf{v}_t$. $\mathbf{v}_t$ is a within-period choice variable that appears in the employment law of motion at t, the value recursion at t, and the value recursion at $t-1$ (through $\beta F \mathbb{E}_{t-1} \mathbf{v}_t$):

$$\boldsymbol{\xi}_t - F^\top \boldsymbol{\xi}_{t-1} - \phi(I - BF)^\top \boldsymbol{\psi}_t = \mathbf{0}. \tag{F$_v$}$$

The lagged multiplier $F^\top \boldsymbol{\xi}_{t-1}$ introduces history dependence: under commitment, current policy internalizes that yesterday's promised continuation values shape today's reallocation margin.

Derivation of the targeting criterion. We now derive equation (52). Pre-multiply (F$_\pi$) by $\mathbf{1}^\top K$:

$$\mathbf{1}^\top K\, D_\pi\, \boldsymbol{\pi}_t + \mathbf{1}^\top K\, \boldsymbol{\lambda}_t - \mathbf{1}^\top K\, \boldsymbol{\lambda}_{t-1} - \mathbf{1}^\top K\, M^\top \boldsymbol{\eta}_t = 0. \tag{C.16}$$

We evaluate each term. Since K and D_π are both diagonal,

$$\mathbf{1}^\top K\, D_\pi\, \boldsymbol{\pi}_t = \sum_i \alpha_i \epsilon_i\, \pi_{it}.$$

From (F$_{\tilde{y}}$), $\mathbf{1}^\top K\, \boldsymbol{\lambda}_t = \tilde{y}_t$ and $\mathbf{1}^\top K\, \boldsymbol{\lambda}_{t-1} = \tilde{y}_{t-1}$. Finally, $\mathbf{1}^\top K M^\top = (MK\mathbf{1})^\top$ and

$$MK\mathbf{1} = K\mathbf{1} - \mathbf{1}\boldsymbol{\alpha}^\top K\mathbf{1} = K\mathbf{1} - \bar{\kappa}\,\mathbf{1},$$

where $\bar{\kappa} \equiv \boldsymbol{\alpha}^\top K\mathbf{1} = \sum_i \alpha_i \kappa_i$, so

$$\mathbf{1}^\top K\, M^\top \boldsymbol{\eta}_t = (MK\mathbf{1})^\top \boldsymbol{\eta}_t = \sum_i (\kappa_i - \bar{\kappa})\, \eta_{it}.$$

Substituting into (C.16):

$$\sum_i \alpha_i \epsilon_i\, \pi_{it} + \tilde{y}_t - \tilde{y}_{t-1} - \sum_i (\kappa_i - \bar{\kappa})\, \eta_{it} = 0,$$

which rearranges to (52).

Under the timeless perspective, the planner commits to the policy rule that would have been optimal had it been adopted in the infinite past. The lagged multipliers $(\boldsymbol{\lambda}_{-1}, \boldsymbol{\xi}_{-1})$ are initialized at their unconditional steady-state values, so that the targeting rule (52) holds at every date including $t = 0$. □

D Quantitative Section Appendix

D.1 Empirical fit of reversibility

Assumption 1 is imposed for tractability. Its observable implication, established in Lemma 2, is that the steady-state transition matrix is reversible:

$$\alpha_i F_{ij} = \alpha_j F_{ji}, \qquad \text{equivalently} \qquad Q \equiv A^{1/2} F A^{-1/2} \text{ is symmetric,}$$

where $A = \text{diag}(\alpha_i)$.

We assess this implication using CPS sector-to-sector transition flows. Figure D.1 shows that reversibility is a close approximation in the data. The asymmetry measure is small,

$$\frac{\|Q - Q^\top\|}{\|Q\|} = 8 \times 10^{-4},$$

and the correlation between $\alpha_i F_{ij}$ and $\alpha_j F_{ji}$ across sector pairs is 0.9962. Thus, while Assumption 1 is not itself directly testable from observed flows, its implied symmetry restriction on labor transitions fits the data closely.

D.2 Calibrating the Mobility Elasticity

We derive the mapping between the mobility elasticity ϕ in our model and the migration dispersion parameter ν estimated by Caliendo et al. (2019).

Our model. The logit (8) under contemporaneous reallocation timing gives, for $i \neq j$:

$$\log(\mu_t^{ij} / \mu_t^{ii}) = \theta\,(V_{jt} - V_{it}) - \theta(\tau^{ij} - \tau^{ii}). \tag{D.1}$$

For a temporary change in destination wage w_{jt}, the continuation value adjusts only through the contemporaneous flow utility: $\Delta V_{jt} \approx \Delta U_{jt}^{\text{flow}}$. From the flow utility and the labor supply condition (6):

$$\frac{\partial U_{jt}^{\text{flow}}}{\partial \log w_{jt}} = \zeta\,\varphi\,\bar{L}^{\varphi} \cdot \frac{\bar{L}}{\varphi} = \zeta\,\bar{L}^{1+\varphi}, \tag{D.2}$$

so the semi-elasticity of relative log flows to log contemporaneous destination wages is

$$\frac{\partial \log(\mu_t^{ij} / \mu_t^{ii})}{\partial \log w_{jt}} = \theta\,\zeta\,\bar{L}^{1+\varphi} = \frac{\phi}{\varphi}, \tag{D.3}$$

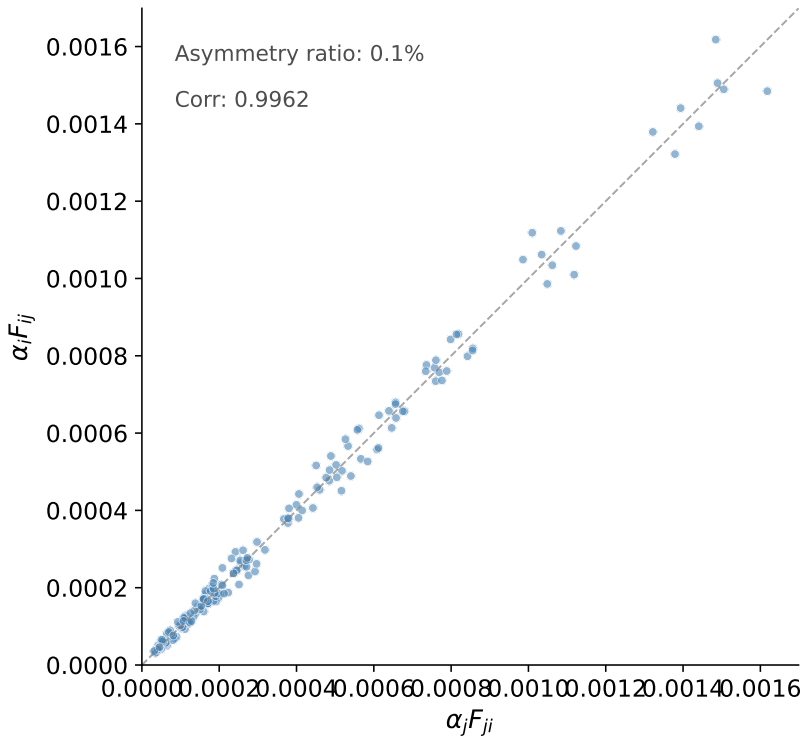

Figure D.1: Empirical fit of reversibility in sectoral labor transitions. The figure plots $\alpha_i F_{ij}$ against $\alpha_j F_{ji}$ using CPS sector-to-sector transition flows. Under reversibility, all observations lie on the 45-degree line. In the data, the asymmetry measure $\|Q - Q^\top\| / \|Q\|$ is 8×10^{-4} and the correlation between $\alpha_i F_{ij}$ and $\alpha_j F_{ji}$ is 0.9962, indicating that the symmetry implication of Lemma 2 is a close approximation empirically.

using $\phi = \varphi\,\theta\,\zeta\,\bar{L}^{1+\varphi}$.

Caliendo et al. (2019) model. In their formulation, workers choose destination at the end of period t and arrive at $t+1$. The flow payoff is $\log(w_j/p_j)$ and idiosyncratic shocks have dispersion ν, so the logit gives:

$$\log(\mu_t^{ij}/\mu_t^{ii}) = \frac{1}{\nu}\left[\beta\,\mathbb{E}_t(V_{j,t+1} - V_{i,t+1}) - (\kappa^{ij} - \kappa^{ii})\right]. \tag{D.4}$$

Since $\partial V_{j,t+1}/\partial \log w_{j,t+1} = 1$ for a temporary wage change, and the destination wage workers care about when choosing μ_t^{ij} is $w_{j,t+1}$, the undiscounted semi-elasticity is $1/\nu$:

$$\frac{\partial \log(\mu_t^{ij}/\mu_t^{ii})}{\partial \log w_{j,t+1}} = \frac{\beta}{\nu}, \qquad \left.\frac{\partial \log(\mu_t^{ij}/\mu_t^{ii})}{\partial \log w_{j,t+1}}\right|_{\text{undiscounted}} = \frac{1}{\nu}. \tag{D.5}$$

The factor β reflects one-period discounting between the decision date and the payoff date in CDP's timing; it is absent in our model because reallocation is contemporaneous.

Mapping. Equating undiscounted semi-elasticities—the sensitivity of flow decisions to the destination wage each flow actually earns:

$$\frac{\phi}{\varphi} = \frac{1}{\nu} \qquad \Longrightarrow \qquad \phi = \frac{\varphi}{\nu}. \tag{D.6}$$

Caliendo et al. (2019) estimate $\nu = 5.34$ at quarterly frequency. With $\varphi = 2$, this implies $\phi = 2/5.34 \approx 0.37$.

D.3 Solving the Ramsey Problem

This appendix assembles the first-order conditions from Appendix C.3 into a linear rational-expectations system suitable for numerical solution via standard Blanchard–Kahn procedures (Blanchard and Kahn, 1980; Klein, 2000).

Unlike χ_t, $\mathbf{n}_t$, and $\mathbf{v}_t$—which the primitive constraints confine to $\mathcal{R}$—sectoral inflation $\boldsymbol{\pi}_t$ carries an endogenous aggregate component $\pi_t^C \equiv \boldsymbol{\alpha}^\top \boldsymbol{\pi}_t$ selected implicitly by the planner through $\tilde{y}_t$. The system therefore retains $\boldsymbol{\pi}_t$ and its co-state $\boldsymbol{\lambda}_t$ in $\mathbb{R}^N$, adding three scalar dimensions and one stable eigenvalue relative to a pure relative-block system.

Variable classification. Four predetermined vectors at date t:

$$\mathbf{x}_t \;\equiv\; \big(\chi_{t-1},\ \mathbf{n}_{t-1},\ \boldsymbol{\lambda}_{t-1},\ \boldsymbol{\xi}_{t-1}\big), \qquad \dim \mathbf{x}_t = 4(N-1)+1.$$

Three live in $\mathcal{R}$ $(\boldsymbol{\chi}_{t-1}, \mathbf{n}_{t-1}, \boldsymbol{\xi}_{t-1})$; the fourth, $\boldsymbol{\lambda}_{t-1} \in \mathbb{R}^N$, carries the aggregate Ramsey commitment $\lambda^C_{t-1} \equiv \mathbf{1}^\top \boldsymbol{\lambda}_{t-1}$.

Eight forward-looking jumps:

$$\mathbf{z}_t \;\equiv\; \big(\boldsymbol{\pi}_t,\ \boldsymbol{\chi}_t,\ \mathbf{n}_t,\ \mathbf{v}_t,\ \boldsymbol{\lambda}_t,\ \boldsymbol{\eta}_t,\ \boldsymbol{\psi}_t,\ \boldsymbol{\xi}_t\big), \qquad \dim \mathbf{z}_t = 8(N-1)+2,$$

with $\boldsymbol{\pi}_t, \boldsymbol{\lambda}_t \in \mathbb{R}^N$ and the remaining six jumps in $\mathcal{R}$; the aggregate components are $\pi^C_t = \boldsymbol{\alpha}^\top \boldsymbol{\pi}_t$ and $\lambda^C_t = \mathbf{1}^\top \boldsymbol{\lambda}_t$. The output gap is then pinned down by $\mathrm{F}_{\tilde{y}}$: $\tilde{y}_t = \mathbf{1}^\top K \boldsymbol{\lambda}_t$. Stacking, $\mathbf{w}_t \equiv (\mathbf{x}_t, \mathbf{z}_t)$ has dimension $12(N-1)+3$.

Canonical BK form. The system takes the form

$$\mathcal{A}\,\mathbb{E}_t \mathbf{w}_{t+1} \;=\; \mathcal{B}\,\mathbf{w}_t \;+\; \mathcal{C}_a\,\Delta\hat{\mathbf{a}}_t, \tag{D.7}$$

assembled from twelve block-rows listed in Table D.1: rows 1–4 are state-identification rows, rows 5–8 are the primitive constraints (C.12)–(C.15), and rows 9–12 are the first-order conditions $\mathrm{F}_\pi, \mathrm{F}_\chi, \mathrm{F}_n, \mathrm{F}_v$ from Appendix C.3, with $\tilde{y}_t = \mathbf{1}^\top K \boldsymbol{\lambda}_t$ substituted throughout.

Row	Equation (at date t, with $\tilde{y}_t = \mathbf{1}^\top K\boldsymbol{\lambda}_t$ substituted)	Source	Type	Dim
1	$\mathbb{E}_t \boldsymbol{\chi}_t = \boldsymbol{\chi}_t$	state	static	$N-1$
2	$\mathbb{E}_t \mathbf{n}_t = \mathbf{n}_t$	state	static	$N-1$
3	$\mathbb{E}_t \boldsymbol{\lambda}_t = \boldsymbol{\lambda}_t$	state	static	N
4	$\mathbb{E}_t \boldsymbol{\xi}_t = \boldsymbol{\xi}_t$	state	static	$N-1$
5	$\beta \mathbb{E}_t \boldsymbol{\pi}_{t+1} = \boldsymbol{\pi}_t + (1+\varphi)K\boldsymbol{\chi}_t + \varphi K \mathbf{n}_t - (\gamma+\varphi)K\mathbf{1}\mathbf{1}^\top K\boldsymbol{\lambda}_t$	(C.12)	forward	N
6	$\mathbf{0} = \boldsymbol{\chi}_t - \boldsymbol{\chi}_{t-1} - M\boldsymbol{\pi}_t - \Delta\hat{\mathbf{a}}_t$	(C.13)	static	$N-1$
7	$\mathbf{0} = \mathbf{n}_t - B\mathbf{n}_{t-1} - \phi(I-BF)\mathbf{v}_t$	(C.14)	static	$N-1$
8	$\beta F \mathbb{E}_t \mathbf{v}_{t+1} = \mathbf{v}_t + \boldsymbol{\chi}_t + \mathbf{n}_t$	(C.15)	forward	$N-1$
9	$\mathbf{0} = D_\pi \boldsymbol{\pi}_t + \boldsymbol{\lambda}_t - \boldsymbol{\lambda}_{t-1} - M^\top \boldsymbol{\eta}_t$	(F_π)	static	N
10	$\beta \mathbb{E}_t \boldsymbol{\eta}_{t+1} = MA\big[(1+\varphi)\boldsymbol{\chi}_t + \varphi \mathbf{n}_t\big] + (1+\varphi)MK\boldsymbol{\lambda}_t + \boldsymbol{\eta}_t + \boldsymbol{\xi}_t$	(F_χ)	forward	$N-1$
11	$\beta B^\top \mathbb{E}_t \boldsymbol{\psi}_{t+1} = \varphi MA(\boldsymbol{\chi}_t + \mathbf{n}_t) + \varphi MK\boldsymbol{\lambda}_t + \boldsymbol{\xi}_t + \boldsymbol{\psi}_t$	(F_n)	forward	$N-1$
12	$\mathbf{0} = \boldsymbol{\xi}_t - F^\top \boldsymbol{\xi}_{t-1} - \phi(I-BF)^\top \boldsymbol{\psi}_t$	(F_v)	static	$N-1$

Table D.1: Rows of the Ramsey BK system (D.7). Rows 3, 5, and 9 are N-dimensional; the rest are $(N-1)$-dimensional. Total: $12(N-1)+3$.

Matrices. With $\mathbf{w}_t$ ordered as $(\boldsymbol{\chi}_{t-1}, \mathbf{n}_{t-1}, \boldsymbol{\lambda}_{t-1}, \boldsymbol{\xi}_{t-1}, \boldsymbol{\pi}_t, \boldsymbol{\chi}_t, \mathbf{n}_t, \mathbf{v}_t, \boldsymbol{\lambda}_t, \boldsymbol{\eta}_t, \boldsymbol{\psi}_t, \boldsymbol{\xi}_t)$ (columns 1–12, block-row sizes $N-1, N-1, N, N-1, N, N-1, N-1, N-1, N, N-1, N-1, N-1$), the block-matrix $\mathcal{A}$ is

$$\mathcal{A} \;=\; \mathrm{diag}\big(I_{N-1},\ I_{N-1},\ I_N,\ I_{N-1},\ \beta I_N,\ \mathbf{0},\ \mathbf{0},\ \beta F,\ \mathbf{0},\ \beta I_{N-1},\ \beta B^\top,\ \mathbf{0}\big).$$

The matrix $\mathcal{B}$ has nonzero blocks

$$
\begin{aligned}
&\mathcal{B}_{1,6} = I_{N-1}, \;\; \mathcal{B}_{2,7} = I_{N-1}, \;\; \mathcal{B}_{3,9} = I_N, \;\; \mathcal{B}_{4,12} = I_{N-1}, \\
&\mathcal{B}_{5,5} = I_N, \;\; \mathcal{B}_{5,6} = (1+\varphi)K, \;\; \mathcal{B}_{5,7} = \varphi K, \;\; \mathcal{B}_{5,9} = -(\gamma+\varphi)K\mathbf{1}\mathbf{1}^\top K, \\
&\mathcal{B}_{6,1} = -I_{N-1}, \;\; \mathcal{B}_{6,5} = -M, \;\; \mathcal{B}_{6,6} = I_{N-1}, \\
&\mathcal{B}_{7,2} = -B, \;\; \mathcal{B}_{7,7} = I_{N-1}, \;\; \mathcal{B}_{7,8} = -\phi(I-BF), \\
&\mathcal{B}_{8,6} = I_{N-1}, \;\; \mathcal{B}_{8,7} = I_{N-1}, \;\; \mathcal{B}_{8,8} = I_{N-1}, \\
&\mathcal{B}_{9,3} = -I_N, \;\; \mathcal{B}_{9,5} = D_\pi, \;\; \mathcal{B}_{9,9} = I_N, \;\; \mathcal{B}_{9,10} = -M^\top, \\
&\mathcal{B}_{10,6} = (1+\varphi)MA, \;\; \mathcal{B}_{10,7} = \varphi MA, \;\; \mathcal{B}_{10,9} = (1+\varphi)MK, \;\; \mathcal{B}_{10,10} = I_{N-1}, \;\; \mathcal{B}_{10,12} = I_{N-1}, \\
&\mathcal{B}_{11,6} = \varphi MA, \;\; \mathcal{B}_{11,7} = \varphi MA, \;\; \mathcal{B}_{11,9} = \varphi MK, \;\; \mathcal{B}_{11,11} = I_{N-1}, \;\; \mathcal{B}_{11,12} = I_{N-1}, \\
&\mathcal{B}_{12,4} = -F^\top, \;\; \mathcal{B}_{12,11} = -\phi(I-BF)^\top, \;\; \mathcal{B}_{12,12} = I_{N-1}.
\end{aligned}
$$

Blocks mapping between $\mathbb{R}^N$ and $\mathcal{R}$ are rectangular: K, D_π embed $\mathcal{R} \hookrightarrow \mathbb{R}^N$; M, MA, MK project $\mathbb{R}^N \twoheadrightarrow \mathcal{R}$. Productivity forcing enters only through row 6: $\mathcal{C}_{a,6} = I_{N-1}$, all other blocks zero.

Determinacy. Blanchard–Kahn requires $(\mathcal{B}, \mathcal{A})$ to have exactly $4(N-1)+1$ stable generalized eigenvalues, one per scalar predetermined entry in $\mathbf{x}_t$. When satisfied, the solution takes the form

$$
\mathbf{z}_t \;=\; \mathbf{\Theta}\,\mathbf{x}_t \;+\; \sum_{j=0}^{\infty} \mathbf{\Phi}_j\, \mathbb{E}_t[\Delta\hat{\mathbf{a}}_{t+j}], \qquad \mathbf{x}_{t+1} \;=\; \mathbf{P}^{\text{Ramsey}}\,\mathbf{x}_t \;+\; (\text{exogenous forcing}), \tag{D.8}
$$

where $\mathbf{\Theta}$ is $[8(N-1)+2] \times [4(N-1)+1]$ and $\mathbf{P}^{\text{Ramsey}}$ is $[4(N-1)+1] \times [4(N-1)+1]$.

Extracting quantities of interest. Denoting blocks of $\mathbf{\Theta}$ by $\Theta_\pi, \Theta_\lambda, \Theta_\eta$, optimal sectoral inflation is $\boldsymbol{\pi}_t^* = \Theta_\pi \mathbf{x}_t$ (with aggregate $\pi_t^{C,*} = \boldsymbol{\alpha}^\top \Theta_\pi \mathbf{x}_t$), the output gap is $\tilde{y}_t^* = \mathbf{1}^\top K \Theta_\lambda \mathbf{x}_t$, and the reallocation wedge $(MK\mathbf{1})^\top \Theta_\eta \mathbf{x}_t$ enters the targeting criterion (52). The implementing nominal rate follows from the IS curve (22). Generically $\pi_t^{C,*} \neq 0$; symmetric stickiness ($\kappa_i = \bar{\kappa}$, so $MK\mathbf{1} = \mathbf{0}$) kills the wedge and restores $\pi_t^{C,*} \equiv 0$.

Timeless-perspective initialization. The physical states $(\boldsymbol{\chi}_{-1}, \mathbf{n}_{-1})$ are exogenous initial conditions; the commitment co-states are initialized at their steady-state values, $\boldsymbol{\lambda}_{-1} = \mathbf{0} \in \mathbb{R}^N$ and $\boldsymbol{\xi}_{-1} = \mathbf{0} \in \mathcal{R}$ (Woodford, 2003).

Numerical implementation. The system (D.7) is solved via generalized Schur decomposition (Klein, 2000). Projection of the relative-block components onto $\mathcal{R}$ uses the orthonormal basis of

Lemma 2 (eigenvectors $\{s_k\}_{k=2}^{N}$ of $Q = A^{1/2}FA^{-1/2}$).